Class 8
MATHEMATICS
Solutions

Table of Content

1.	Number System	1 – 14
2.	Surds	15 – 25
3.	Logarithms	26 – 30
4.	Polynomials	31 – 39
5.	Quadratic Equations	40 – 48
6.	Complex Numbers	49 – 78
7.	Theory Of Equations	79 – 84
8.	Sets	85 – 87
9.	RELATIONS	88 – 90

1. NUMBER SYSTEM

Worksheet – 1

Level – I

1.	A	4.	B	7.	C	10.	A
2.	A	5.	A	8.	D	11.	A
3.	A	6.	B	9.	D		

Level – II

1) B	2) A	3) C	4) C	5) A	6) C
7) D	8) B	9) C	10) B	11) A	12) B
13) C	14) B	15) A	16) C	17) B	18) B
19) A	20) C	21) D	22) C	23) B	24) D

1. $8400 = 2^4 \times 3^1 \times 5^2 \times 7^1$
   Number of divisors $= (4+1)(1+1)(2+1)(1+1) = (5)(2)(3)(2) = 60$
   Number of divisors except $1 = 60 - 1 = 59$.
2. $484 = 2^2 \times 11^2 \Rightarrow d(N) = (a+1)(b+1)$
   Here 11 is odd prime factor
   So, no. of odd factors $= (2+1) = 3$.
3. Conceptual 4. Conceptual
   5. $|x| = 5 \Rightarrow x = \pm 5$ 6. $210 = 2 \times 3 \times 5 \times 7$
7. $22932 \div 13 = 1764 \ \therefore \sqrt{1764} = 42$
8. $24 = 2^3 \times 3^1$ Number of factors $= (3+1)(1+1) = 8$
9. The minimum value of $(3x + 15) = 0 \Rightarrow 3x + 15 = 0 \Rightarrow 3x = -15 \Rightarrow x = -5$
10. $24 = 2^3 \times 3^1$ Sum of factors of $24 = \frac{(2^4-1)(3^2-1)}{(2-1)(3-1)} = \frac{15 \times 8}{1 \times 2} = 60$
11. The maximum value of the expression $27 - |9x - 8|$ will be maximum only when the value of $|9x - 8|$ is minimum but the minimum possible value of any $|k|$ is zero.
    Hence maximum value is
    $27 - |9x - 8| = 27 - 0 = 27$
12. The required numbers are 7,14,21, ……147 which are 21 in number.
13. $36 = 2^2 \times 3^2$
    Number of factors $= (2+1)(2+1) = 3 \times 3 = 9$

14. $90 = 2^1 \times 3^2 \times 5^1$ Here 3,5 are odd primes
    $\therefore$ Number of odd factors $= (2+1)(1+1) = 3 \times 2 = 6$
15. If a number is multiple of 2&5 it is a multiple of 10 , because 2&5 are co-primes so the answer is 10 .
16. $|17x - 8| - 9$ is minimum only when $|17x - 8|$ is zero
    Hence minimum value of $|17x - 8| = 0 - 9 = -9$
17. $270 = 2 \times 3^3 \times 5$
    Sum of factors of $270 = \frac{(2^{1+1}-1)(3^{3+1}-1)(5^{1+1}-1)}{(2-1)(3-1)(5-1)} = \frac{3\times80\times24}{1\times2\times4} = 720.$
18. The product of first 50 odd natural numbers ends with 5 .
19. $180 = 2^2 \times 3^2 \times 5^1$
    Number of factors $= (2+1)(2+1)(1+1) = 18$
    Hence, there are total $\frac{18}{2} = 9$ ways in which 180 can be expressed as a product of two factors
20. First number divisible by 5 is 335
    Last number divisible by 5 is 665
    So, difference is $665 - 335 = 330$
    and divide it with 5 $\Rightarrow \frac{330}{5} = 66$ So $66 + 1 = 67$ numbers
21. $2a - 9 = b + a \Rightarrow 2a - a = b + 9$
    $a = b + 9 \Rightarrow a - b = 9$ (or) $b - a = -9$
    Hence $|a - b| + |b - a| = |9| + |-9| = 18$
22. $7056 = 2^4 \times 3^2 \times 7^2$
    $\therefore$ Number of divisors of $7056 = (4+1)(2+1)(2+1) = 45$
    Product of factors $= (7056)^{45/2} = (84)^{45}$
23. $576 = 2^6 \times 3^2$
    Total number of factors $= (6+1)(2+1) = 21$
    So the number of ways of expressing 576 as a product of two distinct factors $= \left(\frac{21-1}{2}\right) = 10$
24. As we know that sum of three odd numbers cannot be even. i.e. one of the prime is even or 2. Let one of the other prime is p_1. Then the third prime is $p_1 + 36$.
    Now, according to the question, $2 + p_1 + p_1 + 36 = 100 \Rightarrow p_1 = 31$.
    Hence, the primes are 2,31,67.
25. Only for n = 1, the given numbers are primes.

Level – III

26) A,D 27) A 28) B 29) C 30) C

31) a-ps,t; b-q,r; c-p,t; $d-p,t$ 32)A , B, C 33) A 34)A, C, D 35) A

26. Conceptual.
27. Statement II is true. we have $32 = 2^5$. As it is in the form of 2^k, number of odd divisors is 1 . So, statement I is also true.
28. Conceptual
29. If $n = 2,2(2) + 1 = 5$ which is prime
30. Conceptual 31. Conceptual 32. If $n = 2$ then $n^3 - 1 = 7$
31. Conceptual 34. Conceptual 35. Conceptual

Worksheet – 2

Level – I

1.	D	4.	B	7.	D
2.	C	5.	D	8.	C
3.	C	6.	B	9.	A

1. sequence of a prime number is prime number
2. M and N are two co-primes, which of the following is/are true $HCF(M,N) = 1 \Rightarrow$ LCM (M,N)
3. LCM of $\frac{3}{4},\frac{6}{7},\frac{9}{8} \Rightarrow LCM = \frac{LCM \text{ of } Nr}{HCF \text{ of } Dr} = \frac{18}{1} = 18$
4. According to Wilson theorem, the remainder is ' 0 '.
5. $a_1 = b_1(\text{mod}n)$
   $a_2 = b_2(\text{mod}n)$
   $\Rightarrow (1) + (2)$
   $a_1 + a_2 = b_1(\text{mod}n) + b_2(\text{mod}n) \Rightarrow a_1 + a_2 = b_1 + b_2(\text{mod}n)$
6. $(17-1)! + 1$ is divisible by
   $(a-1)! + 1$ then divisible by $a \Rightarrow 16! + 1$
   It is divisible by 7
7. $4^3 = x(\text{mod}3)$ then $x = \cdots \Rightarrow 64 \equiv x(\text{mod}3) \Rightarrow \frac{64-x}{3} = \frac{60}{3} = 20x = 4$
8. $\Rightarrow a_1 + b_1 \equiv a_2 + b_2(\text{mod}n)$ 9. $3^{4k+3} \Rightarrow 7$

Level – II

1.	C	6.	C	11.	D	16.	B	21.	A
2.	C	7.	B	12.	C	17.	B	22.	A
3.	C	8.	A	13.	6	18.	B	23.	C
4.	A	9.	A	14.	B	19.	A	24.	A
5.	C	10.	B	15.	B	20.	C	25.	A,B,C

26. $a \to r; b \to s; c \to s; d \to p$
27. C
28. C
29. A
30. A
31. $a \to s; b \to r; c \to p; d \to q$
32. C,D
33. B,D
34. A

1. $\text{LCM} = \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}} = \frac{1}{100}$.
2. Conceptual.
3. cyclicity of 7 is 4
   unit's place digit in 7^{2012} is 7
4. $\text{HCF} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}} = \frac{1}{72}$
5. Largest scale would be used = HCF (496,624) = 16 m
6. $\text{a} = 17, \text{b} = 8, \text{c} = 3$
   $\text{a} - \text{b} = (17 - 8) = 9; \text{a} - \text{b} = 9, \text{c} = 3\ a - b \equiv 0(\text{mod}c)$
7. $a^2 = 16, b^4 = 16$
   $\text{a} = 4,\ \text{b} = 2\ \text{a} + \text{b} = 6\ 6 \equiv 1(\text{mod}5)$
8. If $\text{a} = 4; 4 + 1 = 5 \Rightarrow b = 5; 4 \equiv -1(\text{mod}5)$
9. cyclicity of $9 - 2 \Rightarrow 9^{2012} = 9^{\text{even}}$ $\Rightarrow$ Unit's place digit-1
10. $\phi(1) = 1; 1$ is only integer ≤ 1 and coprime to 1
    1,3,5,7 are only integer ≤ 8 and coprime to 8
11. $x, 2x, 3x$ are three numbers

$\frac{1}{6} \times 12 \ \therefore x = 2 \ \therefore$ The numbers of 2,4,6

12. $a^{p-1} \equiv 1(\text{mod}p)$. 13. As the cyclicity of ' 6 ' is ' 1 ', the units digit is 6 .

13. 1897 when divided by 15,27,35 and 42 leaves remainder 7

14. $\text{LCM} = 105, \text{HCF} = 1$

15. $10^n \equiv 1(\text{mod}9)$ for all $n \in N \Rightarrow 4333^3 \equiv 4^3(\text{mod}9) \equiv 64(\text{mod}9) \equiv 1(\text{mod}9)$

When 4333^3 is divided by 9 , the remainder is 1 ,

17. Let $\phi(n)$ be the mo, of integers less than η that are relatively prime to η. If η is a prime, then $\phi(n) = \eta - 1$.

if m and n are relatively prime then $\phi(mn) = \phi(m)\phi(n)$

$\phi(2002) = \phi(2 \times 7 \times 1113) = \phi(2)\phi(7)\phi(11)\phi(13) = (1)(6)(10)(12) = 720$

18. Let $n = 1$ then $8^1 - 3^1 = 5$ divisible by 5

19. Divisibility rule ' 8 ' Verification $\frac{1512}{8}$ = remainder ' 0 ' $\therefore x = 1$

20. Divisors of 12 are 1,2,3,4,6,12

$\therefore$ Sum of the cubes of the divisors of 12 are

$1^3 + 2^3 + 3^3 + 4^3 + 6^3 + 12^3 = 2044$

21. $24 \equiv 7(\text{mod}17); 24^{1947} \equiv 7^{1947}(\text{mod}17)$

but by Fermat's theorem $7^{1947} \equiv 1(\text{mod}17)$

$24^{16} \equiv 1(\text{mod}17)$ so $7^{1947} = 7^{16\times121\div1} = 7^{16\times121} \times 7^{11} \equiv 7^{11}(\text{mod}17)$

but $7^2 \equiv -2(\text{mod}17)$ So $7^{11} \equiv (72)^5, 7 \equiv (-2)^5(7) \equiv 14(\text{mod}17)$

Thus remainder is 14 .

22. As $(3,7) = 1, 3^6 \equiv 1(\text{mod}7) \Rightarrow 3^{96} \equiv 1(\text{mod}7)$.

But, $3^4 \equiv 4(\text{mod}7) \Rightarrow 3^{96+4} \equiv 1 \times 4(\text{mod}7) \Rightarrow 3^{100} \equiv 4(\text{mod}7)$.

23. $N = 3600 = 2^4 \times 3^2 \times 5^2 \Rightarrow \phi(N) = \phi(3600) = \phi(2^2 \times 3^2 \times 5^2)$

$= 3600\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = 3600 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 960.$

24. Conceptual.

25. $(LCM)(HCF)$ = product of two numbers

$32 \times 192 = 64 \times 6 \Rightarrow a = 64, b = 96 \Rightarrow a + b = 160$

26. a) $2^3 \equiv 1(\text{mod}7) \Rightarrow 2^{99} \equiv 1(\text{mod}7) \Rightarrow 2^{100} \equiv 2(\text{mod}7)$

b) $3^3 \equiv -1(\text{mod}7) \Rightarrow 3^{99} \equiv -1(\text{mod}7) \Rightarrow 3^{100} \equiv -3(\text{mod}7) \Rightarrow 3^{100} \equiv 4(\text{mod}7)$

c) $4^3 \equiv 1(\text{mod}7) \Rightarrow 4^{99} \equiv 1(\text{mod}7) \Rightarrow 4^{100} \equiv 4(\text{mod}7)$

d) $6 \equiv -1(\text{mod}7) \Rightarrow 6^{100} \equiv 1(\text{mod}7)$

27. $\phi(5 \cdot 3) = \phi(5) \cdot \phi(3)$

28. $\phi(4) = 2, \phi(5) = 4, \phi(3) = 2, \phi(7) = 6 \Rightarrow \phi(4 \cdot 5 \cdot 3 \cdot 7) = 2 \times 4 \times 2 \times 6 = 96$

29. $N = 2^4 \times 3^2 \times 5^2 \Rightarrow \phi(N) = 3600 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 960$

30. As 5 and 7 are co-primes, $5^6 \equiv 1(\text{mod}7) \Rightarrow 15625 \equiv 1(\text{mod}7)$.

31. a) $2^{100} = (2^4)^{25} = (16)^{25}$. Unit's digit is 6 .

    b) $3^{100} = (3^4)^{25} = (81)^{25}$. Unit's digit is 1 .

    c) unit's digit is 5 d) unit's digit is 7

32. $2^{36} - 1 = (2^3)^{12} - (1)^{12} = 8^{12} - 1^{12}$ is divisible by 9 .

    $6 + 8 + a + 1 + 9 + 4 + 7 + 6 + 7 + 3 + 5$ is divisible by 9 . Therefore, $a = 7$.

33. $2^{48} - 1 = (2^{24} - 1)(2^{24} + 1) = (2^{12} - 1)(2^{12} + 1)(2^8 + 1)(2^{16} - 2^8 + 1)$

    $= (2^6 - 1)(2^6 + 1)(2^4 + 1)(2^8 - 2^4 + 1)(2^8 + 1)(2^{16} - 2^8 + 1)$

    $= 63 \times 65 \times 17 \times 241 \times 257 \times (2^{16} - 2 + 1)$.

    So, $(2^{48} - 1)$ is divisible by 63 and 65 which lie between 60 and 70 .

34. Unit digit of 3^{2720} is 1 .

Worksheet – 3

Level – I

1.	B	4.	A	7.	D	10.	C
2.	B	5.	B	8.	B	11.	B
3.	B	6.	A	9.	B	12.	A

1. $\frac{15}{26}$ as 15 and 26 has common factor as " 1 "

2. $\frac{33}{10}$

3. 3. 1

4. Additive inverse of $\left(\frac{-2}{3}\right)$ is $\frac{2}{3}$

5. Additive inverse of $\left(\frac{-a}{b}\right)$ is $\frac{a}{b}$

6. $\frac{-20}{28}$

7. 0{All}

8. 4/5

9. $\sqrt{ab}$

10. a

11. Even prime number $= 2 \quad 2^3 = 8$

12. 8

Level – II

1) A 2)B 3)C 4)C 5)B 6)B 7)A 8)C 9)A 10)D 11)D 12)A 13)A 14)D

15)A 16)C 17)A 18)B 19)B 20)A 21)A 22)B 23)A 24)A 25)A,B,C 26)A,B

27)A 28)B 29)A 30)C 31) a-q, b-r, c-t; d-p 32)A,D 33)A,B,C 34)C

35) a-p, b-q, c-r; d-t 36) a-p,r,s,t, b-p,r, c-p,r; d-p,r

1. N MNKLMRational number between $a\&b = \frac{1}{2}(a+b) = \frac{1}{2}\left[\frac{2}{3}+\frac{3}{4}\right] = \frac{1}{2}\left[\frac{8+9}{12}\right] = \frac{17}{24}$

2. $\frac{5426}{100} = \frac{2713}{50}$

3. $5.\bar{2}$

   Let x = 5.222 .....(1)

   10x = 52.222.....(2)

   10x = 52.222.....

   $-x = 5.222.....$

   $9x = 47$

   $x = \frac{47}{9}$

4. multiplicative inverse of $\frac{8}{3} = \frac{3}{8}$

   additive inverse of $\frac{8}{3} = \frac{-8}{3}$

   sum $= \frac{3}{8} + \left(\frac{-8}{3}\right) = \frac{3}{8} - \frac{8}{3} = \frac{9-64}{24} = \frac{-55}{24}$

5. $\sqrt{625} = 25$ 6. 196

7. $\begin{bmatrix} 7 \\ 7 \\ 7 \end{bmatrix} \begin{vmatrix} 49 \\ 7 \\ 1 \end{vmatrix}$ $= 2 \times 7 = 14$

8. Period=94; periodicity=2; sum=96.

9. Sum of period and periodicity = 94 + 2 = 96.

10. Let x = 3.12323 – – –.

    $100x = 312.323 - - -.$

    100x = 312.323 – – –

    –x = 3.123 – – – –.

    $\overline{99x = 309.2}$

$x = \frac{309.2}{99} = \frac{3092}{990} = \frac{1046}{495}\ \frac{1546}{495}$

10. $\frac{169}{117} = 1.\overline{4}$; Period = 4

11. $\frac{1}{50}$ can be written as $\frac{2}{100} = 0.02, \therefore$ its terminating

12. $0.\overline{54} + 4.\overline{7} = \frac{54}{99} + \frac{47-4}{9} = \frac{54}{99} + \frac{43\times11}{9\times11} = \frac{54+473}{99} = \frac{527}{99}$

13. $10.\overline{36} = \frac{p}{q} = \frac{1036-10}{99} = \frac{1026}{99} = \frac{342}{33} = \frac{114}{11}$

$(p+q)(p-q) =? \ (114+11)(114-11) = 125 \times 103 = 12875$

14. $\sqrt{108.16} = \sqrt{\frac{10816}{100}} = \sqrt{\left(\frac{104}{10}\right)^2} = 10.4$

15. $\sqrt{2.88} \times \sqrt{1.28} = \sqrt{\frac{288}{100} \times \frac{128}{100}} = \frac{192}{100} = 1.92$

16. $3087 = 3 \times 3 \times 7 \times 7 \times 7$ $\therefore$ with 3 we should multiply 3087 to make it as perfect cube.

17. $A = 2^x, B = 4^y, c = 8^z; x = 0.\overline{1}, y = 0.\overline{4}, z = 0.\overline{6}$

$x = \frac{1}{9}, y = \frac{4}{9}, z = \frac{6}{9}; A \times B \times C = 2^{\frac{1}{9}} \times 4^{\frac{4}{9}} \times 8^{\frac{6}{9}} = 2^{\frac{1}{9}} \times 2^{2\left(\frac{4}{9}\right)} \times 2^{3\left(\frac{6}{9}\right)} = 2 = 2^{\frac{27}{9}} = 2^3 = 8$

18. $A + B + C = \frac{87}{32} + \frac{5}{8} + \frac{3}{4} = 2.71875 + 0.625 + 0.75 = 4.09375 = \frac{409375}{100000}$

19. $\sqrt[3]{288} \times \sqrt[3]{432} \times \sqrt[3]{648}$

2	288	2	432	2	648
2	144	2	216	2	324
2	72	2	108	2	162
2	108	2	54	3	81
3	18	3	27	3	27
3	3	9	3	9	
9	3	3	3	3	
3	3	1		1	

$= 3\sqrt{2^{12} \times 3^9} = 2^4 \times 3^3 = 16 \times 27 = 432$

20. $512 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \Rightarrow a^2 = 5/2 = 2^9 = 8^3 \Rightarrow a = 8 \Rightarrow b = 3$

$\sqrt[b]{a} = \sqrt[3]{8} = 2$

21. $\sqrt[3]{1953.125} + \sqrt[3]{42.875} = \sqrt[3]{(12.5)^3} + \sqrt[3]{(3.5)^3} = 12.5 + 3.5 = 16$

22. $a^2 = b^3 = 64$

$a^2 = 64 = 8^2 \quad b^3 = 64$
$a = 4 \qquad\quad = 4^3$

$c^4 = d^2 = 16$

$\underline{c^4 = 2^4\ d^2 = 4^2}$
$\underline{c = 2 \quad d = 4}$

$x + y = 5$

$p + q = g - h = 3$

$m + n = 1$

$x = \dfrac{a^{m+n}}{b^{p+q}} = \dfrac{8^1}{4^3} = \dfrac{8}{8} = \dfrac{1}{8}$

$x^+ + y = 5, y = \dfrac{c^{x+y}}{d^{g-h}} = \dfrac{2^5}{4^3} = \dfrac{2^5}{2^6} = \dfrac{1}{2}$

Rational number between $\frac{1}{8}\&\frac{1}{2} = \frac{1}{2}(a+b) = \frac{1}{2}\left(\frac{1}{8}+\frac{1}{2}\right) = \frac{1}{2}\left(\frac{5}{8}\right) = \frac{5}{32} \quad \frac{5}{16}$

23. $x = 0.\overline{82} \Rightarrow x = \frac{82}{99}; y = 0.\overline{17} \Rightarrow y = \frac{17}{99}$

$A = 0.\overline{7} \Rightarrow A = \dfrac{7}{9}; B = 0.\overline{2} \Rightarrow B = \dfrac{2}{9}$

$[(x+y)+(A+B)]^{(x+y)+(A+B)} = \left[\left(\dfrac{82}{99}+\dfrac{17}{99}\right)+\left(\dfrac{7}{9}+\dfrac{2}{9}\right)\right]^{\left[\left(\frac{82}{99}+\frac{17}{99}\right)+\left(\frac{7}{9}+\frac{2}{9}\right)\right]} = (1+1)^{1+1} = 2^2 = 4$

24. $0.\overline{2} + 0.2\overline{3} = \frac{2}{9} + \frac{23}{99} = \frac{22+23}{99} = \frac{45}{99} = 0.\overline{45}$

25. Conceptual.

26. Terminating decimals

a) $\frac{5}{32} = \frac{5}{2^5}$

b) $\frac{27}{80} = \frac{27}{2^4 \times 5^1}$ [As the denominations of form $2^m \times 5^n$]

27. Conceptual.

28. $\frac{3}{11} = 0.2727 \ldots\ldots = 0.\overline{27}$; Period = 27

29. $\frac{1}{6} = 0.1666 \ldots\ldots = 0.1\overline{6}$

Under the repeating block only one digit 6 is present periodicity = 1

30. $\frac{5}{13} = 0.\overline{384615}$; No of digits under the repeating block = 6

Periodicity = 6.

31. Conceptual.

32. $0.\overline{2} + 0.\overline{3} + 0.\overline{4} + 0.\overline{9} + 0.\overline{39} = \frac{2}{9} + \frac{3}{9} + \frac{4}{9} + \frac{9}{9} + \frac{39}{99}$

multiply by 11 .

$$= \frac{22}{99} + \frac{33}{99} + \frac{44}{99} + \frac{99}{99} + \frac{39}{99} = \frac{22+33+44+99+39}{99}$$

$= \frac{237}{99}$……(a) $\quad = \frac{79}{33} = 2\frac{13}{33}$……(b)

33. Conceptual

34. Conceptual

35. $1.5x = 0.04y$

$$\frac{1.5}{0.04} = \frac{y}{x} \Rightarrow \frac{y}{x} = \frac{75}{2} \Rightarrow \frac{\frac{y}{x} - 1}{\frac{y}{x} + 1} = \frac{\frac{75}{2} - 1}{\frac{75}{2} + 1} = \frac{\frac{73}{2}}{\frac{77}{2}} = \frac{73}{77} \text{ or } \frac{730}{770}$$

36.

$x = 0.\overline{90}$

$= \frac{90}{99} = \frac{10}{11}$, $y = \frac{18}{99} = \frac{2}{11}$

(a) $x \div y = \frac{10}{11} \div \frac{2}{11} = 5$

c) $x - y = \frac{10}{11} - \frac{2}{11} = \frac{8}{11}$

d) $xy = \frac{10}{11} \times \frac{2}{11} = \frac{20}{121}$

Worksheet – 4

Level – I

1.	D	4.	B	7.	C	10.	A
2.	C	5.	C	8.	A	11.	B
3.	B	6.	C	9.	B		

1. $[x] = [2.57] = 2 \Rightarrow [x]^{[x]} = 2^2 = 4$
2. $x = 0.32 \Rightarrow \{x\} = 0.32$
3. $x = -5.35 \Rightarrow [x] = [-5.35] = [-6 + 0.65] = -6$
4. $x = -4.3 \Rightarrow [x] = [-4.3] = -5 \Rightarrow \{[x]\} = \{-5\} = 0$ 5. $[x] + \{x\} = x \Rightarrow \therefore x = 2019.19$
5. $x = 2.71 \Rightarrow [x] = 2 \Rightarrow \{[x]\} = \{2\} = 0 \Rightarrow \therefore |\{[x]\}| = 0$
6. Highest power of 3 in 10! $= \left[\frac{10}{3}\right] + \left[\frac{10}{3^2}\right] = [3.33] + [1.11] = 3 + 1 = 4$
7. Number of zero's at the end of 8! = 1 ⇒ [as 8!=40320]
8. Highest power of 15 in 100 !

   Power of 3 in 100! $= \left[\frac{100}{3}\right] + \left[\frac{100}{3^2}\right] + \left[\frac{100}{3^3}\right] + \left[\frac{100}{3^4}\right] = 33 + 11 + 3 + 1 = 48$

   Power of 5 in 100! $= \left[\frac{100}{5}\right] + \left[\frac{100}{5^2}\right] = 20 + 4 = 24$

   $100! = 3^{48} \times 5^{24} \times \ldots\ldots = 15^{24} \times \ldots..$ Power of 15 in 100! = 24
10. $|3x - 1| = 2$

    $3x - 1 = 2,\ 3x - 1 = -2$

    $x = 1,\ x = \frac{-1}{3}$

11. $|x-2|>6$

$x-2>6$ and $x-2<-6$

$x>8$ and $x<-4$

$\therefore x \in (-\infty,-4) \cup (8,\infty)$

Level – II

1)D	2)A	3)D	4)D	5)B	6)C	7)C
8)C	9)A	10)B	11)B	12) B	13)A	14)B
15) D	16)C	17)A	18)B	19)B	20)B	21)C
22)B	23)B	24) C	25) A,B	26)A,B	27) D	
28) a-t, b-p, c-s; d-t		29) C	30) B	31)C	32)D	33)A,D
34)a-q, b-s, c-p; d-r		35)22	36) A			

1. Conceptual.
2. $|3x-2|=10 \Rightarrow 3x-2=\pm 10 \Rightarrow 3x-2=10,3x-2=-10$

   $x=4, x=\frac{-8}{3} \Rightarrow x=\frac{-8}{3},4$
3. Highest power of 3 in 500 !

   $=\left[\frac{500}{3}\right]+\left[\frac{500}{3^2}\right]+\left[\frac{500}{3^3}\right]+\left[\frac{500}{3^4}\right]+\left[\frac{500}{3^5}\right]=166+55+18+6+2=247$
4. Power of 2 in 33! $=\left[\frac{33}{2}\right]+\left[\frac{33}{2^2}\right]+\left[\frac{33}{2^3}\right]+\left[\frac{33}{2^4}\right]+\left[\frac{33}{2^5}\right]=16+8+4+2+1=31$

   `33 ! is divisible by 2^{31}
5. 2019! Highest power of 10 in 2019 !

   $=\left[\frac{2019}{5}\right]+\left[\frac{2019}{5^2}\right]+\left[\frac{2019}{5^3}\right]+\left[\frac{2019}{5^4}\right]=403+80+16+3=502$
6. $x=0.23 \Rightarrow [x]=0.\{x\}=0.23 \Rightarrow x\cdot[x].|x|\cdot\{x\}=x\times 0\times|x|\times\{x\}=0$
7. $y=(-1)^{[x]} \Rightarrow$ As x is real $\Rightarrow [x]$ can be any integer
8. $[x]\cdot[x]\cdot[x]\cdot\{x\}=[2.57]^3\times\{2.57\}=(2)^3\times(0.57)=8\times 0.57=4.56$

$3\underline{|35}$

$3\underline{|11} \rightarrow 2$ $\quad (1022)_3$

$3\underline{|3} \rightarrow 2$

$3\underline{|1} \rightarrow 0$

$0 \rightarrow 1$

$(17)_{10}$

$2\underline{|17}$

$2\underline{|8} \rightarrow 1$

$2\underline{|4} \rightarrow 0$

$2\underline{|2} \rightarrow 0$ $\quad (10001)_2$

10. $1 \rightarrow 0$

11. No. of zeros 6250 ! end with

$= \left[\frac{6250}{5}\right] + \left[\frac{6250}{5^2}\right] + \left[\frac{6250}{5^3}\right] + \left[\frac{6250}{5^4}\right] + \left[\frac{6250}{5^5}\right] = 1250 + 250 + 50 + 10 + 2 = 1562$

12. $\left[\frac{35}{5}\right] + \left[\frac{35}{5^2}\right] = 7 + 1 = 8.$

13. $x = -7.54 \Rightarrow [x] = [-7.54] = -8 \Rightarrow \{[x]\} = \{-8\} = 0 = \cos 90°$

14. $\therefore x = -3.4$

$|x|\{x\} + x[x] = (3.4)(0.4) + (-3.4)(-4) = 1.36 + 13.6 = 14.96$

15. Conceptual.

16. $|x - 2| > 6 \Rightarrow x - 2 < -6$ and $x - 2 > 6 \Rightarrow x < -4$ and $x > 8$

$x \in (-\infty, -4) \cup (8, \infty)$

17. $(38)_{10}$

$\lfloor 0 \rightarrow 1$

18. $(25)_n \times (31)_n = (1015)_n$

$(2n + 5)(3n + 1) = n^3 + n + 5$ $(25)_n = (2 \times n^1 + 5 \times n^0) = (2n + 5)$

$6n^2 + 17n + 5 = n^3 + n + 5$

$(31)_n = (3 \times n^1 + 1 \times n^0) = (3n + 1)$

$\therefore n^3 - 6n^2 - 16n = 0$

$n[n^2 - 6n - 16] = 0$

$n = 0,8,-2$

$\therefore n = 8$

19. $\left|\frac{8-x}{4}\right| \leq 4$

$\frac{8-x}{4} \geq -4$ or $\frac{8-x}{4} \leq 4$

$8 - x \geq -16\ 8 - x \leq 16$

$24 \geq x\ 8 - 16 \leq x$

$x \leq 24\ x \geq -8$

$x \in [-8,24]$

20. $\{2.4\} + \{-1.6\} + \{3.2\} = 0.4 + 0.4 + 0.2 = 1$

21. If $x = 9.3$

$[x] = [9.3] = 9$

$\{x\} + [x] = x = 9.3$

22. $-(x-9) + (5-x) + x + 9 + 5 + x = -\not{x} + 9 + 5 - \not{x} + \not{x} + 9 + 5 + \not{x} = 28$

23. $\{1.4\} + \{-0.6\} + \{2.6\} + \{-1.4\} + \{3.2\} + \{-2.8\} = 0.4 + 0.4 + 0.6 + 0.6 + 0.2 + 0.2 = 2.4$

24. $[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + [\sqrt{4}] + [\sqrt{5}] + [\sqrt{6}]$

$= 1 + [1.414] + [1.732] + [2] + [2.21] + [2.42] = 1 + 1 + 1 + 2 + 2 + 2 = 9$

25. $\{2.4\} + \{-1.6\} + \{3.2\} = 0.4 + 0.4 + 0.2 = 1 = \text{Sin } 90° = \text{Cos } 0^0$

26. $x = -4.7 \Rightarrow \{x\} = \{-4.7\} = 0.3 \Rightarrow [0.3] = 0 = \cos 90°$

27. Conceptual.

28. Conceptual.

29. $\left[\frac{10}{3}\right] + \left[\frac{10}{3^2}\right] + [+ \cdots \ldots . = 3 + 1 = 4]$ 30. $\left[\frac{100}{5}\right] + \left[\frac{100}{5^2}\right] = 20 + 4 = 24$

30. $\left[\frac{2005}{5}\right] + \left[\frac{2005}{5^2}\right] + \left[\frac{2005}{5^3}\right] + \left[\frac{2005}{5^4}\right] = 401 + 80 + 16 + 3 = 500$

31. Highest power of 2 in 33! $= \left[\frac{33}{2}\right] + \left[\frac{33}{2^2}\right] + \left[\frac{33}{2^3}\right] + \left[\frac{33}{2^4}\right] + \left[\frac{33}{2^5}\right] = 16 + 8 + 4 + 2 + 1 = 31$

$\therefore n = 31$

33. Highest power of 5 in 100! $= \left[\frac{100}{5}\right] + \left[\frac{100}{5^2}\right] + \left[\frac{100}{5^3}\right] = 20 + 4 + 0 = 24$

34. $(101 \mid 110)_2$

$2^2 2^1 2^0 2^2 2^1 2^0$

$4 + 1 = 5; 4 + 2 = 6$

a) $(101 \mid 110)_2 = (56)_8$

$101 \mid 110$

$2^2 2^1 2^0 2^2 2^1 2^0$

$5 \times 10 + 6 \times 1 = 56$

b) $(110 \mid 110) =$

$110 \mid 110$

$2^0 2^2 2^1 2^0 2^2 2^1 2^0$

$100 \times (4 + 0 + 1)(4 + 2 + 0) \mid = (156)_8$

$\times 10$

c) $(101 \mid 101) = (1 \times 2^0)^{100}[0 \times 2^2) + 1 \times 2^1 + 1 \times 2^0] \times 10 + [1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0]$

d) $111 \mid 110 = [1 \times 2^0] \times 100 + [1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0] \times 10 + (1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0)$

$= 100 + 70 + 6 = 176$

35. Exponent of 3 in 50! $= \left[\frac{50}{3}\right] + \left[\frac{50}{3^2}\right] + \left[\frac{50}{3^3}\right] = 16 + 5 + 1 = 22$

36. Exponent of 7 in 400! $= \left[\frac{400}{7}\right] + \left[\frac{400}{7^2}\right] + \left[\frac{400}{7^3}\right]$

$S = 57 + 8 + 1 = 66$

$((3!)!) = (720)!$

Exponent of 3 in $(720)! = \left[\frac{720}{3}\right] + \left[\frac{720}{3^2}\right] + \left[\frac{720}{3^3}\right] + \left[\frac{720}{3^4}\right] + \left[\frac{720}{3^5}\right] + \left[\frac{720}{3^6}\right]$

$= 240 + 80 + 26 + 8 + 2$

$t = 356$

$s + t = 66 + 356 = 422 = \frac{844}{2!}$

2. SURDS

Worksheet – 1

Level – I

1. A
2. D
3. B
4. B
5. A
6. D
7. C
8. C
9. B
10. B

Level – II

1) B 2) D 3) C 4) B 5) D 6) A 7) C

8) A

9) B 10) B 11) C 12) A 13) B 14) B 15) B

16) D 17) C 18) B 19) C

1. $\sqrt[3]{24}+\sqrt[4]{81}-\sqrt[3]{192}+\sqrt[5]{32} \Rightarrow \sqrt[3]{8\times 3}+\sqrt[4]{3^4}-\sqrt[3]{2^6\times 3}+\sqrt[5]{2^5} \Rightarrow 2\cdot\sqrt[3]{3}+3-4\times\sqrt[3]{3}+2 \Rightarrow 5-2\cdot\sqrt[3]{3}$
2. $\sqrt[3]{108}=\sqrt[3]{3^3\times 2^2}=\sqrt[3]{3^3}\times\sqrt[3]{4} \Rightarrow 3\times\sqrt[3]{4}$
3. Pure surd form of $2\cdot\sqrt[3]{5}=\sqrt[3]{2^3\times 5}=\sqrt[3]{8\times 5}=\sqrt[3]{40}$
4. $4\sqrt{3}-3\cdot\sqrt{12}+2\cdot\sqrt{75}$
   $\Rightarrow 4\sqrt{3}-3.2\sqrt{3}+2\cdot\sqrt{25\times 3} \Rightarrow 4\sqrt{3}-6\sqrt{3}+2\cdot 5\cdot\sqrt{3} \Rightarrow 4\sqrt{3}-6\sqrt{3}+10\sqrt{3}$
   $\Rightarrow 8\sqrt{3}$
5. If $3\sqrt{a+1}, 4\sqrt{5}$ are similar surds then $a+1=5 \Rightarrow a=4$
6. $\sqrt[3]{a+1}\times\sqrt[4]{5}$ is a mixed surd then the least possible integral value of a is 7 because $\sqrt[3]{a+1}\times\sqrt[4]{5}=\sqrt[3]{7+1}\times\sqrt[4]{5} \Rightarrow \sqrt[3]{8}\times\sqrt[4]{5}=2\cdot\sqrt[4]{5}$ is a mixed surd
7. $\sqrt{1.69}=\sqrt{(1.3)^2}=1.3 \ \therefore \sqrt{1.69}$ is not a surd
8. $x=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}=\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}\times\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}=\frac{(\sqrt{5}+\sqrt{3})^2}{5-3}=\frac{5+3+2\sqrt{15}}{2}$
   $x=\frac{8+2\sqrt{15}}{2}=\frac{2(4+\sqrt{15})}{2}=4+\sqrt{15} \Rightarrow \frac{1}{x}=\frac{1}{4+\sqrt{15}}=\frac{1}{4+\sqrt{15}}\times\frac{4-\sqrt{15}}{4-\sqrt{15}}$
   $=\frac{4-\sqrt{15}}{16-15}=4-\sqrt{15}$
   $x^2+\frac{1}{x^2}=(4+\sqrt{15})^2+(4-\sqrt{15})^2=2\left(4^2+(\sqrt{15})^2\right)=2(16+15)=62$

9. $\frac{\sqrt{2}-1}{\sqrt{2}+1}=a+b\sqrt{2} \Rightarrow \frac{\sqrt{2}-1}{\sqrt{2}+1}\times\frac{\sqrt{2}-1}{\sqrt{2}-1}=a+b\sqrt{2}\Rightarrow\frac{(\sqrt{2}-1)^2}{2-1}=a+b\sqrt{2}\Rightarrow 2+1-2\sqrt{2}=a+b\sqrt{2}\Rightarrow 3-2\sqrt{2}=a+b\sqrt{2}$
$\therefore a=3,b=-2$

10. $\frac{2}{\sqrt[3]{9}+\sqrt[3]{3}+1}=\frac{2}{3^{\frac{2}{3}}+3^{\frac{1}{3}}+1}=\frac{2}{\left(3^{\frac{1}{3}}\right)^2+3^{\frac{1}{3}}\times 1+(1)^2}$
$\Rightarrow\frac{2\left(3^{\frac{1}{3}}-1\right)}{\left[\left(3^{\frac{1}{3}}\right)^2+3^{\frac{1}{3}}\times 1+(1)^2\right]\times\left(3^{\frac{1}{3}}-1\right)}\Rightarrow\frac{2\left(3^{\frac{1}{3}}-1\right)}{\left(3^{\frac{1}{3}}\right)^3-(1)^3}=\frac{2\left(3^{\frac{1}{3}}-1\right)}{2}\Rightarrow(\sqrt[3]{3}-1)=\sqrt[3]{a}+b$
$\therefore a=3,b=-1\ \ \therefore(a,b)=(3,-1)$

11. $\alpha=\sqrt{7}-\sqrt{5},\beta=\sqrt{13}-\sqrt{11}\Rightarrow\frac{1}{\alpha}=\frac{1}{\sqrt{7}-\sqrt{5}}\times\frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}+\sqrt{5}}=\frac{\sqrt{7}+\sqrt{5}}{7-5}=\frac{\sqrt{7}+\sqrt{5}}{2}$
$\Rightarrow\frac{1}{\beta}=\frac{1}{\sqrt{13}-\sqrt{11}}\times\frac{\sqrt{13}+\sqrt{11}}{\sqrt{13}+\sqrt{11}}=\frac{\sqrt{13}+\sqrt{11}}{13-11}=\frac{\sqrt{13}+\sqrt{11}}{2}\Rightarrow\frac{1}{\beta}>\frac{1}{\alpha}\Rightarrow\beta<\alpha\Rightarrow\alpha>\beta$

12. $\frac{4}{3\sqrt{3}-2\sqrt{3}}+\frac{3}{3\sqrt{3}+2\sqrt{2}}\Rightarrow\frac{4}{3\sqrt{3}-2\sqrt{2}}\times\frac{3\sqrt{3}+2\sqrt{2}}{3\sqrt{3}+2\sqrt{2}}+\frac{3}{3\sqrt{3}+2\sqrt{2}}\times\frac{3\sqrt{3}-2\sqrt{2}}{3\sqrt{3}-2\sqrt{2}}$
$\Rightarrow\frac{12\sqrt{3}+8\sqrt{2}}{27-8}+\frac{9\sqrt{3}-6\sqrt{2}}{27-8}\Rightarrow\frac{12\sqrt{3}+8\sqrt{2}+9\sqrt{3}-6\sqrt{2}}{19}=\frac{21\sqrt{3}+2\sqrt{2}}{19}$
$\Rightarrow\frac{21\times 1.732+2\times 1.414}{19}=\frac{36.372+2.828}{19}=\frac{39.2}{19}=2.0631$

13. $p=\sqrt{32}-\sqrt{24}\Rightarrow\sqrt{16\times 2}-\sqrt{4\times 6}=4\sqrt{2}-2\sqrt{6}$
$q=\sqrt{50}-\sqrt{48}=\sqrt{25\times 2}-\sqrt{16\times 3}=5\sqrt{2}-4\sqrt{3}$
$p=\sqrt{32}-\sqrt{24}\Rightarrow p^2=32+24-2\sqrt{32\times 214}\Rightarrow 56-2\cdot\sqrt{768}$
$q=\sqrt{50}-\sqrt{48}\Rightarrow q^2=50+48-2\sqrt{50\times 48}\Rightarrow 98-2\sqrt{2400}$
$\therefore q^2<p^2\Rightarrow p>q$

14. $x=\sqrt{5},y=\sqrt[4]{10},z=\sqrt[3]{6}$
$\Rightarrow x=\sqrt{5}=5^{\frac{1}{2}\times\frac{6}{6}}=5^{\frac{6}{12}}=\sqrt[12]{5^6}=\sqrt[12]{15625}\Rightarrow y=\sqrt[4]{10}=10^{\frac{1}{4}\times\frac{3}{3}}=10^{\frac{3}{12}}=\sqrt[12]{10^3}=\sqrt[12]{1000}$
$\Rightarrow z=\sqrt[3]{6}=6^{\frac{1}{3}\times\frac{4}{4}}=6^{\frac{4}{12}}=\sqrt[12]{6^4}=\sqrt[12]{1296}\ \ \therefore y<z<x$ 15. $\frac{\sqrt{5}-1}{\sqrt{5}+1}-\frac{\sqrt{5}+1}{\sqrt{5}-1}=a+b\sqrt{5}\Rightarrow\frac{(\sqrt{5}-1)^2-(\sqrt{5}+1)^2}{5-1}=a+b\sqrt{5}\Rightarrow\frac{-4\cdot\sqrt{5}\cdot 1}{4}=a+b\sqrt{5}$
$\Rightarrow 0+\sqrt{5}=a+b\sqrt{5}\ \ \therefore a=0,b=-1$
$\therefore(a+b)(a-b)\div(a^2+b^2)=(0+1)(0-1)\div(0^2+1^2)=\frac{-1}{1}=-1$

16. $\frac{5+2\sqrt{3}}{7+4\sqrt{3}}=a+b\sqrt{3}\Rightarrow\frac{5+2\sqrt{3}}{7+4\sqrt{3}}\times\frac{7-4\sqrt{3}}{7-4\sqrt{3}}=a+b\sqrt{3}$
$\Rightarrow\frac{35-20\sqrt{3}+14\sqrt{3}-8\times 3}{49-16\times 3}=a+b\sqrt{3}\Rightarrow\frac{35-6\sqrt{3}-24}{1}=a+b\sqrt{3}$
$\Rightarrow 11-6\sqrt{3}=a+b\sqrt{3}\ \ \therefore a=11,b=-6$
$\therefore a^2+b^2+2ab=(a+b)^2=(11-6)^2=5^2=25$

17. $x=1+\frac{1}{1+\frac{1}{1+\cdots}}$
$\Rightarrow x=\frac{1}{1}+\frac{1}{x}\Rightarrow\frac{x}{1}=\frac{x+1}{x}\Rightarrow x^2=x+1\Rightarrow x^2-x-1=0\Rightarrow a=1;\mathrm{b}=-1;\mathrm{c}=-1$
$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{1\pm\sqrt{(-1)^2+4\times 1\times 1}}{2\times 1}=\frac{1\pm\sqrt{1+4}}{2}=\frac{1\pm\sqrt{5}}{2}$

18. $x = \frac{5+\sqrt{6}}{5-\sqrt{6}} = \frac{5+\sqrt{6}}{5-\sqrt{6}} \times \frac{5+\sqrt{6}}{5+\sqrt{6}} = \frac{25+6+10\sqrt{6}}{25-6} = \frac{31+10\sqrt{6}}{19}$

$y = \frac{5-\sqrt{6}}{5+\sqrt{6}} = \frac{5-\sqrt{6}}{5+\sqrt{6}} \times \frac{5-\sqrt{6}}{5-\sqrt{6}} = \frac{25+6-10\sqrt{6}}{25-6} = \frac{31-10\sqrt{6}}{19}$

$\therefore x+y = \frac{31+10\sqrt{6}}{19} + \frac{31-10\sqrt{6}}{19} = \frac{31+10\sqrt{6}+31-10\sqrt{6}}{19} = \frac{62}{19}$

$\therefore x-y = \frac{31+10\sqrt{6}}{19} - \left(\frac{31-10\sqrt{6}}{19}\right) = \frac{31+10\sqrt{6}-31+10\sqrt{6}}{19} = \frac{20\sqrt{6}}{19}$

$\therefore (x+y) \div (x-y) = \frac{62}{19} \div \frac{20\sqrt{6}}{19} = \frac{62}{20\sqrt{6}} = \frac{31}{10\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{31\sqrt{6}}{60}$

19. $x = 3 + 3^{\frac{1}{3}} + 3^{\frac{2}{3}} \Rightarrow x - 3 = 3^{\frac{1}{3}} + 3^{\frac{2}{3}}$

cubing on both sides $\Rightarrow (x-3)^3 = \left(3^{\frac{1}{3}} + 3^{\frac{2}{3}}\right)^3 \Rightarrow x^3 - 3 \cdot x^2 \cdot 3 + 3 \cdot x \cdot 3^2 -$

$3^3 = \left(3^{\frac{1}{3}}\right)^3 + \left(3^{\frac{2}{3}}\right)^3 + 3 \cdot 3^{\frac{1}{3}} \cdot 3^{\frac{2}{3}}\left(3^{\frac{1}{3}} + 3^{\frac{2}{3}}\right)$

$\Rightarrow 3 + 3^2 + 3 \times 3(x-3) \left(\because 3^{\frac{1}{3}} + 3^{\frac{2}{3}} = x - 3\right)$

$\Rightarrow x^3 - 9x^2 + 27x - 27 = 12 + 9x - 27 \Rightarrow x^3 - 9x^2 + 27x - 9x = 12$

adding 20 on both sides $\Rightarrow x^3 - 9x^2 + 27x - 9x + 20 = 12 + 20 = 32$

Level – III

20) B,D 21) 5 22) B 23) B 24) C 25) a-p; b-s; c-s; d-q,s 26) B 27) A 28) A 29) a-p; b-q; c-r; d-s 30. a-s, b-t, c-p, d-r, e-q

31) a-q, b-r, c-s, d-s

20. The R.F of $\sqrt[3]{3}$ is $\sqrt[3]{3^{3-1}} = \sqrt[3]{3^2} = \sqrt[3]{9}$ (TheR.F of $\because \sqrt[m]{a^n} = \sqrt[m]{a^{m-n}}$)
21. The entire form of $3 \cdot \sqrt{5}$ is $\sqrt[a]{b}$

$= 3.\sqrt{5} = \sqrt{5 \times 3^2} = \sqrt{45}$

$\Rightarrow \sqrt[a]{b} = \sqrt{45} \ \therefore a = 2, b = 45$

$\therefore \frac{b-5a}{7} = \frac{45-5\times 2}{7} = \frac{45-10}{7} = 5$

22. The order of the surd $\sqrt[3]{a^{\frac{1}{3}}} = 9$

because $\sqrt[3]{a^{\frac{1}{3}}} = \sqrt[3]{\sqrt[3]{a}} = a^{\frac{1}{9}} = \sqrt[9]{a}$

23. The radicand of $6 \cdot \sqrt[5]{8}$ is 8
24. The order of $7.\sqrt{8}$ is 2
25. a) $2 + 3\sqrt{3}$ is binomial
    b) $\sqrt[3]{5}$ is monomial
    c) $\sqrt[4]{2}$ is monomial
    d) $\sqrt{3}$ is monomial & quadratic
26. The R.F of $5 \cdot \sqrt{3} - 3 \cdot \sqrt{2}$ is $5\sqrt{3} + 3\sqrt{2}$

27. The R.F of $\sqrt[3]{1}+\sqrt[3]{2}$ is $\left(1^{\frac{2}{3}}-2^{\frac{1}{3}}+2^{\frac{2}{3}}\right)=1-\sqrt[3]{2}+\sqrt[3]{4} \Rightarrow \left(1^{\frac{1}{3}}+2^{\frac{1}{3}}\right)\left(\left(1^{\frac{1}{3}}\right)^2-1^{\frac{1}{3}}\cdot 2^{\frac{1}{3}}+\left(2^{\frac{1}{3}}\right)^2\right)$

28. The R.F of $1+\sqrt{5}$ is $1-\sqrt{5}$ $(1+\sqrt{5})(1-\sqrt{5})=1^2-(\sqrt{5})^2=1-5=-4$

29. a) $\sqrt{125}=\sqrt{25\times 5}=5.\sqrt{5}$ simplest R.F. of $\sqrt{125}$ is $\sqrt{5}$

30. $\sqrt{5}\times\sqrt{5}=5\times 5=25$
b) $\sqrt[3]{16}=\sqrt[3]{4^2}$ simplest R.F. of $\sqrt[3]{4^2}=\sqrt[3]{4^{3-2}}=\sqrt[3]{4}$
c) $\sqrt[3]{25}=\sqrt[3]{5^2}$ simplest R.F of $\sqrt[3]{5^2}=\sqrt[3]{5^{3-2}}=\sqrt[3]{5}$ d) $\sqrt{32}=\sqrt{16\times 2}=\sqrt{16}\times\sqrt{2}=4\cdot\sqrt{2}$ Rationalising factor of $\sqrt{32}$ is $\sqrt{2}$

30.
a) $5.\sqrt{6}=\sqrt{5^2\times 6}=\sqrt{25\times 6}=\sqrt{150}$
b) $\sqrt[2]{\sqrt[3]{5}}=\sqrt[6]{5}$
c) $\sqrt[3]{4^2}=3\times\sqrt[2]{4^{2\times 2}}=\sqrt[6]{4^4}$
d) $\sqrt[2]{5}\times\sqrt[2]{6}=\sqrt[2]{5\times 6}=\sqrt[2]{30}$
e) $\frac{\sqrt[4]{2}}{\sqrt[4]{3}}=\sqrt[4]{\frac{2}{3}}=\sqrt[4]{\frac{2\times 3}{3\times 3}}=\sqrt[4]{\frac{6}{9}}$

31. a) $\sqrt[4]{3}$ is monomial surd
b) $2\sqrt{3}, 4\sqrt{3}$ are like surds
c) $\sqrt{2}+\sqrt{5}-7$ is a compound surd
d) $\frac{3}{2}\sqrt{8}+1$ is compound surd
e) $7\sqrt{5}, 7\sqrt{6}$ are disimilar surd

Worksheet – 2

Level – I

1. C
2. A
3. B
4. A
5. C
6. C
7. D
8. C

6. The geometric mean of $1+\sqrt{2}$ and $7+5\sqrt{2}=\sqrt{(1+\sqrt{2})(7+5\sqrt{2})}$
$=\sqrt{7+5\sqrt{2}+7\sqrt{2}+10}=\sqrt{17+12\sqrt{2}}=\sqrt{17+2\sqrt{6^2\times 2}}=\sqrt{17+2\sqrt{72}}$
$=\sqrt{(\sqrt{9}+\sqrt{8})^2}=\sqrt{9}+\sqrt{8}=3+2\sqrt{2}$

7. $\sqrt{14+6\sqrt{5}}=\sqrt{14+2\sqrt{3^2\times 5}}=\sqrt{14+2\sqrt{45}}=\sqrt{(\sqrt{9}+\sqrt{5})^2}=\sqrt{9}+\sqrt{5}=3+\sqrt{5}$

8. $\sqrt{13-\sqrt{120}}=\sqrt{13-\sqrt{4\times 30}}=\sqrt{13-2\sqrt{30}}=\sqrt{10+3-2\sqrt{10\times 3}}$
$=\sqrt{(\sqrt{10})^2+(\sqrt{3})^2-2\sqrt{10}\times\sqrt{3}}=\sqrt{(\sqrt{10}-\sqrt{3})^2}=\sqrt{10}-\sqrt{3}$

Level – II

1) 3	2) 3	3) 4	4) 4	5) 1	6) 4	7) 3
8) 3						
9) 3	10) 2	11) 3	12) 4	13) 3	14) 1	15) 4
16) 1						
17) 1,3	18) 1,3,4	19) 1	20) 3	21) 3	22) s,r,q,p	23) 3
24) 3,4						
25) s,r,q,p		26) 5	27) 7	29) 8	28) 3	29) 3
30) 4						

1. $\sqrt{\sqrt{12}+14} = K(\sqrt{3}+1) \Rightarrow \sqrt{2\sqrt{3}+4} = K(\sqrt{3}+1)$
SOBS
$2\sqrt{3}+4 = K^2(3+1+2\sqrt{3}) \Rightarrow 2\sqrt{3}+4 = K^2(4+2\sqrt{3}) \Rightarrow K^2 = 1 \Rightarrow K = 1$

2. $\sqrt{x+1} - \sqrt{x-1} = 1$
squaring on both sides
$x+1+x-1-2\sqrt{x+1}\sqrt{x-1} = 1 \Rightarrow 2x-1 = 2\sqrt{x^2-1}$
squaring on both sides
$(2x-1)^2 = 4(x^2-1) \Rightarrow 4x^2-4x+1 = 4x^2-4$
$1+4 = 4x \Rightarrow 4x = 5$
$x = \frac{5}{4}$

3. $\frac{4}{\sqrt{6+\sqrt{20}}} - \frac{2}{\sqrt{8+\sqrt{60}}} - \frac{2}{\sqrt{4-\sqrt{12}}}$
$= \frac{4}{\sqrt{5}+\sqrt{1}} \times \frac{\sqrt{5}-1}{\sqrt{5}-1} - \frac{2}{\sqrt{5}+\sqrt{3}} \times \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}-\sqrt{3}} - \frac{2}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$
$= \frac{4(\sqrt{5}-\sqrt{1})}{5-1} - \frac{2(\sqrt{5}-\sqrt{3})}{5-3} - \frac{2(\sqrt{3}+1)}{3-1}$
$\sqrt{5}-1-\sqrt{5}+\sqrt{3}-\sqrt{3}-1 = -2$

4. $\sqrt{32+\sqrt{960}} = \sqrt{32+\sqrt{4\times240}} = \sqrt{32+2\sqrt{240}}$
$= \sqrt{(\sqrt{20}+\sqrt{12})^2} = \sqrt{20}+\sqrt{12} = \sqrt{4\times5}+\sqrt{4\times3} = 2\cdot\sqrt{5}+2\cdot\sqrt{3}$ 5.
$\sqrt{78+\sqrt{1728}} = \sqrt{78+\sqrt{4\times432}} \Rightarrow \sqrt{78+2\sqrt{432}} = \sqrt{(\sqrt{72}+\sqrt{6})^2} = \sqrt{72}+\sqrt{6}$

6. $x = 7+5\sqrt{2} \Rightarrow \sqrt[3]{7+5\sqrt{2}} = x+\sqrt{y}$
cubing on both sides
$\sqrt[3]{7+5\sqrt{2}} = (x+\sqrt{y})^3 = (x^3+3xy)+\sqrt{y}(3x^2+y) \Rightarrow x^3+3xy = 7$
$\sqrt[3]{7+5\sqrt{2}} = x+\sqrt{y}$ ...
$(2) \Rightarrow \sqrt[3]{7-5\sqrt{2}} = x-\sqrt{y}$
$(2)\times(3) \Rightarrow (\sqrt[3]{7+5\sqrt{2}})(\sqrt[3]{7-5\sqrt{2}}) = (x+\sqrt{y})(x-\sqrt{y})$
$\sqrt[3]{7^2-(5\sqrt{2})^2} = x^2-y$
$\sqrt[3]{49-50} = x^2-y \Rightarrow \sqrt[3]{-1} = x^2-y \Rightarrow x^2-y = -1 \Rightarrow y = x^2+1$
$1 \Rightarrow x^3+3xy = 7 \Rightarrow x^3+3x(x^2+1) = 7 (\because y = x^2+1)$
$x^3+3x^3+3x = 7 \Rightarrow 4x^3+3x = 7$

Put x = 1

$4(1)^3 + 3(1) = 7 \Rightarrow 7 = 7 \Rightarrow y = x^2 + 1 = 1^2 + 1 = 2$

$\sqrt[3]{7 + 5\sqrt{2}} = x + \sqrt{y} = 1 + \sqrt{2} \Rightarrow \sqrt[3]{x} - 1 = 1 + \sqrt{2} - 1 = \sqrt{2}$

7. $\sqrt[3]{72 - 32\sqrt{5}} = x - \sqrt{y}$

cubing on both sides

$72 - 32\sqrt{5} = (x - \sqrt{y})^3 = x^3 - y\sqrt{y} - 3x^2\sqrt{y} + 3x \cdot y$

$= (x^3 + 3xy) - \sqrt{y}(y + 3x^2)$

$x^3 + 3xy = 72 \ldots . (1) \Rightarrow \sqrt[3]{72 - 32\sqrt{5}} = x - \sqrt{y} \ldots (2) \Rightarrow \sqrt[3]{72 + 32\sqrt{5}} = x + \sqrt{y} \ldots$

$(2) \times (3)(\sqrt[3]{72 - 32\sqrt{5}})(\sqrt[3]{72 + 32\sqrt{5}}) = (x - \sqrt{y})(x + \sqrt{y})$

$\sqrt[3]{(72)^2 - (32\sqrt{5})^2} = x^2 - y \Rightarrow \sqrt[3]{5184 - 1024 \times 5} = x^2 - y \Rightarrow \sqrt[3]{5184 - 5120}$

$= x^2 - y$

$\sqrt[3]{64} = x^2 - y \Rightarrow x^2 - y = 4 \Rightarrow y = x^2 - 4$

$(1) \Rightarrow x^3 + 3xy = 72$

$x^3 + 3x(x^2 - 4) = 72 (\because y = x^2 - 4)$

$x^3 + 3x^3 - 12x = 72$

$4x^3 - 12x = 72 \Rightarrow$ If $x = 3$ then $4(3)^3 - 12(3) = 72$

$108 - 36 = 72 \Rightarrow 72 = 72$

If x = 3 then $y = x^2 - 4 = 3^2 - 4 = 9 - 4 = 5 \Rightarrow \sqrt[3]{72 - 32\sqrt{5}} = x - \sqrt{y} = 3 - \sqrt{5}$

8. $\sqrt{19 - 4\sqrt{x}} = \sqrt{12} - \sqrt{7}$

Squaring on both sides

$19 - 4\sqrt{x} = 12 + 7 - 2\sqrt{84} \Rightarrow 19 - 4\sqrt{x} = 19 - 2\sqrt{84} \Rightarrow 4\sqrt{x} = 2\sqrt{84}$

sobs

$16 \times x = 4 \times 84 \Rightarrow x = \frac{84}{4} = 21$

9. $\sqrt{21 - 4 \cdot \sqrt{5} + 8 \cdot \sqrt{3} - 4\sqrt{15}}$

$= \sqrt{(\sqrt{5})^2 + (\sqrt{4})^2 + (\sqrt{12})^2 - 2(\sqrt{5})(\sqrt{4}) + 2(\sqrt{4})(\sqrt{12}) - 2(\sqrt{5})(\sqrt{12})}$

$= \sqrt{(-\sqrt{5} + \sqrt{4} + \sqrt{12})^2} = -\sqrt{5} + 2 + \sqrt{12} = -\sqrt{5} + 2 + 2\sqrt{3}$

10. $\sqrt{\sqrt{50} + \sqrt{48}} = K(\sqrt{3} + \sqrt{2}) \Rightarrow \sqrt{\sqrt{2 \times 25} + 2 \times \sqrt{24}} = K(\sqrt{3} + \sqrt{2})$

$\sqrt{\sqrt{2}}(\sqrt{\sqrt{25} + \sqrt{24}}) = K(\sqrt{3} + \sqrt{2}) \Rightarrow 2^{\frac{1}{4}}(\sqrt{5 + 2\sqrt{6}}) = K(\sqrt{3} + \sqrt{2})$

$2^{\frac{1}{4}}\left(\sqrt{(\sqrt{3} + \sqrt{2})^2}\right) = K(\sqrt{3} + \sqrt{2}) \Rightarrow 2^{\frac{1}{4}}(\sqrt{3} + \sqrt{2}) = K(\sqrt{3} + \sqrt{2}) \Rightarrow K = 2^{\frac{1}{4}}$

11. $\sqrt{x + 1} - \sqrt{x - 1} = 1$

sobs

$x + 1 + x - 1 - 2 \cdot \sqrt{(x + 1)(x - 1)} = 1 \Rightarrow 2x - 1 = 2\sqrt{x^2 - 1}$

sobs

$(2x - 1)^2 = 4(x^2 - 1) \Rightarrow 4x^2 - 4x + 1 = 4x^2 - 4 \Rightarrow 1 + 4 = 4x \Rightarrow 5 = 4x \Rightarrow x = \frac{5}{4}$

12. $a = 7 + 5\sqrt{2} \Rightarrow \sqrt[3]{a} = \sqrt[3]{7 + 5\sqrt{2}} = x + \sqrt{y}$

cobs

$7+5\sqrt{2}=(x^3+3xy)+\sqrt{y}(3x^2+y)\Rightarrow x^3+3xy=7\ldots$

$\sqrt[3]{7+5\sqrt{2}}=x+\sqrt{y}\ldots\ldots(2)\Rightarrow\sqrt[3]{7-5\sqrt{2}}=x-\sqrt{y}\ldots$

$(2)\text{x}(3)\Rightarrow\sqrt[3]{7+5\sqrt{2}}\sqrt[3]{7-5\sqrt{2}}=(x+\sqrt{y})(x-\sqrt{y})\Rightarrow\sqrt[3]{7^2-(5\sqrt{2})^2}=x^2-y$

$\sqrt[3]{49-50}=x^2-y\Rightarrow\sqrt[3]{-1}=x^2-y\Rightarrow-1=x^2-y\Rightarrow y=x^2+1$

$x^3+3xy=7\Rightarrow x^3+3x(x^2+1)=7\Rightarrow x^3+3x^3+3x=7$

$4x^3+3x=7$ Put $x=1$ then $4(1)^3+3(1)=7$

$7=7\Rightarrow y=x^2+1=1^2+1=2\Rightarrow\sqrt[3]{7+5\sqrt{2}}=x+\sqrt{y}=1+\sqrt{2}$

$\sqrt[3]{a}-1=1+\sqrt{2}-1=\sqrt{2}$

13. $10+6\sqrt{3}=(x^3+3xy)+\sqrt{y}(x^2+3xy)\Rightarrow x^3+3xy=10.$

$\sqrt[3]{10+6\sqrt{3}}=x+\sqrt{y}\ldots..(2)\Rightarrow\sqrt[3]{10-6\sqrt{3}}=x-\sqrt{y}\ldots$

$(2)\text{x}(3)\Rightarrow\sqrt[3]{10+6\sqrt{3}}\sqrt[3]{10-6\sqrt{3}}=(x+\sqrt{y})(x-\sqrt{y})$

$\sqrt[3]{10^2-36\times3}=x^2-y\Rightarrow\sqrt[3]{100-108}=x^2-y$

$\sqrt[3]{-8}=x^2-y\Rightarrow-2=x^2-y\Rightarrow y=x^2+2\Rightarrow x^3+3xy=10\Rightarrow x^3+3x(x^2+2)=10$

$x^3+3x^3+6x=0\Rightarrow4x^3+6x=10\Rightarrow$ Put x = 1 then $4(1)^3+6(1)=0$

$4+6=10\Rightarrow10=10\Rightarrow y=x^2+2=1^2+2=3$

$\sqrt[3]{10+6\sqrt{3}}=x+\sqrt{y}=1+\sqrt{3}\Rightarrow\sqrt[3]{10-6\sqrt{3}}=x-\sqrt{y}=1-\sqrt{3}$

$\Rightarrow\sqrt[3]{10+6\sqrt{3}}+\sqrt[3]{10-6\sqrt{3}}=1+\sqrt{3}+1-\sqrt{3}=2$

14. $\sqrt{11+\sqrt{112}}=\sqrt{11+\sqrt{4\times28}}=\sqrt{7+4+2\sqrt{7\times4}}=\sqrt{(\sqrt{7}+\sqrt{4})^2}=\sqrt{7}+\sqrt{4}=\sqrt{7}+2$

15. $\sqrt{\frac{7+4\sqrt{3}}{4}}=\sqrt{\frac{7+2\sqrt{4\times3}}{4}}=\sqrt{\frac{(\sqrt{4}+\sqrt{3})^2}{4}}=\frac{1}{2}(\sqrt{4}+\sqrt{3})=(2+\sqrt{3})\left(\frac{1}{2}\right)$

16. $x=\frac{1}{2-\sqrt{3}}=\frac{1}{2-\sqrt{3}}\times\frac{2+\sqrt{3}}{2+\sqrt{3}}=\frac{2+\sqrt{3}}{2^2-(\sqrt{3})^2}=\frac{2+\sqrt{3}}{4-3}=2+\sqrt{3}$

$$y=\frac{1}{2+\sqrt{3}}=\frac{1}{2+\sqrt{3}}\times\frac{2-\sqrt{3}}{2-\sqrt{3}}=\frac{2-\sqrt{3}}{4-3}=2-\sqrt{3}$$

$xy=(2+\sqrt{3})(2-\sqrt{3})=2^2-3=1$

$7x^2+11xy-7y^2$

$=7(2+\sqrt{3})^2+11(1)-7(2-\sqrt{3})^2=7(4+3+4\sqrt{3})+11-7(4+3-4\sqrt{3})$

$=7(7+4\sqrt{3})+11-7(7-4\sqrt{3})=49+28\sqrt{3}+11-49+28\sqrt{3}=11+56\sqrt{3}$

17. $\sqrt{x+y-2\sqrt{xy}}\ x>0,y>0$

$=\sqrt{(\sqrt{x})^2+\sqrt{(y)}^2-2\sqrt{x}\sqrt{y}}=\sqrt{(\sqrt{x}-\sqrt{y})^2}$ or $\sqrt{(\sqrt{y}-\sqrt{x})^2}=\sqrt{x}-\sqrt{y}$ or $\sqrt{y}-\sqrt{x}$ 18. $x=\sqrt{a\sqrt{a\sqrt{a\ldots\ldots\ldots\infty}}}(a\neq0)\Rightarrow x=\sqrt{a.x}\Rightarrow x^2=ax\Rightarrow x=a$ or $x=|a|$ or $x=\sqrt{a^2}$

19. $\sqrt{3+2\sqrt{2}}=(\sqrt{2})^2+(\sqrt{1})^2+2\sqrt{2}\cdot1=\sqrt{(\sqrt{2}+\sqrt{1})^2}=\sqrt{2}+\sqrt{1}=\sqrt{2}+1$

20.. $\sqrt{19-4\sqrt{21}}=\sqrt{19-2\sqrt{84}}=\sqrt{(\sqrt{12}-\sqrt{7})^2}=\sqrt{12}-\sqrt{7}$

21. $\sqrt{7-2\sqrt{12}}=\sqrt{a}-b\sqrt{3}\Rightarrow\sqrt{(\sqrt{4}-\sqrt{3})^2}=\sqrt{a}-b\sqrt{3}$

$\sqrt{4}-\sqrt{3}=\sqrt{a}-b\sqrt{3}\Rightarrow \text{a}=4,-\sqrt{3}=-b\sqrt{3}\Rightarrow b=1$

$3\sqrt{a-b}=3\sqrt{4-1}=3\sqrt{3}$

22.

a) $\sqrt[3]{7+\sqrt{50}}=x+\sqrt{y}$

COBs

$7+\sqrt{50}=(x^3+3xy)+\sqrt{y}(3x^2+xy)\quad --(1)$

$x^3+3xy=7\quad --(2)$

$\sqrt[3]{7+\sqrt{50}}=x+\sqrt{y}\quad --(3)$

$(2)\times(3)\Rightarrow\sqrt[3]{7+\sqrt{50}}\cdot\sqrt[3]{7-\sqrt{50}}=(x+\sqrt{y})(x-\sqrt{y})$

$\sqrt[3]{49-50}=x^2-y\Rightarrow\sqrt[3]{-1}=x^2-y\Rightarrow-1=x^2-y\Rightarrow y=x^2+1$

$x^3+3xy=7$

$x^3+3x(x^2+1)=7$

$x^3+3x^3+3x=7\Rightarrow x^3+3x^3+3x=7$

Put $x=1\Rightarrow 4(1)^3+3(1)=7\Rightarrow 7=7\Rightarrow y=x^2+1=1^2+1=2$

$\sqrt[3]{7+\sqrt{50}}=x+\sqrt{y}=1+\sqrt{2}$

b) $\sqrt[3]{10+2\sqrt{27}}=x+\sqrt{y}$

Cubing on bothsides

$10+2\sqrt{27}=(x^3+3xy)+\sqrt{y}(3x^2+xy)$

$x^3+3xy=10$

$\sqrt[3]{10+2\sqrt{27}}=x+\sqrt{y}$

$\sqrt[3]{10-2\sqrt{27}}=x-\sqrt{y}$

$(2)\times(3)\Rightarrow\sqrt[3]{10+2\sqrt{27}}\cdot\sqrt[3]{10-2\sqrt{27}}=(x+\sqrt{y})(x-\sqrt{y})$

$\sqrt[3]{10^2-(2\sqrt{27})^2}=x^2-y\Rightarrow\sqrt[3]{100-4\times27}=x^2-y\Rightarrow\sqrt[3]{100-108}=x^2-y$

$\sqrt[3]{-8}=x^2-y\Rightarrow-2=x^2-y\Rightarrow x^3+3x(x^2+2)=10\Rightarrow x^3+3x^3+6x=10$

$4x^3+6x=10\Rightarrow$ Put x = 1 $4(1)^3+6(1)=10\Rightarrow 10=10\Rightarrow y=x^2+2=1^2+$

$2=3\Rightarrow\sqrt[3]{10+2\sqrt{27}}=x+\sqrt{y}=1+\sqrt{3}$

c) $\sqrt[3]{25+2\sqrt{242}}=x+\sqrt{y}$ COBS

$25+2\sqrt{242}=(x^3+xy)+\sqrt{y}(3x^2+xy)\,x^{33}+3xy=25$

$\sqrt[3]{25+2\sqrt{242}}=x+\sqrt{y}$

$\sqrt[3]{25-2\sqrt{242}}=x-\sqrt{y}$

$(2)\times(3)\Rightarrow(\sqrt[3]{25+2\sqrt{242}})(\sqrt[3]{25-2\sqrt{242}})=(x+\sqrt{y})(x-\sqrt{y})$

$\sqrt[3]{25^2-(2\sqrt{242})^2}=x^2-y\Rightarrow\sqrt[3]{625-968}=x^2-y\Rightarrow\sqrt[3]{-343}=x^2-y$

$\sqrt[2]{(-7)^3} = x^2 - y \Rightarrow -7 = x^2 - y \Rightarrow y = x^2 + 7 \Rightarrow x^3 + 3xy = 25$
$x^3 + 3x(x^2 + 7) = 25 \Rightarrow x^3 + 3x^3 + 21x = 25 \Rightarrow 4x^3 + 21x = 25$
Put x = $1 \Rightarrow y = x^2 + 7 = 1^2 + 7 = 8$
$\sqrt[3]{25 + 2\sqrt{242}} = x + \sqrt{y} = 1 + \sqrt{8} = 1 + 2\sqrt{2}$

d) $\sqrt[3]{37 + 15\sqrt{12}} = x + \sqrt{y}$
COBS
$37 + 15\sqrt{2} = (x^3 + 3xy) + \sqrt{y}(3x^2 + xy) \Rightarrow\ x^3 + xy = 37$
$\sqrt[3]{37 + 15\sqrt{12}} = x + \sqrt{y}$
$\sqrt[3]{37 - 15\sqrt{12}} = x - \sqrt{y}$
(2) × (3) $\Rightarrow \sqrt[3]{37 + 15\sqrt{12}}\sqrt[3]{37 - 15\sqrt{12}} = (x + \sqrt{y})(x - \sqrt{y})$
$\sqrt[3]{37^2 - (15\sqrt{12})^2} = x^2 - y \Rightarrow \sqrt[3]{1369 - 225 \times 12} = x^2 - y$
$\sqrt[3]{1369 - 2700} = x^2 - y \Rightarrow \sqrt[3]{-1331} = x^2 - y$
$\sqrt[3]{(-11)^3} = x^2 - y \Rightarrow x^2 - y = -11$
$x^2 + 11 = y \Rightarrow x^3 + 3xy = 37 \Rightarrow x^3 + 3x(x^2 + 11) = 37$
$4x^3 + 3x^3 + 33x = 37 \Rightarrow 4x^3 + 33x = 37$
Put x = 1 then
$4(1)^3 + 33(1) = 37 \Rightarrow 37 = 37 \Rightarrow y = x^2 + 11 = 1^2 + 11 = 12$
$x^2 + 11 = 12 \Rightarrow x^2 = 1 \Rightarrow x = 1 \Rightarrow \sqrt[3]{37 + 15\sqrt{12}} = 1 + \sqrt{12} = 1 + 2\sqrt{3}$

23. $\sqrt{8 - 2\sqrt{x}} = \sqrt{6} - \sqrt{2}$ SOBS $8 - 2\sqrt{x} = 6 + 2 - 2\sqrt{12} \Rightarrow 8 - 2\sqrt{x} = 8 - 2\sqrt{12} \Rightarrow x = 12$
Sum of digits in x = 1 + 2 = 3

24. $\sqrt{3\sqrt{3\sqrt{3 \ldots \ldots}}}$ 6 time $= 3^{\left(\frac{2^6-1}{2^6}\right)} = 3^{1-\frac{1}{2^6}} = 3^{\frac{63}{64}}$ ∵ $\sqrt{a\sqrt{a\sqrt{a \ldots \ldots}}}$ n times $= a^{\frac{2^n-1}{(2)^n}}$

a) $\sqrt{3 + 2\sqrt{2}} = \sqrt{(\sqrt{2} + \sqrt{1})^2} = 1 + \sqrt{2}$

b) $\sqrt{5 + 2\sqrt{6}} = \sqrt{(\sqrt{3} + \sqrt{2})^2} = \sqrt{3} + \sqrt{2}$

c) $\sqrt{12 - 2\sqrt{20}} = \sqrt{((\sqrt{10}) - (\sqrt{2}))^2} = \sqrt{10} - \sqrt{2}$

d) $\sqrt{8 - 2\sqrt{20}} = \sqrt{(\sqrt{6} - \sqrt{2})^2} = \sqrt{6} - \sqrt{2}$

26. K $= \sqrt{6 - \sqrt{6 - \sqrt{6} \ldots \ldots .}} \Rightarrow$ K $= \sqrt{6 - K}$
SOBS
$K^2 = 6 - K \Rightarrow K^2 + K - 6 = 0 \Rightarrow K^2 + 3K - 2K - 6 = 0$
$K(K + 3) - 2(K + 3) = 0 \Rightarrow (K + 3)(K - 2) = 0$
K + 3 = 0 (or) K − 2 = 0 ⇒ K = 2 (or) K = −3 ⇒ 2 K + 1 = 2 × 2 + 1 = 5

27. $\sqrt{a - \sqrt{b}} = 3 + \sqrt{2}$ SOBS
$a - \sqrt{b} = 9 + 2 + 6\sqrt{2} \Rightarrow$ a $= 1 - \sqrt{b} = 6\sqrt{2} \Rightarrow \sqrt{b} = -6\sqrt{2} \Rightarrow b = 36 \times 2 = 72$
$\sqrt{a^2 - b} = \sqrt{11^2 - 72} = \sqrt{121 - 72} = \sqrt{49} = 7$

28. $x = 45 + 29\sqrt{2} \Rightarrow \sqrt[3]{x} = \sqrt[3]{45 + 29\sqrt{2}} = x + \sqrt{y}$

COBS

$45+29\sqrt{2}=(x^3+3xy)+\sqrt{y}(3x^2+xy)$

$x^3+3xy=45$

$\sqrt[3]{45+29\sqrt{2}}=x+\sqrt{y}$

$\sqrt[3]{45-29\sqrt{2}}=x-\sqrt{y}$

$(2)\times(3)\Rightarrow\sqrt[3]{45-29\sqrt{2}}\sqrt[3]{45-29\sqrt{2}}=(x+\sqrt{y})(x-\sqrt{y})$

$\sqrt[3]{45^2-29^2\times2}=x^2-y\Rightarrow\sqrt[3]{2025-841\times2}=x^2-y\Rightarrow\sqrt[3]{2025-1682}=x^2-y$

$\sqrt[3]{343}=x^2-y\Rightarrow\sqrt[3]{7^3}=x^2-y\Rightarrow x^2-y=7\Rightarrow y=x^2-7$

$x^3+3xy=45\Rightarrow x^3+3x(x^2-7)=45\Rightarrow x^3+3x^3-21x=45\Rightarrow 4x^3-21x=45$

If $x=3$ then $4(3)^3-21\times3=45\Rightarrow108-63=45\Rightarrow45=45$

$y=x^2-7=3^2-7=9-7=2$

$\sqrt[3]{x}=3\sqrt{45+29\sqrt{2}}=x+\sqrt{y}=3+\sqrt{2}\Rightarrow\sqrt[3]{x}-3=3+\sqrt{2}-3=\sqrt{2}$

29. $\sqrt[3]{-26+15\sqrt{3}}=x+\sqrt{y}$COBS

$-26+15\sqrt{3}=(x^3+3xy)+\sqrt{y}(3x^2+xy)$

$x^3+3xy=26$

$\sqrt[3]{-26+15\sqrt{3}}=x+\sqrt{y}$

$\sqrt[3]{-26-15\sqrt{3}}=x-\sqrt{y}$

$(2)\times(3)\Rightarrow(\sqrt[3]{-26+15\sqrt{3}})(\sqrt[3]{-26-15\sqrt{3}})=(x+\sqrt{y})(x-\sqrt{y})$

$\sqrt[3]{(-26)^2-(15\sqrt{3})^2}=x^2-y\Rightarrow\sqrt[3]{676-675}=x^2-y$

$\sqrt[3]{1}=x^2-y\Rightarrow x^2-y=1\Rightarrow y=x^2-1\Rightarrow x^3+3xy=-26\Rightarrow x^3+3x(x^2-1)=-26$

$x^3+3x^3-3x=-26\Rightarrow4x^3-3x=-26\Rightarrow$ If x $=-2$ then $4(-2)^3+3\times2=-26$

$-32+6=-26\Rightarrow-26=-26$

$y=x^2-1-(-2)^2-1=4-1=3\Rightarrow\sqrt[3]{-26+15\sqrt{3}}=x+\sqrt{y}=-2+\sqrt{3}$

30. $\sqrt[3]{37+30\sqrt{3}}=x+\sqrt{y}$ COBS

$37+30\sqrt{3}=(x^3+3xy)+\sqrt{y}(3x^2+xy)$

$x^3+3xy=37-----(1)\ \sqrt[3]{37+30\sqrt{3}}=x+\sqrt{y}$

$(2)\times(3)\Rightarrow3\sqrt{37+30\sqrt{3}}3\sqrt{37-30\sqrt{3}}=(x+\sqrt{y})(x-\sqrt{y})$

$\sqrt[3]{(37)^2-(30\sqrt{3})^2}=x^2-y\Rightarrow\sqrt[3]{1369-900\times3}=x^2-y\Rightarrow\sqrt[3]{1369-2700}=x^2-y$

$\sqrt[3]{-1331}=x^2-y\Rightarrow\sqrt[3]{(-11)^3}=x^2-y\Rightarrow x^2-y=-11\Rightarrow y=x^2+11$

$x^3+3xy=37\Rightarrow x^3+3x(x^2+11)=37\Rightarrow x^3+3x^3+33x=37$

$4x^3 + 33x = 37$ Put x = 1 than $4(1)^3 + 33(1) = 37$

$37 = 37$

$y = x^2 + 11 = 1^2 + 11 = 12 \Rightarrow \sqrt[3]{37 + 30\sqrt{3}} = x + \sqrt{y} = 1 + \sqrt{12} = 1 + \sqrt{4 \times 3}$

$= 1 + 2\sqrt{3}$

$\sqrt[3]{37 - 30\sqrt{3}} = x - \sqrt{y} = 1 - \sqrt{12} = 1 - \sqrt{4 \times 3} = 1 - 2\sqrt{3}$

Given $\sqrt[3]{37 + 30\sqrt{3}} = \sqrt[3]{37 - 30\sqrt{3}}x\sqrt{3} \Rightarrow 1 + 2\sqrt{3} - (1 - 2\sqrt{3}) = x\sqrt{3}$

$1 + 2\sqrt{3} - 1 + 2\sqrt{3} = x\sqrt{3} \Rightarrow 4\sqrt{3} = x\sqrt{3} \Rightarrow$ x = 4

3. LOGARITHMS

Worksheet – 1

Level – I

1) A 2) D 3) B 4) B 5) B 6) D 7) C
8) C 9) A 10) A

Level – II

1) A 2) A 3) D 4) C 5) A 6) C 7) C
8) C 9) 9 10) D 11) A 12) B 13) C 14) C
15) 375 16) A 17) C 18) C 19) A 20) C
21) A 22) B 23) D 24) C 25) C

1. $x = 27, y = \log_3 4 \Rightarrow x^y = 27^{\log_3^4} = 3^{3\log_3 4} = 64$
2. $\log (a^2b^3c^{-2})^{1/12} = \log \left(a^{1/6}b^{1/4}c^{-1/6}\right) = \frac{1}{6}\log a + \frac{1}{4}\log b - \frac{1}{6}\log c$
3. $\log_9 \sqrt{3} = \log_{3^2} 3^{1/2} = \frac{1/2}{2}\log_3 3 = \frac{1}{4} \times 1 = \frac{1}{4}$
4. $\log_{2\sqrt{3}} 1728 = \log_{(12)^{1/2}} (12)^3 = \frac{3}{1/2}\log_{12} 12 = 6 \times 1 = 6$
5. $\frac{1}{\log_x xy} + \frac{1}{\log_y xy} = \log_{xy} x + \log_{xy} y = \log_{xy} xy = 1$
6. $\log_3 x = 4^{1/2} \Rightarrow \log_3 x = 2 \Rightarrow x = 3^2 = 9$
7. $x = \log_7 11, y = \log_8 5 \Rightarrow \log_7 11 > \log_8 5$
8. $\log_c \sqrt{c\sqrt{c}} = \log_c \sqrt{c^{3/2}} = \log_{c^1} c^{3/4} = \frac{3}{4}\log_c c = \frac{3}{4}$
9. $4^{\log_2 3} = 2^{2\log_2 3} = 2^{\log_2 3^2} = 9$
10. $3^{\sqrt{\log_3 7}} - 7^{\sqrt{\log_7 3}} = \frac{7}{2} - \frac{3}{2} = 2$
11. $\frac{\log_4 17}{\log_9 23} - \frac{\log_2 17}{\log_3 23} = \frac{2\log_2 17}{2\log_3 23} - \frac{\log_2 17}{\log_3 23} = 0.$
12. $2^{2-\log_2 5} = 2^2 \cdot 2^{\log_2 5^{-1}} = 4 \cdot \frac{1}{5} = \frac{4}{5}$
13. $\sqrt{10^{\frac{5}{2}\log_{10} 16}} = \sqrt{16^{5/2}} = (2^4)^{5/4} = 32.$
14. $\log_{10} x = -\log_{10} a \Rightarrow x = a^{-1}$
15. $3^{3\log_3 5} \cdot (27)^{1/3} = 125.3 = 375$
16. $1 + \log_b \{1 + \log_a x\} = 1 \Rightarrow \log_b \{1 + \log_a x\} = 0 \Rightarrow 1 + \log_a x = 1 \Rightarrow \log_a x = 0 \Rightarrow x = a^0 = 1.$
17. $\log_{30} 3 = a, \log_{30} 5 = b; \log_{30} 32 = \log_{30} 2^5 = 5\log_{30} 2 = 5\log_{30} \left(\frac{30}{15}\right)$
    $= 5[\log_{30} 30 - \log_{30} 15] = 5[1 - (a + b)] = 5[1 - a - b]$
18. $3^{\frac{4}{\frac{2}{2}\log_2 3}} + 27^{\frac{1}{\frac{2}{2}\log_6 3}} + 81^{\frac{1}{\log_5 3}} = 3^{4\log_3 2} + 3^{3\log_3 6} + 3^{4\log_3 5} = 16 + 216 + 625 = 857$
19. $\frac{\log 2}{\log 3} \cdot \frac{\log 3}{\log 2} \cdot \frac{\log 4}{\log 6} \cdot \frac{\log 6}{\log 4} = 1$

20. $\log_{\sqrt{2}}[\log_2\{\log_4(x-15)\}]=0$
$\log_2\{\log_4(x-15)\}=1\Rightarrow\log_4(x-15)=2\Rightarrow x-15=16\Rightarrow x=31$
21. $\log_y\sqrt{x}\cdot\log_z y^3\cdot\log_x z^{\frac{2}{3}}=\frac{1}{2}\log_y x\cdot 3\log_z y\cdot\frac{2}{3}\log_x z=\frac{1}{2}\cdot 3\cdot\frac{2}{3}=1$
22. If x, y, z are Consecutive natural numbers, then $1+xz=y^2\Rightarrow\log(1+xz)=2\log y$
23. $7^n>10^5\Rightarrow n\log_{10}7>5\Rightarrow n(2.5353)>15$ The least value of ' n ' is 6 .
24. $(3.7)^a=(0.37)^b=1000\Rightarrow a\log 3.7=3; b\log 0.37=3\Rightarrow\frac{3}{a}+1=\frac{3}{b}+2\Rightarrow\frac{1}{a}-\frac{1}{b}=\frac{1}{3}$
25. $x^{\frac{1}{3}}+y^{\frac{1}{3}}+z^{\frac{1}{3}}=0\Rightarrow x^{\frac{1}{3}}+y^{\frac{1}{3}}=-z^{\frac{1}{3}}$
$\Rightarrow x+y+z=3x^{\frac{1}{3}}y^{\frac{1}{3}}z^{\frac{1}{3}}\Rightarrow\log\left(\frac{x+y+z}{3}\right)=\frac{1}{3}\log x+\frac{1}{3}\log y+\frac{1}{3}\log z$

Level – III

26) C,D 27) D 28) C 29) C 30) A

31) a-p,r,s;b-q,t;c-q,s;d-p,t 32) B 33) C 34) A,B

35) a-t; b-s; -q; d-p

26. By verification or by using definition.
27. $A.M\geq GM$. By substituting we have to verify.
28. $x^{\log_x(x+3)^2}=16\Rightarrow(x+3)^2=16$. It has two solutions.
29. $3^{\log_5 2}-2^{\log_5 x}=64$. By verification we get x = 3125
30. $2^{\log_6(-4x)}=\log_7(2401)=\log_7^{7^4}\Rightarrow 2^{\log_6(-4x)}=4=2^2\Rightarrow\log_6(-4x)=2\Rightarrow -4x=36\Rightarrow x=-9$. Only one solution.
31. a) $\log_{0.5}0.0625=4$
b) $\log_2 8=3$
c) $\frac{\log 3}{\log 4}\cdot\frac{\log 4}{\log 5}\cdot\frac{\log 5}{\log 3}=1$
d) $\log_3\log_2 512=\log_3 9=2$
32. $\log_2\log_3\left(8+\log_3\log_2(x^2+4)\right)=1$
$\log_3\left(8+\log_3\log_2(x^2+4)\right)=2\Rightarrow\left(8+\log_3\log_2(x^2+4)\right)=9$
$\Rightarrow\log_3\log_2(x^2+4)=1\Rightarrow\log_2(x^2+4)=3\Rightarrow(x^2+4)=8\Rightarrow x=\pm 2$
33. $\log(a-b)=\log\left(\frac{a}{b}\right)\Rightarrow a-b=\frac{a}{b}\Rightarrow a-\frac{a}{b}=b\Rightarrow a\left[1-\frac{1}{b}\right]=b\Rightarrow a=\frac{b^2}{b-1}$
34. $\log(x+y)=\log\left(\frac{3x-3y}{2}\right)\Rightarrow x+y=\frac{3x-3y}{2}\Rightarrow x=5y.$
$\therefore\log x-\log y=\log\left(\frac{x}{y}\right)=\log\left(\frac{5y}{y}\right)=\log 5(\text{ or })\log 10-\log 2$
35. $\log_8 128=\log_{2^3}2^7=\frac{7}{3}$; b) $\log_6 216\sqrt{6}=\log_6 6^{\frac{7}{2}}=\frac{7}{2}$
c) $\log_{2\sqrt{3}}1728=\log_{(12)^{\frac{1}{2}}}(12)^3=6$ d) $\log_{243}729=\log_{3^5}3^6=\frac{6}{5}$

Worksheet – 2

Level – I

1) 1 2) 1 3) 2 4) 3 5) 1 6) 4 7) 1 8) 1 9) 4 10) 3

Level – II

1) 2 2) 4 3) 1 4) 4 5) 1 6) 1 7) 3 8) 1

9)3 10)4 11) 1 12) 2 13) 1 14) 1 15) 2

16) 1 17) 4 18) 1 19) 2 20) 3 21) 2 22) 2

23) 1 24) 2 25) 2 26) 3 27) 1 28) 1 29) 1,4

30) 1 31) 2 32)2

33) $a\square q; b\square r; c\square p; d\square s$ 34) 1 35) 4 36) a-r; b-p; c-s,r; d-p,t

1. $x = y\log_{10} e$
2. $\log 9^{100} = \log 3^{200} = 200\log 3 = 200(0.4771) = 95.2$

   The number of digits in $9^{100} = 96$.
3. $\log_{10} 25 = \log_{10}\left(\frac{100}{4}\right) = \log_{10} 100 - \log_{10} 4 = 2 - 2\log_{10} 2 = 2 - 2(0.3010) = 1.398$
4. $\log 81^{21} = 21\log 3^4 = 84(\log 3) = 84(0.4771) = 40.0764$. Number of digits = 41.
5. $\log_{10} 3^3 3^{-2} = \frac{-2}{3}\log_{10} 3 = \frac{-2}{3}(0.4771)$. Characteristic is -1 .
6. By definition
7. $100\log 27.31 = 100[\log 2731 - \log 100] = 100[\log 2731 - 2] = 100[3.4364 - 2] =$ 143.64 Number of digits = 144.
8. $25\log(108) = 25\log(4 \times 27) = 25[2\log 2 + 3\log 3] = 25[0.6020 + 1.4313] =$ 50.8325 Number of digits = 51.
9. $\log(2^{10} \times 3^5) = 10\log 2 + 5\log 3 = 10(0.3010) + 5(0.4771) = 5.3955$

   Number of digit is = 6.
10. $5\log x = 5(\overline{3}.7781) = \overline{18}.8905$. $11.3\log_{10} 3 = 1.4313; 3^n > 10^4$, least value is 9 .
11. $\log_e x(1+x) = 0 \Rightarrow x(1+x) = 1 \Rightarrow x^2 + x = 1$
12. $\log_5 1250 = \log_5(5^4 \times 2) = 4\log_5 5 + \log_5 2 = 3 + \frac{1}{x}$

13. $3^3\log\ 3 = 27(0.4771) = 12.8817$. Number of digits = 13.

14. $\log_e\ (x+1)x(x-1) = 0 \Rightarrow x^3 - x = 1 \Rightarrow x^3 - x - 1 = 0$

15. $5^x = 400 \Rightarrow x = \log_5\ 400 = \log_5\ (25 \times 16) = 2\log_5\ 5 + 4\log_5\ 2 = 3.72$

16. $\log_{10}\ 1 + \log_{10}\ 10 + \log_{10}\ 100 + \log_{10}\ 1000 + - - - - - +$

$\log_{10}\ 10000000000 = 0 + 1 + 2 + \cdots \ldots + 10 = 55$

17. $\frac{1}{\log_3\ e} + \frac{1}{\log_3\ e^2} + \frac{1}{\log_3\ e^4} + \cdots$

$= \log_e\ 3 + \frac{1}{2}\log_e\ 3 + \frac{1}{4}\log_e\ 3 + \cdots \ldots . = \log_e\ 3\left[1 + \frac{1}{2} + \frac{1}{4} + \cdots \ldots \ldots .\right] = \log_e\ 3[2]$

$= \log_e\ 9$

19. $\log\ 0.0075 = \log\ \frac{75}{10^4} = 2\log\ 5 + \log\ 3 - 4 = \bar{2}.1249$

20. $\log\ (0.5)^{100} = 100\log\ (0.5) = \overline{30} \cdot 1029$. Number of zeros = 30.

21. $2^x \cdot 3^{2x} = 100 \Rightarrow 18^x = 100 \Rightarrow x\log\ 18 = 2 \Rightarrow x = \frac{2}{\log\ 2 + 2\log\ 3} = \frac{2}{1.2553} = 1.59$

22. $\log_{10}(100x) = 2 + \log_{10} x = 2 + 1.9675 = 3.9675$ 23. $\log_{\sqrt{2}} 7 = 2\log_2 7 = 2\frac{\log 7}{\log 2} = 5.614$

24. $3^{x-2} = 5 \Rightarrow (x-2)\log\ 3 = \log\ 5 \Rightarrow x - 2 = \frac{\log\ 5}{\log\ 3} \Rightarrow x = 2 + \frac{\log\ 5}{\log\ 3} \Rightarrow 2\log\ 5 =$

0.6990

25. $a\log_{0.1}\ x = b\log_{0.01}\ x \Rightarrow a\log_{10}\ x = \frac{1}{2}b\log_{10}\ x \Rightarrow 2a = b$

26. $\log\ (0.005292)^{10} = 10(\log\ 5292 - \log\ 10^6) = 10(2\log\ 2 + 3\log\ 3 + 2\log\ 7 -$

$6) = \overline{23}.235$

The number of zeros is 22 .

27. $\log_\pi\ \left(\tan\ \frac{25\pi}{100}\right) = \log_\pi\ 1 = 0.$

28. $e^{f(x)} = \frac{10+x}{10-x} \Rightarrow f(x) = \log\ \left(\frac{10+x}{10-x}\right)$

$$f\left(\frac{200x}{100+x^2}\right) = \log\left(\frac{10 + \frac{200x}{100+x^2}}{10 - \frac{200x}{100+x^2}}\right) = \log\left(\frac{1000 + 10x^2 + 200x}{1000 + 10x^2 - 200x}\right) = \log\left(\frac{10+x}{10-x}\right)^2 = 2f(x)$$

$$\Rightarrow \frac{1}{2}f\left(\frac{200x}{100+x^2}\right) = f(\mathrm{x}). \therefore k = \frac{1}{2}$$

29. $\log_{10} 3 = 0.4771$

30. $\log_{10} 875 = 2.9421$

$\log (875)^{10} = 10\log (875) = 10 \times 2.9421 = 29.421$; Number of digits = 30.

31. let $A = (81)^{11} \Rightarrow \log A = 11\log (81) = 44(\log 3) = 20.99.$; Number of digits = 21.

32. $\log 2^{100} = 100\log 2 = 100 \times 0.3010 = 30.10.$ Number of digits = 31.

33. Conceptual

34. $50\log 2^4 = 200\log 2 = 200(0.3010) = 60.2$; Number of digits = 61

35. $2\log_{10} x - \log_x 10^{-2} = 2\log_{10} x + 2\log_x 10 = 2[\log_{10} x + \log_x 10] \geq 2.2 = 4.$

36. a) $\log (2^{10} \times 3^5)^{20} = 200\log 2 + 100\log 3 = 107.91$

b) $100\log (27.31) = 100[\log 2731 - 2] = 143.1$; Number of digits = 144.

c) $100\log 9 = 100 \times 2\log 3 = 200 \times 0.4771 = 94.12$; Number of digits = 95

d) $5\log 5^3 = 15\log 5 = 15(0.699) = 10.48$; Number of digits = 11.

4. POLYNOMIALS

Worksheet – 1

Level – I

1.	C	5.	B	9.	A
2.	A	6.	A	10.	A
3.	D	7.	D	11.	A
4.	B	8.	B		

Level – II

1) B	2) D	3) A	4) A	5) B	6) C	7) B
8) B	9) C	10) A	11) D	12) A	13) D	14) D
15) C	16) B	17) D	18) C	19) A	20) C	21) C
22) A	23) B					

1. $x^2 - 15x + 56 \Rightarrow x^2 - 7x - 8x + 56 \Rightarrow x(x-7) - 8(x-7) \Rightarrow (x-7)(x-8)$
2. $x^2 - 36 = x^2 - 6^2 = (x+6)(x-6)$
3. $x - 2 = 0 \Rightarrow x = 2 \Rightarrow 2^3 - 2^2 - 5(2) + 6 = 8 - 4 - 10 + 6 = 0$
4. $f(2) = 3(2) - 2 = 6 - 2 = 4$
5. $x = 0 \Rightarrow f(0) = 0^3 + 3(0)^2 + 3(0) + 1 = 1$
6. $x = a \Rightarrow a^3 - a(a)^2 + 6a - a = a^3 - a^3 + 6a - a = 5a$
7. Let $p(x) = x^{2020} - 1 \Rightarrow p(1) = 1^{2020} - 1 = 0$
8. $f(x) = 5x^3 - x^2 + 6x - 2 \Rightarrow 1 - 5x = 0 \Rightarrow 5x = 1$

   $x = \frac{1}{5} \Rightarrow f\left(\frac{1}{5}\right) = 5\left(\frac{1}{5}\right)^3 - \left(\frac{1}{5}\right)^2 + 6\left(\frac{1}{5}\right) - 2$

   $= 5\left(\frac{1}{125}\right) - \frac{1}{25} + \frac{6}{5} - 2 = \frac{6}{5} - 2 = \frac{6-10}{5} = \frac{-4}{5}$
9. $x = 2\ f(2) = 0$

   $2(2)^3 - 11(2)^2 + b(2) - 6 = 0 \Rightarrow 16 - 44 + 2b - 6 = 0 \Rightarrow 16 - 50 + 2b = 0$

   $-34 + 2b = 0 \Rightarrow 2b = 34 \Rightarrow b = 17$
10. $p(x) = 16x^4 - 8x^3 - 8x + 4 \Rightarrow p\left(\frac{1}{2}\right) = 16\left(\frac{1}{2}\right)^4 - 8\left(\frac{1}{2}\right)^3 - 8\left(\frac{1}{2}\right) + 4 = 0$
11. Let $p(x) = x^3 - 3x^2 - 7x + 7$ then $p(1) = 1^3 - 3(1)^2 - 7(1) + 7 = -2$
12. Let $p(x) = x^4 + 3x^3 - 3x + 5 \Rightarrow p(-1) = (-1)^4 + 3(-1)^3 - 3(-1) + 5 = 6$
13. The variable contains rational index ∴ Degree of the polynomial does not exist
14. Let $p(x) = x^3 + x^2 + x + 1$

    $\Rightarrow p(1) = 1^3 + 1^2 + 1 + 1 = 4 \Rightarrow p(2) = 2^3 + 2^2 + 2 + 1 = 8 + 4 + 3 = 15$

15. $x = -2 \Rightarrow 4(-2)^3 + 3(-2)^2 + 10(-2) + 11$

    $4(-8) + 3(4) - 20 + 11 = -32 + 12 - 20 + 11 = -32 - 8 + 11 = -29$

16. $2x - 1 = 0 \Rightarrow x = \frac{1}{2} \Rightarrow f\left(\frac{1}{2}\right) = 16\left(\frac{1}{2}\right)^4 - 8\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 - 8\left(\frac{1}{2}\right) + 4$

    $= 1\emptyset\left(\frac{1}{-}\right) - \emptyset\left(\frac{1}{-}\right) + \frac{1}{4} - 4 + 4 \Rightarrow$

17. degree of $1 + x^3 = 3$; Degree of $9 - ssx^2$ is 2 ; Sum $= 3 + 2 = 5$

18. $f(x) = 2x^3 - ax^2 - 5x + 6 \Rightarrow f(3) = 2(3)^3 - a(3)^2 - 5(3) + 6$

    $0 = 2(27) - a(9) - 15 + 6 \Rightarrow 0 = 54 - 9a - 9 \Rightarrow 45 - 9a = 0 \Rightarrow 9a = 45$

    $a = 5 \Rightarrow 2x^3 - 5x^2 - 5x + 6$

19. $1 + 2i, 2 - \sqrt{3}$ and 5 are the zeroes of the polynomial f(x), then $1 - 2i, 2 + \sqrt{3}$ are also the zeroes of f(x) $\therefore$ The least value of n is 5

20. $p(x) = 25x^2 + 70x + 49 \Rightarrow p(-1) = 25(-1)^2 + 70(-1) + 49 = 4$

21. put $x = -1 \Rightarrow (-1)^3 + 6(-1)^2 - (-1) - 6 \Rightarrow -1 + 6 + 1 - 6 = 0$

    $x = 6$

    $x = 2$

    $x = 3$ are not factors

22. $x^2 - 7x + 12 \Rightarrow x^2 - 3x - 4x + 12 \Rightarrow x(x-3) - 4(x-3) \Rightarrow (x-3)(x-4) \Rightarrow x = 3,4$

23. $x^2 - 2ax + a^2 - b^2 \Rightarrow (x-a)^2 - b^2 \Rightarrow (x - a + b)(x - (a + b))$

Level – III

24) A,B 25) A 26) B 27) B 28) a-p,q; b-p,q; c-r,s; d-r,s 29) C

30) A,B 31) A 32) B 33) B,D 34) D 35) A 36) C 37) B

24. $x^2 - 3x + 2 = (x-1)(x-2)$

25. $f(x) = a + bx = 0 \Rightarrow f\left(\frac{-a}{b}\right) = a + b\left(\frac{-a}{b}\right) = 0$

26. $f(x) = x^4 - 10x^3 + 36x^2 - 49x + 24 \Rightarrow f(1) = 2$

27. $f(x) = x^2 + 2020x + 2021 \Rightarrow f(0) = 2021$

28. Degree = Highest power of the term

29. The number of zeroes is equal to the degree of the polynomial

30. $p(x) = x^2 - 5x + 6 = (x-2)(x-3)$

31. Sum of the zeroes $= \frac{-\text{ Coefficient of } x^2}{\text{Coefficient of } x^3} = 1$

32. Degree = Highest power of the term

33. $2x-1=0 \Rightarrow x=\frac{1}{2} \Rightarrow 16\left(\frac{1}{2}\right)^4-8\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^2-8\left(\frac{1}{2}\right)+4$

    $1\emptyset\left(\frac{1}{}\right)-$

34. $(x-a)(x-b)=x^2+x+4 \Rightarrow x^2-(a+b)x+ab=x^2+x+4$

    $\therefore a+b=-1, ab=4 \Rightarrow \frac{1}{a^2}+\frac{1}{b^2}=\frac{a^2+b^2}{(ab)^2}=\frac{(a+b)^2-4ab}{(ab)^2}=\frac{1-16}{16}=\frac{-15}{16}$

35. $\left(\frac{a}{b}-\frac{b}{a}\right)^2=\left(\frac{a^2-b^2}{ab}\right)^2=\left(\frac{(a+b)(a-b)}{ab}\right)^2=\frac{-15}{16}$

36. $a^3+b^3=(a+b)(a^2-ab+b^2)$

37. Product of the zeroes $=\frac{\text{constant term}}{\text{Coefficient of } x^3}=2$

Worksheet – 2

Level – I

1.	B	5.	C	9.	D
2.	A	6.	B	10.	B
3.	A	7.	A		
4.	B	8.	A		

Level – II

1)B	2) A	3) C	4) A	5) D	6) D	7) A	8) B
9) A	10) C	11) A	12) B	13) A	14) C	15) A	16) B
17) A	18) A	19) D	20) B	21) B	22) B	23) A	24) C
25) B							

1. $(x^2-1)\div(x-1)=(x-1)(x+1)\div(x-1)=(x+1)$ Quotient is $x+1$
2. let $p(x)=x^{2019}-1 \Rightarrow p(1)=1^{2019}-1=0$
3. Let $p(x)=x^{2020}+1 \Rightarrow p(-1)=(-1)^{2020}+1=2$
4. Let $p(x)=x^{5051}+1 \Rightarrow p(-1)=(-1)^{5051}+1=0$
6.

$$
\begin{array}{l}
(x-1)\,|\,x^3-27x^2+8x+18\,|\,x^2-26x-18 \\
x^3-x^2 \\
\quad -26x^2+8x+18 \\
\quad -26x^2+26x \\
\quad\quad -18x+18 \\
\quad\quad -18x+18 \\
\hline
\quad\quad\quad 0
\end{array}
$$

7. $x^2 + 2x - 8 = (x + 4)(x - 2)$ and $x^2 + 4x = x(x + 4)$

   L.C.M of $x^2 + 2x - 8$ and $x^2 + 4x$ is $x(x + 4)(x - 2)$

8. $a^2 + 4a - 12 = (a + 6)(a - 2)$

   $a^2 - 5a + 6 = (a - 2)(a - 3)$

   $\therefore$ LCM is $(a - 3)(a - 2)(a + 6)$

9. $2x^2 + 5xy + 7y^2)\overline{14x^4 + 45x^3y + 78x^2y^2 + 45xy^3 + 14y^4}$

$$\begin{array}{r} 14x^4 + 35x^3y + 49x^2y^2 \\ 10x^3y + 29x^2y^2 + 45xy^3 \\ \hline 10x^3y + 25x^2y^2 + 35xy^3 \\ 4x^2y^2 + 10xy^3 + 14y^4 \\ 4x^2y^2 + 10xy^3 + 14y^4 \end{array}$$

   $\therefore$ Quotient $= 7x^2 + 5xy + 2y^2$; Remainder is zero.

10. $x^4 + x^2 + 1 = x^4 + x^2 + x^2 - x^2 + 1 = x^4 + 2x^2 + 1 - x^2$

    $= (x^2 + 1)^2 - x^2 = (x^2 + 1 - x)(x^2 + 1 + x)$

11. $10(x^2 - y^2), 20(x^3 + y^3) \Rightarrow 10(x^2 - y^2) = 2 \times 5 \times (x + y)(x - y)$

    $20(x^3 - y^3) = 2 \times 2 \times 5(x + y)(x^2 - xy + y^2)$

    L.C. $M = 2 \times 2 \times 5(x + y)(x - y)(x^2 - xy + y^2) = 40(x + y)(x - y)(x^2 - xy + y^2)$

13. $3k^3 - k + 1)\overline{6k^5 - 15k^4 + 4k^3 + 7k^2 - 7k + 2}$

$$\begin{array}{r} 6k^5 - 0 - 2k^3 + 2k^2 \\ -15k^4 + 6k^3 + 5k^2 - 7k \\ -15k^4 + 0 + 5k^2 - 5k \\ 6k^3 - 2k + 2 \\ 6k^3 - 2k + 2 \end{array}$$

    $\therefore$ Quotient $= 2k^2 - 5k + 2$; Remainder is zero

14. $\frac{a^9 - b^9}{a^2 + ab + b^2} = \frac{(a^3)^3 - (b^3)^3}{a^2 + ab + b^2} = \frac{(a^3 - b^3)(a^6 + a^3b^3 + b^6)}{a^2 + ab + b^2}$

    $= \dfrac{(a - b)(a^2 + ab + b^2)(a^6 + a^3b^3 + b^6)}{a^2 + ab + b^2} = (a - b)(a^6 + a^3b^3 + b^6)$

    $\therefore$ Quotient is $(a - b)(a^6 + a^3\ b^3 + b^6)$; Remainder is zero.

15. $14x^2y = 2 \times 7 \times x \times x \times y$

    $21x^2y^2 = 3 \times 7 \times x \times x \times y \times y \Rightarrow 35x^3y^3 = 5 \times 7 \times x \times x \times x \times y \times y \times y$

    $\therefore$ HCF of $14x^2y, 21x^2y^2$ and $35x^3y^3$ is $7x^2y$

16. HCF of $12(x^4 - x^3)$ and $(x^4 - 3x^3 + 2x^2) \Rightarrow 12(x^4 - x^3) = 12x^3(x - 1)$

    and $x^4 - 3x^3 + 2x^2 = x^2(x^2 - 3x + 2) = x^2(x - 1)(x - 2)$ $\therefore$ HCF $= x^2(x - 1)$

17. $p(x) = p(1 - x) \Rightarrow x^2 + 3x - 1 = (1 - x)^2 + 3(1 - x) - 1$

    $\Rightarrow x^2 + 3x - 1 = 1 + x^2 - 2x + 3 - 3x - 1 \Rightarrow x^2 + 3x - 1 = x^2 - 5x + 3$

    $\Rightarrow 8x = 4 \Rightarrow x = \dfrac{1}{2}$

18. $\frac{x-y}{x^{\frac{3}{4}}+x^{\frac{1}{2}}y^{\frac{1}{4}}} \div \frac{x^{1/2}+y^{1/2}}{x^{\frac{1}{2}}y^{\frac{1}{4}}+x^{\frac{1}{4}}y^{\frac{1}{2}}}$

Let $p = x^{1/4}$ and $q = y^{1/4}$

Then $p^2 = x^{1/2}$

$p^3 = x^{3/4}$

$p^4 = x$

$q^2 = y^{1/2}$

$q^3 = y^{3/4}$

$q^4 = y$

Hence $\frac{p^4-q^4}{p^3+p^2q} \div \frac{p^2+q^2}{p^2q+pq^2}$

$= \frac{(p^2-q^2)(p^2+q^2)}{p^2(p+q)} \times \frac{pq(p+q)}{p^2+q^2} = \frac{q(p^2-q^2)}{p} = \frac{y^{1/4}\left(x^{1/2}-y^{1/2}\right)}{x^{1/4}} = x^{1/4} \cdot y^{1/4} - y^{3/4} \cdot x^{-1/4}$

19. Given $x + y + z = -3a$, then we have

$2x - y - z = 3x - x - y - z = 3(x + a)$

$2y - z - x = 3y - x - z - y = 3(y + a)$

$2z - x - y = 3z - x - y - z = 3(z + a)$

Now $(2x - y - z)(2y - z - x)(2z - x - y) \div a^2 + a(x + y) + xy$

$= \frac{(2x-y-z)(2y-z-x)(2z-x-y)}{(x+a)(y+a)} = \frac{3(x+a)3(y+a)3(z+a)}{(x+a)(y+a)}$ from (1), (2) & (3)

$= 27(z + a)$ is required quotient.

20. The factors of $24(x^4 - 2ax^3 - 8a^2x^2) = 2^3 \times 3 \times x^2(x - 4a)(x + 2a)$

Factors of $54(x^5 - ax^4 - 6a^2x^3) = 3^3 \times 2 \times x^3(x - 3a)(x + 2a)$

from (1) & (2) H.C.F of given expressions

$2 \times 3 \times x^2(x + 2a) = 6x^2(x + 2a)$

21. $x^2 - 42x - 343 = (x - 49)(x + 7)$

zeroes of $(x - 49)(x + 7)$ are $x = 49, -7$

Is $a = -7$ then $x^2 + ax - 98 = x^2 - 7x - 98 = x^2 - 14x + 7x - 98 = (x - 14)(x + 7)$

$\therefore x + 7$ is the HCF $\therefore a = -7$

22. $ax^3 + bx^2 + (x + d) = (x^2 + 7x + 12)(x + 5)$

$= x^3 + 7x^2 + 12x + 5x^2 + 35x + 60 = x^3 + 12x^2 + 47x + 60$

23. $9x^4 - 28x^2 + 3 = 9x^4 - 27x^2 - x^2 + 3$

$= 9x^2(x^2 - 3) - 1(x^2 - 3) = (x^2 - 3)(9x^2 - 1)$

$= (x^2 - 3)(3x - 1)(3x + 1)$

$27x^4 - 12x^2 + 1 = 27x^4 - 9x^2 - 3x^2 + 1$

$= 9x^2(3x^2 - 1) - 1(3x^2 - 1) \Rightarrow = (3x^2 - 1)(9x^2 - 1)$

$= (3x^2 - 1)(3x - 1)(3x + 1)$

$27x^4 + 6x^2 - 1 = 27x^4 + 9x^2 - 3x^2 - 1$

$= 9x^2(3x^2 + 1) - 1(3x^2 + 1)$

$= (3x^2 + 1)(3x - 1)(3x + 1)$

$x^4 - 6x^2 + 9 = x^4 - 3x^2 - 3x^2 + 9 = (x^2 - 3)^2$

∴ L.C.M of the given expressions

$(x^2 - 3)^2(3x^2 - 1)(3x^2 + 1)(3x - 1)(3x + 1) = (x^2 - 3)^2(9x^4 - 1)(9x^2 - 1)$

24. $\frac{x^3+\frac{1}{27}y^3}{x-\frac{1}{3}y}$

Arrange dividend and divisor in descending powers of ' x ' $x^3 + 0 + 0 + 1/27y^3$ and $x - 1/3y$

$$\begin{array}{r} 1 + 1/3 + 1/9 \\ 1 - 1/3 \\ \hline 1 + 0 + 0 + 1/27 \\ 1/3 - 1/9 \\ \hline 1/9 + 1/27 \\ 1/9 - 1/27 \end{array}$$

∴ The complete quotient is $x^2 + 1/3xy + 1/9y^2$

25. LCM of $(x - b), (x - c)$ is $x^2 - 9$

$\Rightarrow x^2 - 9 = (x - b)(x - c) \Rightarrow (x + 3)(x - 3) = (x - b)(x - c)$

∴ $b = 3, c = -3$

$2b + 4c = 2(3) + 4(-3)$

$= 6 - 12 = -6$

Level – III

26) A,B 27) A 28) A 29) C 30) A 31) a-s; b-r; c-s; d-p 32.1,3

33) 1 34) a-r; b-t; c-q; d-p 35) 80 36) A,B

26.

$$\begin{array}{r} 1/4a^2 - 3ax + 9x^2 \\ 1/2a - 3x \overline{)\, 1/8a^3 - 9/4a^2x + 27/2ax^2 - 27x^3} \\ 1/8a^3 - 3/4a^2x \\ \hline -6/4a^2x + 27/2ax^2 \\ -3/2a^2x + 9ax^2 \\ \hline 9/2ax^2 - 27x^3 \\ 9/2ax^2 - 27x^3 \end{array}$$

∴ Quotient: $1/4a^2 - 3ax + 9x^2$; Remainder is zero

$6/25a^3 - 3/5a^2c + 3/2ac^2$

$1/5a^2 + 1/2ac \overline{)\, 6/125a^5 + 3/4a^2c^3}$

$\frac{6/125a^5 + 3/25a^4c}{-3/25a^4c + 3/4a^2c^3}$

$-3/25a^4c - 3/10a^3c^2$

$3/10a^3c^2 + 3/4a^2c^3$

$3/10a^3c^2 + 3/4a^2c^3$

∴ Quotient. $6/25a^3 - 3/5a^2c + 3/2ac^2$; Remainder is zero.

(i) H.C.F = (H.C.F. of numerical coefficients) × (Each common factor raised to lowest power)

(ii) L.C.M = (L.C.M of numerical coefficients) × (Each factor raised to highest power)

28.

Based on the above data answer the following questions.

(28-29) We have: $x^2 + xy = x(x + y)$ and $x^2 - y^2 = (x + y)(x - y)$.

∴ H.C.F. = (x + y).

L.C. M = x(x + y)(x − y).

30 Resolving each given polynomial into factors, we get: $2x^2 + 4x = 2x(x + 2)$,

$x^2 - 4 = x^2 - 2^2 = (x - 2)(x + 2)$

and $x^2 + 4x + 4 = x^2 + 2^2 + 2 \times x \times 2 = (x + 2)^2$.

∴ H.C.F of given polynomials = (x + 2)

L.C.M of given polynomials = $2x(x - 2)(x + 2)^2$.

31. a) $4 - 2x^{-1} + x^{-2})\overline{8 + 4x^{-1} + 2x^{-2}}$

$$\begin{array}{r} 8 - 4x^{-1} + 2x^{-2} \\ \hline 16x^{-1} + 0 + 0 + 2x^{-4} \\ 16x^{-1} - 8x^{-2} + 4x^{-3} \\ \hline 8x^{-2} - 4x^{-3} + 2x^{-4} \\ 8x^{-2} - 4x^{-3} + 2x^{-4} \end{array}$$

∴ Quotient is $2 + 4x^{-1} + 2x^{-2}$; Remainder is zero.

b) $3x^4 - x^3 + 2x^2 - 2x - 4 \div x + 2$

-2	3	-1	2	-2	-4
	0	-6	14	-32	68
	3	-7	16	-34	64

∴ Quotient is $3x^3 - 7x^2 + 16x - 34$ and Remainder is 64 .

c) $x^4 - 11x^3 + 44x^2 - 76x + 48$ by $x^2 - 7x + 12$

7	1	-11	44	-76	48
-12	0	7	-28	28	0
	0	0	-12	48	-48
	1	-4	4	0	0

∴ Quotient is $x^2 - 4x + 4$ and Remainder is zero

d) $(x^4 - 4x^2 + 13x - 4) \div (x^2 - 2x + 3)$

2	1	0	-4	13	-4
-3	0	2	4	-6	0
	0	0	-3	-6	-9
	1	2	-3	1	-13

∴ Quotient is $x^2 + 2x - 3$ and Remainder is $x - 13$.

32. $\frac{x^2+11x+35}{x+5}$

$$1+5\overline{)1+11+35}\quad\begin{matrix}1+6\\ \\ \\ \end{matrix}$$

$1+5$

$6+35$

$6+30$

5

∴ The complete quotient is $(x+6)+\frac{5}{x+5}$

34. Resolving each of the given polynomials into factors, we get:

$$(2a^2 + a - 6) = 2a^2 + 4a - 3a - 6 = 2a(a+2) - 3(a+2) = (a+2)(2a-3)$$

$(2a-3)^2 = (2a-3)^2(4a^2-9) = (2a)^2 - 3^2 = (2a-3)(2a+3)$,

∴ H.C.F of given polynomials $= (2a-3)$.

L.C.M of given polynomials $= (a+2)(2a-3)^2(2a+3)$.

c) $x^2 - x - 6 = x^2 - 3x + 2x - 6 = x(x-3) + 2(x-3) = (x-3)(x+2)$.

$2x^2 + 5x - 3 = 2x^2 + 6x - x - 3 = 2x(x+3) - (x+3) = (x+3)(2x-1)$.

$x^2 + 5x + 6 = x^2 + 3x + 2x + 6 = x(x+3) + 2(x+3) = (x+3)(x+2)$.

$$\therefore \frac{x-3}{x^2-x-6} + \frac{2x-1}{2x^2+5x-3} - \frac{2x+5}{x^2+5x+6}$$

$$= \frac{(x-3)}{(x-3)(x+2)} + \frac{(2x-1)}{(x+3)(2x-1)} - \frac{(2x+5)}{(x+3)(x+2)}$$

$$= \frac{1}{(x+2)} + \frac{1}{(x+3)} - \frac{(2x+5)}{(x+3)(x+2)}$$

$$= \frac{(x+3)+(x+2)}{(x+2)(x+3)} - \frac{(2x+5)}{(x+3)(x+2)} = \frac{(2x+5)-(2x+5)}{(x+2)(x+3)} = 0$$

d) We have: $x^2 - x - 6 = x^2 - 3x + 2x - 6$

$= x(x-3) + 2(x-3) = (x-3)(x+2) \cdot x^2 - 9 = (x+3)(x-3)$

$x^2 - 9(x+3)(x-3)$.

$x^2 + 2x - 24 = x^2 + 6x - 4x - 24 = x(x+6) - 4(x+6) = (x+6)(x-4)$

$x^2 - x - 12 = x^2 - 4x + 3x - 12 = x(x-4) + 3(x-4) = (x-4)(x+3)$.

$\therefore \dfrac{x^2-x-6}{x^2-9} + \dfrac{x^2+2x-24}{x^2-x-12} = \dfrac{(x-3)(x+2)}{(x+3)(x-3)} + \dfrac{(x+6)(x-4)}{(x-4)(x+3)}$

$= \frac{(x+2)}{(x+3)} + \frac{(x+6)}{(x+3)} = \frac{(x+2)+(x+6)}{(x+3)} = \frac{2(x+4)}{(x+3)}$.

35. SO1 L.C.M of numerical coefficients = L.C.M of 16 and 20 = 80.

L.C.M of literals = product of each literal raised to the highest power = x^3y^4z.

$\therefore$ L.C.M of given monomials = $80x^3y^4z$.

36. Arrange dividend and divisor in descending powers of

$x, 8x^3 - 18xyz - (27y^3 + z^3)$ and $4x^2 + (6y + 2z)x + (9y^2 + z^2 - 3yz)$

$$
\begin{array}{r}
2x - 3y - z \\
4x^2 + (6y+2z)x + (9y^2 + z^2 - 3yz) \overline{)\, 8x^3 + 0 - 18yzx - (27y^3 + z^3)} \\
8x^3 + (12y + 4z)x^2 + 2(9y^2 + z^2 - 3yz)x \\
-(12y + 4z)x^2 - 2(9y^2 + z^2 + 6yz)x - (27y^3 + z^3) \\
-(12y + 4z)x^2 - 2(9y^2 + z^2 + 6yz)x - (27y^3 + z^3)
\end{array}
$$

$\therefore$ Quotient is $2x - 3y - z$ and Remainder is zero.

Alter: Using $(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = a^2+b^3+c^3-3abc$

$\therefore 8x^3 - 27y^3 - z^3 - 18xyz = (2x)^3 + (-3y)^3 + (-z)^3 - 3(2x)(-3y)(-z)$

$= (2x - 3y - z)[(4x^2 + (-3y)^2 + (-z)^2 - (2x)(-3y) - (-3y)(-z) - (2x)(-z)]$

Thus $\frac{8x^3-27y^3-z^3-18xyz}{4x^2+9y^2+z^2+6xy+2xz-3yz}$

$$= \frac{(2x - 3y - z)(4x^2 + 9y^2 + z^2 + 6xy + 2xz - 3yz)}{4x^2 + 9y^2 + z^2 + 6xy + 2xz - 3yz}$$

$$= 2x - 3y - z$$

5. QUADRATIC EQUATIONS

Worksheet – 1

Level – I

1) D	2) A	3) B	4) C	5) B	6) C	7) D
8) C	9) D					

Level – II

1)A	2)B	3)B	4)B	5)C	6)A	7)C
8)A	9)A	10)C	11)A	12)B	13)B	14)A
15)C	16) B	17)D	18)A	19)A	20)B	21)B
22) C	23)D	24)A	25)30	26)A,D	27)B	28)D
29)a-p,r;b-s;c-p,q;d-s		30)5	31)B	32)A		
33)a-p,t;b-q,r;c-p;d-q,r		34)a-q;b-r;c-p;d-s				

1. $\alpha \times \frac{1}{\alpha} = \frac{2k}{5} = k = \frac{5}{2}$
2. $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(-1)^2-2(1)}{1} = 1-2 = -1$
3. $\alpha - \beta = 1 \Rightarrow \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = 1 \Rightarrow p^2 - 4q = 1$
4. $8 + i + i - i = -p \ \therefore p = -16$ and $64 + 1 = q \ \therefore q = 65$
5. $(1-\mathrm{p})^2 + p(1-\mathrm{p}) + (1-\mathrm{p}) = 0 \Rightarrow (1-\mathrm{p})[1-p+p+1] = 0 \Rightarrow p = 1 \therefore x^2 + x = 0$
6. $\alpha^2 + \beta^2 = 10 \Rightarrow (\alpha+\beta)^2 - 2\alpha\beta = 10 \Rightarrow p^2 - 2(-3) = 10 \Rightarrow \mathrm{p}^2 = 4$
7. $\alpha, \alpha^2 = 1 \Rightarrow \alpha^3 = -k, \alpha(1+\alpha) = 1 \Rightarrow \mathrm{k}^2 - 4k - 1 = 0 \Rightarrow k = \frac{4\pm\sqrt{20}}{2} = 2 \pm \sqrt{5}$
8. $16 + 4p + 12 = 0 \Rightarrow p = -7 \Rightarrow$ and $4p - 4q = 0 \Rightarrow q = \frac{49}{4}$
9. $1 + \alpha + \beta + \alpha\beta \Rightarrow 1 + p - p + c = 1 + c$
10. $\frac{-b}{a} = \frac{c}{a} \ \therefore -k - 3 = 3k + 5 \Rightarrow -8 = 4k \Rightarrow k = -2$
11. verify

    $\alpha + \beta = 7$

    $\alpha^2 + \beta^2 = 25$
12. $\sin\theta + \cos\theta = \frac{-b}{a} \Rightarrow \sin\theta \cdot \cos\theta = \frac{c}{a} \Rightarrow 1 + 2\sin\theta \cdot \cos\theta = \frac{b^2}{a^2} \ \therefore 1 + \frac{2c}{a} = \frac{a^2}{b^2}$
13. $\alpha + \beta = \frac{27}{9} = 3 \Rightarrow \alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = 7$
14. $\alpha(\alpha+\beta) + \alpha\beta \Rightarrow \frac{c}{a}\left(\frac{-b}{a} + \frac{c}{a}\right) = \frac{-bc+ac}{a^2}$

15. roots are $1, \frac{c}{a} \Rightarrow 1 = \frac{c(a-b)}{a(b-c)} \Rightarrow b = \frac{2ac}{a+c}$

16. $b^2 - 4ac = 0 \Rightarrow 2m^2 - m - 1 = 0 \Rightarrow m = 1, \frac{-1}{2}$

17. $\alpha - \beta = \gamma - 8 \Rightarrow (\alpha+\beta)^2 - 4\alpha\beta = (\gamma+8)^2 - 4\gamma\beta \Rightarrow b^2 - 4c = c^2 - 4b \Rightarrow b + c = -4$

18. roots are $1, \frac{c}{a} \Rightarrow 1 = \frac{a-b}{b-c} \ \therefore 2b = a + c$

19. $\left[\frac{\alpha^2 - \beta^2}{\alpha\beta}\right]^2 = \frac{b^2}{c^2}\left(\frac{b^2 - 4ac}{a^2}\right)$

20. $\frac{\alpha^3 + \beta^3}{\frac{1}{\alpha^3} + \frac{1}{\beta^3}} = (\alpha\beta)^3 = \frac{c^3}{a^3}$

21. $(x_1 + x_2)^2 - 4x_1x_2 = 1 \Rightarrow p^2 - 4(12) = 1 \Rightarrow p = \pm\sqrt{49}$

22. $(\alpha+\beta)^2 - 4\alpha\beta = 1 \Rightarrow k^2 - 4(2) = 1 \Rightarrow k = \pm 3$

23. $2(1 + \alpha + \beta + \alpha\beta) \Rightarrow 2(1 + p - p + c) = 2(1 + c)$

24. $x = \sqrt{6+x} \Rightarrow x^2 - x - 6 = 0 \Rightarrow x^2 - 3x + 2x - 6 = 0 \Rightarrow x(x-3) + 2(x-3) = 05$
$x = 3, x = -2$

25. $x^2 - 13x + q = 0 \Rightarrow 30 = q \Rightarrow x^2 - 13x + 30 = 0 \Rightarrow x = -10, -3$

26. $k^2 - 8k = 0 \Rightarrow k = 0(\text{ or }) 8$

27. $x^2 - 10x - 3x + 30 = 0 \Rightarrow x = 10,3$

28. $2^2 - 2p + 2 = 0 \Rightarrow 2p = 6 \Rightarrow p = 3$

29. a) $D = 16 - 12 = 4 > 0$. Hence, the roots are real and unequal
b) $D = 1 - 4 = -3 < 0$. Hence, the roots are complex
c) $D = 4 - 4 = 0$. Hence, the roots are real and equal.
d) $D = 4 - 12 = -8 < 0$. Hence the roots are complex

30. $\alpha - \beta = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \sqrt{(-1)^2 - 4(-1)} = \sqrt{5}$

31. $2b = a + c = (2b)^2 - 4ac = (a+c)^2 - 4ac = (a-c)^2 \geq 0$

32. Both statements are true. 33. a-p,t;b-q,r;c-p;d-q,r

33. a) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = \frac{b^2 - 2ac}{a^2}$
b) $\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = \frac{3abc - b^3}{a^3}$
c) $\alpha^{-2} + \beta^{-2} = \frac{\alpha^2+\beta^2}{(\alpha\beta)^2} = \frac{b^2-2ac}{c^2}$ d) $\alpha^{-3} + \beta^{-3} = \frac{\alpha^3+\beta^3}{(\alpha\beta)^3} = \frac{3abc-b^3}{c^3}$

Worksheet – 2

Level – I

1) B 2) B 3) B 4) A 5) C 6) B 7) B
8) C 9) A 10) C

Level – II

1. Put $x = 4\alpha \Rightarrow \alpha = \frac{x}{4} \Rightarrow \left(\frac{x}{4}\right)^2 + 2\left(\frac{x}{4}\right) + 1 = 0$
2. $\frac{\alpha}{3} = x \Rightarrow \alpha = 3x \Rightarrow 2(9x^2) + 3x + 5 = 0$
3. roots $\alpha + 1, \beta + 1$

   Let $x = \alpha + 1 \Rightarrow \alpha = x - 1 \Rightarrow 2(x-1)^2 - 3(x-1) + 1 = 0$
4. Let roots be $\alpha, \alpha^2 \Rightarrow \alpha + \alpha^2 = -1, \alpha^3 = -k \Rightarrow \alpha(1+\alpha) = -1$ cubing on both sides

5. $(\alpha+\beta)^2 - 2\alpha\beta = \left(\frac{5}{3}\right)^2 - 2\left(\frac{7}{3}\right) = \frac{25}{9} - \frac{14}{3}$ 6. $\alpha\beta(\alpha+\beta) \Rightarrow 2(1) = 2$

7. $1 + a + b = 0 \Rightarrow 1 + c + d = 0 \Rightarrow a + b = -1, c + d = -1$
8. $4 + 2a + b = 0 \Rightarrow 4 + 2c + d = 0 \Rightarrow 2(a - c) = (d - b)$
9. $p = 2, \alpha = 2$ 10. $\frac{\frac{1}{\alpha^3}+\frac{1}{\beta^3}}{\alpha^3+\beta^3} = \left(\frac{1}{\alpha\beta}\right)^3 = \frac{1}{-q^3}$
10. $\alpha^2 - c\alpha + d = 0 \Rightarrow b - \frac{ac}{2} + d = 0 \Rightarrow 2(b+d) = ac$
11. $a(\alpha+\beta) + 2b = a\left(\frac{-b}{a}\right) + 2b$
12. $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{\alpha} = \frac{\left(\frac{4}{3}\right)^3 - 3\left(\frac{1}{3}\right)\left(\frac{4}{3}\right)}{\frac{1}{3}}$
13. $(6+9)^2 = (-36a - 9a)(3a - 8a) \Rightarrow 225 = (-45a)(-5a) \Rightarrow a = \pm 1$

    common root $= \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} = \frac{15}{-9} = -3$
15. $(ab - ac)^2 = (2c^2 - 2b^2)(2ab - 2ac) \Rightarrow b + c = \frac{-a}{4}$
16. $\sqrt{(x_1 + x_2)^2 - 4x_1x_2} = 1 \Rightarrow \sqrt{p^2 - 4(12)} = 1 \Rightarrow p = \pm 7$
17. roots be $\alpha, 6\alpha \Rightarrow 7\alpha = \frac{14}{p}, 6\alpha^2 = \frac{8}{p} \Rightarrow P = 3$
18. $a\alpha^2 + b\alpha + c = 0 \Rightarrow a\alpha + b = -\frac{c}{\alpha}, a\beta + b = \frac{-c}{\beta} \Rightarrow \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2} = \frac{b^2 - 2ac}{c^2a^2}$
19. $\alpha\beta = \frac{c}{a} > 0 \Rightarrow c < 0, a < 0$ or $c > 0, a > 0 \Rightarrow \alpha + \beta = \frac{-b}{a} > 0 \Rightarrow -b > 0, a > 0$
20. $\alpha - \beta = \gamma - \delta \Rightarrow \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \sqrt{(\gamma+\delta)^2 - 4\gamma\delta}$
21. $x = 1, 1 + \frac{1}{1} = 2$
22. $\alpha\beta = \frac{c}{a} > 0 \Rightarrow c < 0, a < 0$ or $c > 0, a > 0 \Rightarrow \alpha + \beta = \frac{-b}{a} > 0 \Rightarrow -b > 0, a > 0$
23. $D_1 = b^2 - 4ac \Rightarrow D_2 = q^2 - 4pr \Rightarrow \frac{\alpha}{\beta} = \frac{\gamma}{\delta} \Rightarrow \frac{\alpha+\beta}{\alpha-\beta} = \frac{\gamma+\delta}{\gamma-\delta}$
24. $\alpha^2 = x \Rightarrow \alpha = \sqrt{x} \Rightarrow 2\sqrt{x} = 7 - 5x$ Squring on both sides
25. $\sqrt[3]{x^2} + 2(\sqrt[3]{x}) - 4 = 0 \Rightarrow x^{2/3} = 4 - 2\left(x^{1/3}\right) \Rightarrow$ cubing on both sides

Level – III

26. $(\sqrt{x})^2 + 5\sqrt{x} - 9 = 0 \Rightarrow 5\sqrt{x} = 9 - x \Rightarrow$ Squring on both sides
27. conceptual
28. $x^{2/3} + 3x^{1/3} + 2 = 0 \Rightarrow x^{2/3} + 3x^{1/3} = -2 \Rightarrow$ Cubing on both sides
29. conceptual
30. a) $\frac{\alpha}{\alpha+1} = x \Rightarrow \alpha = \frac{x}{1-x}$ sub in $ax^2 + bx + c = 0$

    b) $\frac{\alpha-1}{\alpha+1} = x \Rightarrow \alpha = \frac{x+1}{1-x}$ sub in $ax^2 + bx + c = 0$

    c) $\alpha - 1 = x \Rightarrow \alpha = x + 1$

    sub in $ax^2 + bx + c = 0$

Worksheet – 3

Level – I

1) A 2)A 3)B 4)B 5)A 6)A 7)C 8)C 9) A

1. $7\mathbf{x}^2 + 2x - 8$

   a = 7 b = 2 c = −8

   a = 7 > 0

   ∴ minimum value $= \frac{4ac-b^2}{4a} = \frac{a(7)(-8)-2^2}{4(7)} = \frac{4(7)(-8)-2^2}{4(7)} = \frac{4(-56-1)}{4(7)} = \frac{-57}{7}$
2. $4 - x^2 > 0 \Rightarrow x^2 - 4 < 0 \Rightarrow (x+2)(x-2) < 0 \Rightarrow -2 < x < 2$
3. $-7 < x < 3 \Rightarrow (x+7)(x-3) < 0 \Rightarrow x^2 + 4x - 21 < 0$ ∴ $x^2 + 4x - 21$ is negative
4. $2x - 7 - 5x^2 \Rightarrow$ a = −5 b = 2c = −7 $\Rightarrow a = -5 < 0$

   ∴ Maximum value $= \frac{4ac-b^2}{4a} = \frac{4(-5)(-7)-2^2}{4(-50} = \frac{4(35-1)}{4(-5)} = \frac{-34}{5}$.
5. $3x^2 + 2x + 11$

   a = 3 b = 2 c = 11

   a = 3 > 0

   $x = \frac{-b}{2a} = \frac{-1}{3}$
6. $2x^2 + 3x - 5 < 0$

   $2x^2 + 3x - 5 < 0 = x(2x+5) - 1(2x+5) < 0 = (2x+5)(x-1) < 0 = \frac{-5}{2} < x < 1$
7. $x^2 - 5x + 6 > 0 \Rightarrow \mathrm{x}^2 - 5\mathrm{x} + 6 > 0$

$(x-2)(x-3) > 0 \Rightarrow x < 2$ or $x > 3 \Rightarrow r - [2,3]$

8. $x^2 - 2x + 7 < 0 \Rightarrow \Delta = 4 - 28 < 0$

9. $\Delta = (-8)^2 - 4(1)(15) = 4 > 0$

Level – II

1)A	2)B	3)A	4)A	5)B	6)C
7)D	8)D 9)A	10)D	11)D	12)A	13)A
14)D	15)C 16)D	17)C	18)C	19)D	20)D
21)B					

1. $2x^2 + x - 1 \leq 0 \Rightarrow 2x^2 + 2x - 1x - 1 \leq 0 \Rightarrow 2x(x+1) - 1(x+1) \leq 0$

   $\Rightarrow (x+1)(2x-1) \leq 0 \Rightarrow -1 \leq x \leq \frac{1}{2}$

2. Since α, β are roots of $ax^2 + ba + c > 0 (\alpha < \beta)$ and $\alpha x^2 + ba + c > 0$

   $\Rightarrow x < \alpha q\, x > \beta$ and $d > 0$

3. $\alpha < x < \beta$

4. $\alpha < x < \beta$

5. $-4 < x < 8 \Rightarrow (x+4)(x-8) < 0 \Rightarrow x^2 - 4x - 32 < 0$

6. $2x^2 + x - 1$

   $a = 2\ b = 1\ c = -1$

   $a = 2 > 0$

   minimum value $= \frac{4(2)(-1)-1^2}{4(2)} = \frac{-8-1}{8} = \frac{-9}{8}$

7. $c + 2bx - x^2$

   $a = -1 < 0$

   maximum value $= \frac{4(-1)(c)-(2b)^2}{4(-1)} = \frac{-4c-4b^2}{-4} = c + b^2$

8.

   $2x - 7 - 5x^2$ has maximum at $x = a$

   $\therefore a = \frac{-2}{2(-5)} = \frac{1}{5}$

9. $2x^2 + 4x + 7 \Rightarrow \alpha = \frac{-b}{2a} = \frac{-4}{2(2)} = \frac{-4}{4} = -1$

   $m = \frac{4ac - b^2}{4a} = \frac{4(2)(7) - 4^2}{4(2)} = \frac{4(14-4)}{4(2)} = \frac{10}{2} = 5 \Rightarrow (\alpha, m) = (-1,5)$

10. $a^2 - abx - b^2x^2$

minimum value $= \frac{4(-b^2)(a^2)-(ab)^2}{4(-b^2)} = \frac{b^2(-4a^2-a^2)}{4(-b^2)} = \frac{-5a^2}{-4} = \frac{5a^2}{4}$

11. $5 + 2x - x^2$

maximmum value $= \frac{4(-1)(5)-2^2}{4(-1)} = \frac{-20-4}{-4} = \frac{-24}{-4} = 6$

12. $\frac{4ac-b^2}{4a} = \frac{8}{4} = 2$

13. $a \in R, b \in R, x^2 - abx - a^2 = 0 \Rightarrow \alpha \cdot \beta = \frac{-a^2}{1} < 0$ one root positive and one root negative

14. $\frac{x}{x^2-5x+9} = y \Rightarrow yx^2 - 5yx + 9y = x \Rightarrow yx^2 - 5yx - x + 9y = x \Rightarrow yx^2 - (5y+1)x + 9y = 0 \ \forall x \in R, \Delta \geq 0$

$\Rightarrow (5y+1)^2 - 4(y)(9y) \geq 0 \Rightarrow 25y^2 + 10y + 1 - 36^2 \geq 0 \Rightarrow -11y^2 + 10y + 1 \geq 0$

$\Rightarrow 11y^2 - 10y - 1 \leq 0 \Rightarrow 11y^2 - 11y + 1y - 1 \leq 0 \Rightarrow 11y(y-1) + 1(y-1) \leq 0$

$\Rightarrow (11y+1)(y-1) \leq 0 \Rightarrow \frac{-1}{11} \leq y \leq 1$

15. $\frac{x^2-x+1}{x^2+x+1} = y \Rightarrow yx^2 + yx + y = x^2 - x + 1$

$\Rightarrow (y-1)x^2 + x(y+1) + (y-1) = 0 \Rightarrow \Delta \geq 0 \Rightarrow (y+1)^2 - 4(y-1)^2 \geq 0$

$\Rightarrow y^2 + 2y + 1 - 4(y^2 - 2y + 1) \geq 0 \Rightarrow y^2 + 2y + 1 - 4y^2 + 8y - 4 \geq 0$

$\Rightarrow -3y^2 + 10y - 3 \geq 0$

16. $\frac{x^2-1}{x^2+1} = y \Rightarrow yx^2 + y = x^2 - 1 \Rightarrow x^2(y-1) + (y+1) = 0$

$\Delta \geq 0$

$\Rightarrow 0^2 - 4(y-1)(y+1) \geq 0 \Rightarrow y^2 - 1 \leq 0 \Rightarrow (y-1)(y+1) \leq 0$

$y \in [-1,1]$

17. $y = x^2 - 4x + 4 \Rightarrow y = (x-2)^2$ $a = 1 > 0$

18. α, β

$\therefore$ Touches the x-axis and lies entirely above the x-aris.

19. $x^2 - 2x + 3 > 0 \Rightarrow \forall x \in R, x^2 - 2x + 3 > 0$

$\because \Delta < 0$

$2x^2 + 4x + 3 > 0 \Rightarrow \Delta = 16 - 24 = -8 < 0$

$a = 2 > 0 \ \therefore \forall x \in R,$

$2x^2 + 4x + 3 > 0$

$\therefore x \in (-\alpha, \alpha)$

20. $\frac{x+2}{2x^2+3x+6} = y$

$2yx^2 + 3yx + 6y = x + 2 \Rightarrow 2yx^2 + (3y-1)^x + (6y-2) \geq 0$

$\Delta \geq 0 \Rightarrow (3y-1)^2 - 4(2y)(6y-2) \geq 0 \Rightarrow 9y^2 - 6y + 1 - 48y^2 + 16y \geq 0$

$\Rightarrow 39y^2 + 10y + 1 \geq 0 \Rightarrow 39y^2 - 10y - 1 \geq 0$

$\Rightarrow 39y^2 - 13y + 3y - 1 \leq 0 \Rightarrow 13y(3y-1) + 1(3y-1) \leq 0 \Rightarrow (13y+1)(3y-1) \leq 0$

$y \in \left[\frac{-1}{13}, \frac{1}{3}\right]$

21. $\frac{x^2+34x-71}{x^2+2x-7} = y \Rightarrow x^2 + 34x - 71 = yx^2 + 2yx - 7y$

$\Rightarrow (y-1)x^2 + (2y-34)x - 7y + 71 = 0 \Rightarrow \Delta \geq 0$

$\Rightarrow (y-1)^2 - 4(y-1)(-7y+71) \geq 0 \Rightarrow 4y^2 - 136y + 1156 - 4(-7y^2 + 71y + 7y - 71) \geq 0$

$\Rightarrow 4(y^2 - 34y + 289) - (-7y^2 + 78y - 71) \geq 0 \Rightarrow y^2 - 34y + 289 + 7y^2 - 78y + 71 \geq 0$

$\Rightarrow 8y^2 - 112y + 360 \geq 0 \Rightarrow y^2 - 14 + 45 \geq 0$

$(y-5)(y-9) \geq 0 \Rightarrow y \leq 5$ or $y \geq 9$

Level – III

22)A,B,C,D 23)C 24)A 25)0 26)B

27)B 28)A 29)a-s;b-p;c-t;d-q 30)A,C 31)A

32)A 33)A 34)A 35)A

22. $2x^2 - 3x + 1 \leq 0$

$2x^2 - 2x - 1x + 1 \leq 0 \Rightarrow 2x(x-1) - 1(x-1) \leq 0 \Rightarrow (2x-1)(x-1) \leq 0$

$\Rightarrow \frac{1}{2} \leq x \leq 1$

23. S-1: $2 < X < 4 \Rightarrow (X-2)(X-4) < 0 \Rightarrow x^2 - 6x + 8 < 0$ $\therefore S-1$ is true.

$S-1: x \notin [2,4], x^2 - 6x + 8 > 0$ $\therefore S-2$ is true.

24. $x^2 - 5x + 6 < 0 \Rightarrow (x-2)(x-3) < 0 \Rightarrow x \in (2,3)$

Both S-1,S-2 are true and S-II is correct explanation for I

25. $2x^2 + 4x + 7 \Rightarrow n = \frac{-b}{2a} = -\frac{4}{2} = -2$

$m = \frac{4ac - b^2}{4a} = \frac{4(2)(7) - 4^2}{4(2)} = \frac{4[28-4]}{4(2)} = \frac{10}{2} = 5$ $\therefore m + n = -2 + 5 = 3$

26. As, $a = 1 > 0$, the minimum height of any point on the curve is $\frac{4ac-b^2}{4a} = \frac{24-16}{4} = 2$

27. As, $a = 1 > 0$, the minimum height of any point on the curve is $\frac{4ac-b^2}{4a} = \frac{24-16}{4} = 2$

28. As, $a = -1 < 0$, the maximum height of any point on the curve is $\frac{4ac-b^2}{4a} = \frac{20-36}{-4} = 4$.

29. $x^2 - 5x + 6 = 0 \Rightarrow (x-2)(x-3) = 0 \Rightarrow x = 2$ or $x = 3$

(a) $x^2 - 5x + 6 > 0 \Rightarrow x < 2$ or $x > 3$ (s)

(b) $x^2 - 5x + 6 < 0 \Rightarrow 2 < x < 3$ (P)

(c) $x^2 - 5x + 6 = 0 \Rightarrow x = 2$ or $x = 3$ (t)

(d) $x^2 - 5x + 6 \leq 0 \Rightarrow 2 \leq x \leq 3$ (q)

30. $x^2 - 11x + 10 > 0 \Rightarrow (x-1)(x-10) > 0 \Rightarrow x < 1$ or $x > 10$

31. $x^2 + x + 1 > 0 \Rightarrow \Delta = 1 - 4 = -3 < 0$ $a = 1 > 0$ $\therefore \forall x \in R, x^2 + x + 1 > 0$

$\therefore$ Both A and R are correct and R is correct explansion of A

$\Rightarrow a^2 + 12a - 6a + 72 = 0 \Rightarrow (a+12)(a-6) = 0 \Rightarrow a - 6 = 0$ or $a + 12 = 0$

$x^2 + 5x - 6 = 0$ or $x^2 + 5x + 12 = 0 \Rightarrow x = \frac{-5 \pm \sqrt{25+24}}{2} = \frac{-5 \pm 7}{2} = 1, -6$

32. In the 1st figure α, β both are positive $\Rightarrow \alpha + \beta > 0, \alpha.\beta > 0$

$-\frac{b}{a} > 0, \frac{c}{a} > 0 \Rightarrow a, c$ have same sign which os opposite to b

In the 2nd figure α is negative β is positive $\alpha\beta < 0$

$\frac{c}{a} < 0$ a,c have opposite sign $\therefore$ Both A and B are true

33. $\frac{x}{x^2-5x+9} = y \Rightarrow yx^2 - 5yx + 9y = x \Rightarrow yx^2 - (5y-1)x + 9y = 0$

$\Delta \geq 0$

$(5y-1)^2 - 4(y)(9y) \geq 0 \Rightarrow 25y^2 - 10y + 1 - 36y^2 \geq 0 \Rightarrow -11y^2 - 10y + 1 \geq 0$

$\Rightarrow 11y^2 + 10y - 1 \leq 0 \Rightarrow 11y^2 + 11y - 1y - 1 \leq 0 \Rightarrow 11y(y+1) - 1(y+1) \leq 0$

$\Rightarrow (y+1)(11y-1) \leq 0 \Rightarrow -1 \leq y \leq \frac{1}{11}$

34. $\sin\alpha, \cos\alpha$ are roots of $ax^2 + bx + c = 0$

$\sin\alpha + \cos\alpha = -\frac{b}{a}$ $\sin\alpha\cos\alpha = \frac{c}{a}$

$(\sin\alpha + \cos\alpha)^2 = \frac{b^2}{a^2}$ $1 + 2\sin\alpha\cos\alpha = \frac{b^2}{a^2}$

$1 + \frac{2c}{a} = \frac{b^2}{a^2} \Rightarrow \frac{a+2c}{a} = \frac{b^2}{a^2} \Rightarrow a^2 + 2ac = b^2$

$\Rightarrow b^2 - a^2 = 2ac \Rightarrow a^2 - b^2 + 2ac = 0$

35. $x(x+2)(x+3)(x+5) = 72$

$(x^2+5x)(x^2+5x+6) = 72 \Rightarrow$ let $x^2 + 5x = a \Rightarrow a(a+6) = 72$

$a^2 + 6a - 72 = 0 \Rightarrow a^2 + 12a - 6a - 72 = 0 \Rightarrow (a+12)(a-6) = 0$

$a - 6 = 0$ or $a + 12 = 0$

$\Rightarrow x^2 + 5x - 6 = 0$ or $x^2 + 5x + 12 = 0$

$x = \frac{-5 \pm \sqrt{25 + 24}}{2}$ $x = \frac{-5 \pm \sqrt{25 + 48}}{2} \Rightarrow = \frac{-5 \pm 7}{2} = \frac{-5 \pm \sqrt{-25}}{2}$ $1, -6$

---

6. COMPLEX NUMBERS

Worksheet – 1

Level – I

1.	A	4.	C	7.	C	10.	B
2.	A	5.	A	8.	B	11.	B
3.	C	6.	B	9.	A		

1. $i^{125} = i^{125} = i^{4\times31+1} = i^1 = i$
2. $i^{120} + i^2$

   $i^{120} = i^{4\times30+0} = i^0 = 1$

   $i^{120} + i^2 = 1 - 1 = 0$
3. $\sqrt{-a} \times \sqrt{-b} \Rightarrow \sqrt{-a} \times \sqrt{-b} = \sqrt{ai} \times \sqrt{bi} = -\sqrt{ab}$
4. If $z_1 = a + 3i, z_2 = a - 3i$ then $z_1 + z_2$ is

   $z_1 + z_2 = a + 3i + a - 3i = 2a$
5. If $z_1 = -2 + 2i, z_2 = -2 - 2i$ then $z_1 - z_2$ is

   $z_1 - z_2 = -2 + 2i = (-2 - 2i) = -2 + 2i + 2 + 2i = 4i$
6. If $z_1 = 1 + i, z_2 = 1 - i$ then $z_1 z_2$ is

   $z_1 z_2 = (1 + i)(1 - i) = 1 - i^2 = 1 + 1 = 2$
7. The Additive inverse of $-3 + 5$i

   Additive Inverse of $-3 + 5i$ is $3 - 5i$
8. The multiplicative Inverse of (3,4)

   Multiplicative inverse of (a, b) is $\left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$

   Multiplicate inverse of (3,4) is $\left(\frac{3}{3^2+4^2}, \frac{-4}{3^2+4^2}\right) = \left(\frac{3}{25}, \frac{-4}{25}\right)$
9. The Conjugate Complex of $4 + 5$i is

   Conjugate Complex of $4 + 5i$ is $4 - 5i$
10. If $z + \bar{z} = 0$ then

    Let

    $z = x + iy \& \bar{z} = x - iy$

    $z + \bar{z} = x + iy + x - iy = 2x$ is possible only if x = 0 i.e Re (z)

    $z + \bar{z} = 0$
11. $z\bar{z} = 0$ if and only if

    Let $z = x + iy \& \bar{z} = x - iy$ $z\bar{z} = (x + iy)(x - iy) = x^2 + y^2$

    $z\bar{z} = 0$ is possible only when both real and imaginary parts of z are zero i.e $z = 0$

Level – II

1) B 2) C 3) C 4) B 5) A 6) B 7) 0 8) D

9) C 10) C 11) A 12) A 13) B 14) C 15) C 16) B

17) A 18) B 19) A 20)145/63

1. If $z = 2 - 3i$ then the value of $z^2 - 4z + 13$ is

   $z = 2 - 3\text{i}$

   $z^2 = (2 - 3i)^2 = 2^2 + (3i)^2 - 2 \times 2 \times 3i$

   $= 4 - 9 - 12i = -5 - 12i$

   $z^2 - 4z + 13 = -5 - 12i - 4(2 - 3i) + 13$

   $= -5 - 12i - 8 + 12i + 13$

   $= -13 + 13 = 0$

2. $\sqrt{8 - 6i}$

   $$\left\{\sqrt{a - ib} = \pm\left\{\sqrt{\frac{1}{2}\left(\sqrt{a^2 + b^2} + a\right)} - i\sqrt{\frac{1}{2}\left(\sqrt{a^2 + b^2} - a\right)}\right\}\right.$$

   a = 8, b = 6

   $$= \pm\left\{\sqrt{\frac{1}{2}\left(\sqrt{8^2 + 6^2} + 8\right)} - i\sqrt{\frac{1}{2}\left(\sqrt{8^2 + 6^2 - 8}\right\}\right.}$$

   $$= \pm\left\{\sqrt{\frac{1}{2}(\sqrt{100} + 8)} - i\sqrt{\frac{1}{2}(\sqrt{100 - 8}\right\} = \pm\left\{\sqrt{\frac{1}{2} \times 18} - i\sqrt{\frac{1}{2} \times 2}\right\} = \pm(3 - i)$$

3. $1 + i^2 + i^4 + i^6 + \cdots \ldots + i^{2n} =$

   $1 + i^2 + i^4 + i^6 + \cdots \ldots + i^{2n} = 1 - 1 + 1 - 1 + \cdots . + 1 - 1 = 0$

4. If $i = \sqrt{-1}$ and n is a possitive integer, then $i^n + i^{n+1} + i^{n+2} + i^{n+3}$ is equal to n is a possitive integer then $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$

5. The Additive Inverse of $(1 + 2\text{i})(3 - 4\text{i})$ is

   $(1 + 2\text{i})(3 - 4\text{i}) = 3 - 4i + 6i - 8i^2$

   $= 3 - 4i + 6i - 8 = 11 + 2i$

6. The Conjugate of $z + 2\bar{z}$ is

   Conjugate of $z + 2\bar{z}$ is $z + 2\bar{z} = \bar{z} + 2\overline{(\bar{z})} = \bar{z} + 2z$

7. If $i^2 = -1$ then the value of $\sum_{i=1}^{200} i^n$ is

   $\sum_{i=1}^{200} i^n = i + i^2 + i^3 + i^4 + \cdots \ldots + i^{200}$

   $= \text{i} - 1 - \text{i} + 1 + \text{i} - 1 - \text{i} + 1 = 0$

8. $(1-i)^6+(1-i)^3=$

$(1-i)^6+(1-i)^3=(1-i)^3[(1-i)^3+1]$

$=[1-i^3-3\times1^2\times i+3\times1\times i^2][1-i^3-3\times1^2\times i+3\times1\times i^2+1]$

$=(1+i-3i-3)(1+i-3i-3+1)$

$=(-2-2i)(-1-2i)$

$=2+4i+2i+4i^2=2+6i-4$

$=-2+6i$

9. The Smallest Possitive integer n for which $\left(\frac{1+i}{1-i}\right)^n=1$ is

$\left(\frac{1+i}{1-i}\right)^n=\left(\frac{1+i}{1-i}\times\frac{1+i}{1-i}\right)^n=\left(\frac{1+i^2+2i}{1-i^2}\right)^n=\left(\frac{1-1+2i}{1+1}\right)^n=\left(\frac{2i}{2}\right)^n=i^n\Rightarrow i^n=1$

Smallest Possitive integer of n is 4

10. $\left(\frac{1+i}{\sqrt{2}}\right)^{8n}+\left(\frac{1-i}{\sqrt{2}}\right)^{8n}$

$\left[\left(\frac{1+i}{\sqrt{2}}\right)^2\right]^{4n}+\left[\left(\frac{1-i}{\sqrt{2}}\right)^2\right]^{4n}$

$=\left(\frac{1+i^2+2i}{2}\right)^{4n}+\left(\frac{1+i^2-2i}{2}\right)^{4n}=\left(\frac{1-1+2i}{2}\right)^{4n}+\left(\frac{1-1-2i}{2}\right)^{4n}=\left(\frac{2i}{2}\right)^{4n}+\left(\frac{-2i}{2}\right)^{4n}$

$=i^{4n}+(-i)^{4n}=i^{4n}+i^{4n}=1+1=2$

11. Express $\left(\frac{2+3i}{2-3i}\right)-\left(\frac{2-3i}{2+3i}\right)$ in the form of $a+ib$

$\left(\frac{2+3i}{2-3i}\right)-\left(\frac{2-3i}{2+3i}\right)=\frac{(2+3i)^2-(2-3i)^2}{(2-3i)(2+3i)}=\frac{4(2)(3i)}{2^2-(3i)^2}=\frac{24i}{4+9}=\frac{24}{13}i$

12. $\sum_{n=0}^{\infty}\left(\frac{2i}{3}\right)^n\Rightarrow\sum_{n=0}^{\infty}\left(\frac{2i}{3}\right)^n=1+\frac{2i}{3}+\left(\frac{2i}{3}\right)^2+\left(\frac{2i}{3}\right)^3+\cdots\ldots$ are in G.P

Common ratio, r = $\frac{2i}{3}$

Sum to Infinite Terms,$S_\infty=\frac{a}{1-r}$

$\sum_{n=0}^{\infty}\left(\frac{2i}{3}\right)^n=\frac{1}{1-\frac{2i}{3}}=\frac{1}{\frac{3-2i}{3}}=\frac{3}{3-2i}\times\frac{3+2i}{3+2i}=\frac{9+6i}{9+4}=\frac{9+6i}{13}$

13. $\sum_{n=1}^{13}(i^n+i^{n+1})=$

$\sum_{n=1}^{13}(i^n+i^{n+1})=\sum_{n=1}^{13}i^n(1+i)=(1+i)\sum_{n=1}^{13}i^n$

$=(1+i)[i+i^2+i^3+i^4+\cdots\ldots i^{13}]=(1+i)\times i=i+i^2=i-1$

14. If x = a + ib is a Complex Number such that $x^2=3+4i$ and $x^3=2+11i$ where $i=\sqrt{-1}$ then (a + b) =

$x^2=3+4i$; $x^3=2+11i$

$\frac{x^3}{x^2}=\frac{2+11i}{3+4i}\times\frac{3-4i}{3-4i}$

$x = \frac{6 - 8i + 33i + 44}{9 + 16} = \frac{50 + 25i}{25}$

$= \frac{25(2 + i)}{25}$

$x = 2 + i = a + ib$

a = 2, b = 1

$a + b = 2 + 1 = 3$

15. If $3 + ix^2y$ and $x^2 + y + 4i$ are Conjugate Complex Numbers then $x^2 + y$ The Conjugate of $3 + ix^2y = 3 - ix^2y$

$3 - ix^2y = x^2 + y + 4i$

Compare Real & Imaginary Parts

$x^2 + y = 3 \; -x^2y = 4; x^2 + y = 3$

16. The Value of $\frac{i^5+i^6+i^7+i^8+i^9}{1+i}$ is equal to

$\frac{i^5 + i^6 + i^7 + i^8 + i^9}{1 + i} = \frac{i^5[1 + i + i^2 + i^3 + i^4]}{1 + i}$

$= \frac{i[1 + i - 1 - i + 1]}{1 + i} = \frac{i}{1 + i} \times \frac{1 - i}{1 - i}$

$= \frac{i + 1}{1 - i^2} = \frac{1}{2}(1 + i)$

17. If n is an Odd integer, i = $\sqrt{-1}$ then $(1 + i)^{6n} + (1 - i)^{6n}$ is equal to

$(1 + i)^{6n} + (1 - i)^{6n} = [(1 + i)^2]^{3n} + [(1 - i)^2]^{3n}$

$= (1 - 1 + 2i)^{3n} + [(1 - 1 - 2i)]^{3n} = (2i)^{3n} + (-2i)^{3n}$

n is add integer ; $= (2i)^{3n} - (2i)^{3n} = 0$

18. If $z(1 + a) = b + ic, a^2 + b^2 + c^2 = 1$ then $\frac{1+iz}{1-iz} =$

$a^2 + b^2 + c^2 = 1 \Rightarrow a^2 + b^2 = 1 - c^2$

$(a + ib)(a - ib) = (1 + c)(1 - c)$

$a + ib = \frac{(1 + c)(1 - c)}{a - ib}$

$z(1 + a) = b + ic \Rightarrow z = \frac{b + ic}{1 + a}$

$iz = \frac{i(b + ic)}{1 + a} = \frac{-c + ib}{1 + a}$

$$\frac{1+iz}{1-iz}=\frac{1+\left(\frac{-c+ib}{1+a}\right)}{1-\left(\frac{-c+ib}{1+a}\right)}$$
$$=\frac{1+a-c+ib}{1+a+c-ib}=\frac{(1-c)+a+ib}{1+c+(a-ib)}$$
$$=\frac{(1-c)+(a+ib)}{1+c+\frac{(1+c)(1-c)}{a+ib}}=\frac{(1-c)+(a+ib)}{\frac{(1+c)(a+ib)+(1+c)(1-c)}{a+ib}}$$
$$=\frac{(a+ib)[1-c+a+ib]}{(1+c)[a+ib+1-c]}=\frac{a+ib}{1+c}$$

19. $x=2+\sqrt{-3}\Rightarrow x=2+\sqrt{3}i$

$$x^2=(2+\sqrt{3}i)^2=4-3+4\sqrt{3}i=1+4\sqrt{3}i$$
$$x^4=(1+4\sqrt{3}i)^2=1-48+8\sqrt{3}i=-47+8\sqrt{3}i$$
$$x^4-4x^2+8x+35=-47+8\sqrt{3}i-4(1+4\sqrt{3}i)+8(2+\sqrt{3}i)+35$$
$$=-47+8\sqrt{3}i-4-16\sqrt{3}i+16+8\sqrt{3}i+35=-51+51=0$$

20. $x-y=7i,\ x^3-y^3=198$

$$(x-y)^3=(7i)^3$$
$$x^3-y^3-3xy(x-y)=-343i$$

$198-3xy(7i)=-343i\Rightarrow 198+343i=21xyi\Rightarrow xy=\frac{198+343i}{21i}\times\frac{i}{i}$

$xy=\frac{-343+198i}{-21}\Rightarrow xy=\frac{343-198i}{21}=\frac{343}{21}-\frac{198}{21}i=a+ib$

$a=\frac{343}{21},b=\frac{-198}{21}\Rightarrow\frac{a+b}{3}=\frac{\frac{343}{21}-\frac{198}{21}}{3}=\frac{145}{21\times 3}=\frac{145}{63}$

Level – III

21) A,B,C 22) B 23) A 24) C 25) a-r,b-p,c-s,d-q 26) A 27) A,D

28) B 29) D 30) C 31) A 32)a-s,b-r,c-q,d-p 33) A,B 34) B

35) A 36) C 37) a-s,b-r,c-p,d-b

21. sum of two complex numbers may be a pure real or pure imaginary or complex number.

22. $\frac{i^{365}+i^{246}+i^{95}+i^{35}}{i^{355}+i^{236}+i^{85}+i^{25}}-1$

$$\frac{i^{10}(i^{355}+i^{236}+i^{85}+i^{25})}{i^{355}+i^{236}+i^{85}+i^{25}}-1=i^{10}-1=i^{4\times 2+2}-1=i^2-1=-1-1=-2$$

23. $i^{19}=i^{4\times 4+3}=i^3=-1$

$$\frac{1}{i^{25}} = \frac{1}{i^{4\times6+1}} = \frac{1}{i} \times \frac{i}{i} = \frac{i}{-1} = -i \Rightarrow \left(i^{19} + \left(\frac{1}{i}\right)^{25}\right)^2 = (-i-i)^2 = (-2i)^2 = 4i^2 = -4$$

24. $i^{17} = i^{4\times4+1} = i^1 = i$

$$\frac{1}{i^{34}} = \frac{1}{i^{4\times8+2}} = \frac{1}{i^2} = -1 \Rightarrow \left[i^{17} - \left(\frac{1}{i}\right)^{34}\right]^2 = (i+1)^2 = i^2 + 1 + 2i = -1 + 1 + 2i = 2i$$

25. a) $= \frac{i+1+i-1}{(i-1)(i+1)} = \frac{2i}{i^2-1} = \frac{2i}{-2} = -i$

b) $i^n + i^{n+1} + i^{n+2} + i^{n+3} + 4 = 0 + 4 = 4$

c) $= \frac{i^2(i+i^4+i^7)}{i^2(1+i^2-i^4)} = \frac{i+1+i^{4\times1+3}}{1-1-1} = \frac{i+1+i^3}{-1} = \frac{i+1-i}{-1} = -1$

d) $= i - 1 - i + 1 = 0$

26. Statement - 1 : Conjugate of $-12 + 13i$ is $-12 - 13i$

Statement - 2: The conjugate of $a + ib$ is $a - ib$

27.

$a + b = 5,$ $\quad -(3a + 4b) = 2$
$a = 5 - b$ $\quad -3a - 4b = 2$
$-3(5 - b) - 4b = 2$
$-15 + 3b - 4b = 2$
$-15 - b = 2 \Rightarrow b = -17$
$\therefore a = 5 - (-17) = 22$

28. The additive inverse of $-3 + 4i$ is $3 - 4i$

29. The additive inverse of $(1 + 2i)(3 - 4i)$ is

$(1 + 2i)(3 - 4i) = 3 - 4i + 6i + 8 = 11 + 2i$

additive inverse of $11 + 2i$ is $-11 - 2i$

30. $(a - ib)^2 = x + iy$

$a^2 + (ib)^2 - 2abi = x + iy$

$a^2 - b^2 - 2abi = x + iy$

compare real and imaginary parts

$x = a^2 - b^2,\ y = -2ab$

$x - y = a^2 - b^2 + 2ab + 2b^2 - 2b^2$

$= a^2 + b^2 + 2ab - (\sqrt{2}b)^2 = (a+b)^2 - (\sqrt{2}b)^2 = (a + b + \sqrt{2}b)(a + b - \sqrt{2}b)$

31. Statement - 1 : The value of $\sqrt{-25} + 3\sqrt{-4} + 2\sqrt{-9}$

$\sqrt{-25} + 3\sqrt{-4} + 2\sqrt{-9} = 5i + 3 \times 2i + 2 \times 3i$

Statement - 2 : The value of $(-\sqrt{-1})^{4n+3}, n \in \mathrm{N}$

$(-\sqrt{-1})^{4n+3} = (-1)^{4n+3} \times i^{4n+3} = -1 \times i^3 = -1\mathrm{x} - i = i$

32. a) $(3y - 2) + i(7 - 2x) = 0 + i.0 \Rightarrow 7 - 2x = 0 \Rightarrow 7 = 2x \Rightarrow x = \frac{7}{2}$

b) $(3y - 2) + i(7 - 2x) = 0 + i.0 \Rightarrow 3y - 2 = 0 \Rightarrow 3y = 2 \Rightarrow y = \frac{2}{3}$

c) $4x = 3 \Rightarrow x = \frac{3}{4}$

d) $3x - y = -6$

$3x\frac{3}{4} - y = -6 \Rightarrow \frac{9}{4} + 6 = y$

$\frac{9 + 24}{4} = y$

$\Rightarrow y = \frac{33}{4}$

33. conceptual
34. conceptual
35. conceptual
36. conceptual
37. conceptual

Worksheet – 2

Level – I

1)C 2)A 3)B 4)B 5)C 6)B 7)B 8)A 9)B 10)A

1. The modulus of (1,2) is

   Ans: Modulus of (x, y) is $\sqrt{x^2 + y^2}$

   Modulus of (1,2) is $\sqrt{1^2 + 2^2} = \sqrt{5}$

2. Modulus of $3 + 4! = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

3. Principal vlaue of argument $-\sqrt{3} + i$ is …

   Principal argument $= \pi - \tan^{-1}\left(\frac{y}{|x|}\right)$

   $= \pi - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

4. The amplitude of $1 + \cos\,\theta + i\sin\,\theta$

   $1 + \cos\,\theta + i\sin\,\theta = 1 + 2\cos^2\frac{\theta}{2} - 1 + i2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2} = 2\cos\frac{\theta}{2}\left[\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right]$

   the amplitude of $1 + \cos\,\theta + i\sin\,\theta$ is $\frac{\theta}{2}$

5. $\arg(z_1) + \arg(z_2) =$

   $\arg(\bar{z_1}) = \frac{\pi}{5} \Rightarrow \arg(z_1) = \frac{-\pi}{5}$

   $\arg(z_2) = \frac{\pi}{3}$

   $\arg(z_1) + \arg(z_2) = \frac{-\pi}{5} + \frac{\pi}{3} = \frac{2\pi}{15}$

6. $\arg(z_1) = \frac{\pi}{3}, \arg(\bar{z_1}) = \frac{\pi}{4} \Rightarrow \arg(z_1, z_2) = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$

7. Polar form of $1 + \sqrt{3}i$

$$z = r\text{cis}\ \theta$$
$$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$$
$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \tan^{-1}\left(\frac{\pi}{3}\right) \Rightarrow \theta = \frac{\pi}{3} \Rightarrow z = 2\text{cis}\ \frac{\pi}{3}$$

8. mod- amplitude form of $-\sqrt{3} + i \Rightarrow -\sqrt{3} + i = 2\left[\frac{-\sqrt{3}}{2} + \frac{1}{2}i\right] = 2\text{cis}\ \frac{5\pi}{6}$

9. The value of $\log\ i = \log_e\ i = \log_e\ \text{cis}\ \frac{\pi}{2} = \log_e\ e^{i\pi/2} = i\pi/2$

10. $\left|\frac{2+5i}{3-2i}\right| = \frac{|2+5i|}{|3-2i|} = \sqrt{\frac{2^2+5^2}{3^2+(-2)^2}} = \sqrt{\frac{4+25}{9+4}} = \sqrt{\frac{29}{13}}$

Level – II

1)A	2)A	3)D	4)B	5)A	6)B	7)C	8)B
9)B	10)A	11)C	12)A	13)A	14)C	15)C	16)9
17)A	18)B	19)D	20)A	21)B	22)A	23)A	

1. $|z| = 1$

Multiplicative inverse of z is $\frac{1}{z} \Rightarrow \frac{1}{z} \times \frac{\bar{z}}{\bar{z}} = \frac{\bar{z}}{z \cdot \bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{\bar{z}}{1^2} = \bar{z}$

2. $\arg(\bar{z}) = \frac{2\pi}{3}, \arg(z) = \frac{\pi}{6}$

$$\arg(z) = \frac{-2\pi}{3} \Rightarrow \arg\left(\frac{z_1}{z_2}\right) = \arg_1 - \arg_2 = \frac{-2\pi}{3} - \frac{\pi}{6} = \frac{-5\pi}{6}$$

3. $z = 1 + \sqrt{3}i \Rightarrow |\arg\ z| + |\arg\ \bar{z}| = \bar{z} = 1 - \sqrt{3}i$

$$\text{Arg}(z) = \text{Arg}(1 - i\sqrt{3}) = \frac{\pi}{3} \Rightarrow \text{Arg}(\bar{z}) = \text{Arg}(1 - i\sqrt{3}) = \frac{-\pi}{3}$$
$$\therefore |\text{Arg}\ z| + |\text{Arg}(\bar{z})| = \frac{\pi}{3} + \left|\frac{-\pi}{3}\right| = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

4. The mod amplitude form of $-\sqrt{3} - i$ is

$$-\sqrt{3} - i = 2\left[\frac{-\sqrt{3}}{2} - \frac{1}{2}i\right] = 2\text{cis}\ \theta \Rightarrow \theta = -\pi + \tan^{-1}\left(\left|\frac{y}{x}\right|\right) = 1 - \pi + \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\pi + \frac{\pi}{6}$$
$$= \frac{-5\pi}{6}$$

Mod- amplitude form $= 2\cos\left[\frac{-5\pi}{6}\right]$

5. $\frac{1-\cos\ \theta + i\sin\ \theta}{1+\cos\ \theta - i\sin\ \theta} = \frac{2\sin^2\frac{\theta}{2} + i\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2} - i2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}}$

$$= \frac{2\sin\frac{\theta}{2}\left[\sin\frac{\theta}{2} + i\cos\frac{\theta}{2}\right]}{2\cos\frac{\theta}{2} - \left[\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\right]} = \frac{i\sin\frac{\theta}{2}\left[\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\right]}{\cos\frac{\theta}{2}\left[\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\right]} = i\tan\frac{\theta}{2}$$

6. $|z_1, z_2 \ldots \ldots z_n| = |z_1||z_2| \ldots |z_n|$

$|(a_1 + ib_1)(a_2 + b_2) \ldots (a_n + ib_n)| = |A + iB|$

$\sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2} + \cdots \ldots \cdot \sqrt{a_n^2 + b_n^2} = \sqrt{A^2 + B^2}$

squaring on both sides

$(a_1^2 + b_1^2)(a_2^2 + b_2^2) \ldots \ldots (a_n^2 + b_n^2) = A^2 + B^2$

7. If $x + iy = \sqrt{\frac{3+4i}{5+12i}}$

$$|x + y| = \sqrt{\left|\frac{2 + 4i}{5 + 12i}\right|} \Rightarrow \sqrt{x^2 + y^2} = \sqrt{\sqrt{\frac{3^2 + 4^2}{5^2 + 12^2}}} \Rightarrow \sqrt{x^2 + y^2} = \sqrt{\sqrt{\frac{25}{169}}}$$

$$x^2 + y^2 = \frac{5}{13} \Rightarrow (x^2 + y^2)^2 = \frac{25}{169} \Rightarrow 169(x^2 + y^2)^2 = 25$$

8. $(1 + i)^{100} = 2^{49}(x + iy)$ then $x^2 + y^2$

$|(1 + i)^{100}| = |2^{49}(x + iy)|$

$\Rightarrow \sqrt{2}^{100} = 2^{49}\sqrt{x^2 + y^2} \Rightarrow 2^{50} = 2^{49}\sqrt{x^2 + y^2} \Rightarrow \sqrt{x^2 + y^2} = 2 \Rightarrow x^2 + y^2 = 4$

9. $\left|\frac{\alpha+i\beta}{\beta+i\alpha}\right| = \sqrt{\frac{\alpha^2+\beta^2}{\beta^2+\alpha^2}} = 1$

10. $z = 5 + 3\sqrt{2}i \Rightarrow z\bar{z} = |z|^2 = 5^2 + (3\sqrt{2})^2 = 25 + 9 \times 2 = 25 + 18 = 43$

11. The amplitude $\frac{1+\sqrt{3}i}{\sqrt{3}+i}$

$$\frac{1 + \sqrt{3}i}{\sqrt{3} + i} = \frac{1 + \sqrt{3}i}{\sqrt{3} + i} \times \frac{\sqrt{3} - i}{\sqrt{3} - i} = \frac{2\sqrt{2} + 2i}{3 + 1} = \frac{\sqrt{3}}{2} + \frac{i}{2} = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6}$$

12. Polar form of $\left(\frac{1+3i}{1-2i}\right)$

$$\Rightarrow \frac{1 + 3i}{1 - 2i} = \frac{(1 + 3i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{1 + 2i + 3i + 6i^2}{1 + 4} = \frac{1 + 5i - 6}{5} = \frac{-5 + 5i}{5} = -1 + i$$

Polar form of $z = rcis\theta$

$r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

$$\arg(x) = \pi - \tan^{-1}\left(\frac{b}{|a|}\right) = \pi - \tan^{-1}\left(\frac{1}{1}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Polar form of $\left(\frac{1+3i}{1-2i}\right) = \sqrt{2}\text{cis}\,\frac{3\pi}{4} = \sqrt{2}\left[\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right]$

13. $\left|\frac{z_1-3z_2}{3-z_1\bar{z}_2}\right| = 1$

$n > 1^2 = |3 - z_1z_2|^2$

$|z_1 - 3z_2| = |3 - z_1\bar{z}_2|$

equating on both sides

$|z_1 - 3z_2|^2 = (3 - z_1\bar{z}_2)^2$

$\Rightarrow |z_1|^2 + |3z_2|^2 - 2\text{Re}\,(z_1\bar{z}_2) = |z|^2 + |z_1\bar{z}_2|^2 - 2\text{Re}\,|z_1 \cdot \bar{z}_2|$

$\Rightarrow |z_1|^2 + |3z_2|^2 = 9|z_1|^2 + |\bar{z}_2|^2 \Rightarrow |z_1|^2 + 9|z_2|^2 = 9 + |z_1|^2 + |\bar{z}_2|^2$

$\Rightarrow |z_1|^2 + 9|z_2|^2 - 9 - |z_1|^2 + |\bar{z}_2|^2 = 0 \Rightarrow |z_1|^2(1 - |z_2|^2) - 9(|1 - |\bar{z}_2|^2) = 0$

$(|z_1|^2 - 9)(1 - |z_2|^2) = 0$

$|z_1|^2 = 9 = 0$ of $1 - |z_2|^2 = 0$

since $|z_1| \neq 3$ then $|z_2|^2 = 1$

$|z_2| = 1$

14. $\bar{z}_1 = z_2$ and $\bar{z}_3 = z_4 \Rightarrow z_1 = \bar{z}_2$ and $z_3 = \bar{z}_4$

$\arg\left(\frac{z_1}{z_3}\right) + \arg\left(\frac{z_2}{z_4}\right) = \arg\left(\frac{z_1 z_2}{z_4 z_3}\right)$

$= \arg\left(\frac{z_1\bar{z}_1}{z_3\bar{z}_3}\right) = \arg\left(\frac{|z_1|^2}{|z_3|^2}\right) = \arg\left(\left|\frac{z_1}{z_3}\right|^2 + 0i\right) \arg(2) = \tan^{-1}\left(\frac{0}{\frac{z_1}{z_3}}\right) = 0$

15. If $(1 + \text{i})(1 + 2\text{i})(1 + 3\text{i}) \ldots\ldots + (1 + \text{ni}) = \text{x} + \text{iy}$ then

$2 \times 5 \times 10 \ldots (1 + \text{n}^2) \ldots\ldots$

$|z_1 \cdot z_2 \ldots\ldots z_n|^2 = |z_1|^2|z_2|^2 \cdot |z_3|^2 \ldots\ldots |z_n|^2$

$\Rightarrow 2 \times 5 \times 1 \ldots\ldots (1 + n^2) = x^2 + y^2$

16. If $(\sqrt{8} + i)^{50} = 3^{49}(a + \lambda b)$ then the value of $\text{a}^2 + \text{b}^2$

$\left|(\sqrt{8} + i)^{50}\right| = |3^{49}(a + ib)|$

$\Rightarrow \left(\sqrt{(\sqrt{8})^2 + b^2}\right)^{50} = 3^{49}\sqrt{a^2 + b^2} \Rightarrow (\sqrt{8 + 1})^{50} = 3^{49}\sqrt{a^2 + b^2}$

$\Rightarrow 3^{50} = 3^{49}\sqrt{a^2 + b^2} \Rightarrow 3 = \sqrt{a^2 + b^2} \Rightarrow a^2 + b^2 = 9$

17. Mod-amplitude form of $1 + \cos\,\theta - i\sin\,\theta$

$1 + \cos\,\theta - i\sin\,\theta = 2\cos^2\frac{\theta}{2} - i2\sin\frac{\theta}{2} \cdot \cos\frac{\theta}{2}$

$= 2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\right) = 2\cos\frac{\theta}{2} \cdot \cos\left(-\frac{\theta}{2}\right)$

18. $x = \cos\,\alpha + i\sin\,\alpha, y = \cos\,\beta + i\sin\,\beta$ then $\frac{x}{y} - \frac{y}{x}$

$\frac{x}{y} - \frac{y}{x} = \frac{\cos\,\alpha + i\sin\,\alpha}{\cos\,\beta + i\sin\,\beta} - \frac{\cos\,\beta + i\sin\,\beta}{\cos\,\alpha + i\sin\,\alpha}$

$\cos(\alpha - \beta) + i\sin(\alpha - \beta) - \cos(\alpha + \beta) + i\sin(\alpha - \beta) \Rightarrow 2i\sin(\alpha - \beta)$

19. if $|z_1 + z_2| = |z_1 - z_2|$ then $A\text{rg}\,z_1 - A\text{rg}\,z_2$

Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$

$|z_1 + z_2| = |z_2 + z_1|$

$|(x_1 + x_2) + i(\text{y}_1 + \text{y}_2)| = |(x_1 - x_2) + i(\text{y}_1 - \text{y}_2)|$

$(x_1 + x_2)^2 + (y_1 + y_2)^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$

$x_1^2 + x_2^2 + 2x_1x_2 + y_1^2 + y_2^2 + 2y_1y_2 = x_1^2 + x_2^2 - 2x_1x_2 + y_1^2 + y_2^2 - 2y_1y_2 = 4x_1x_2 + 4y_1y_2 = 0$

$x_1x_2 + y_1y_2 = 0 \Rightarrow x_1x_2 = -y_1y_2$

$\left(\frac{y_1}{x_1}\right)\left(\frac{y_2}{x_2}\right) = -1$

$\tan\,\theta_1 \cdot \tan\,\theta_2 = -1 \Rightarrow \tan\,\theta_1 = -\cot\,\theta_2$

$\tan\,\theta_1 = \left(\tan\,\frac{\pi}{2} + \theta_2\right) \Rightarrow \theta_1 = \left(\frac{\pi}{2} + \theta_2\right)$

$\theta_1 - \theta_2 = \frac{\pi}{2}$

$\text{Arg}\,(z_1) - \text{Arg}\,(z_2) = \frac{\pi}{2}$

20. $1 + i\tan\,\theta = r(\cos\,\theta + i\sin\,\theta) \Rightarrow r = \sqrt{1 + \tan^2\,\theta} = \sqrt{\sec^2\,\theta} = \sec\,\theta$ mod-amplitude form of $1 + i\tan\,\theta = \sec\,\theta.\,\text{cis}\,\theta$

21. If $|z^2 - 3| = 3|z|$ then the maximum value of $|z|$ is $|z^2 - 3| = 3|z|$

$|z_1 - z_2| \geq ||z_1| - |z_2||\ \therefore |z_1^2 - 3| \geq ||z_1^2| - |3||$

$|z_1^2 - 3| \leq |z^2 - 3|$

$|z_1^2 - 3| \leq 3|z|$

$= |a^2 - 3| \leq 3a, a \geq 0 = -3a \leq a^2 - 3 \leq 3a = a^2 - 3a - 3 \leq 0$ and $a^2 + 3a - 3 \geq 0$

$= a \in \left(0, \frac{3+\sqrt{21}}{2}\right]$ and $a \in \left[\frac{3-\sqrt{21}}{2}, \alpha\right) = a \in \left[\frac{-3+\sqrt{21}}{2}, \frac{3+\sqrt{21}}{2}\right]$

22. If $\frac{z-1}{z+1}$ is purely imaginary then $|z| =$

Let $z = x + iy$

$$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1} = \frac{(x-1)+iy}{(x+1)+iy} * \frac{(x+1)-iy}{(x+1)-iy} = \frac{(x-1)(x+1) - i^2y^2 + iy(x+1-(x-1))}{(x+1)^2 + i^2y^2}$$

$$= \frac{x^2 - 1 + y^2 + i2y}{(x+1)^2 + y^2} = \frac{x^2 - 1 + y^2}{(x+1)^2 + y^2} + i\frac{2y}{(x+1)^2 + y^2}$$

$\frac{z-1}{z+1}$ is purely imaginary, $RE\left(\frac{z-1}{z+1}\right) = 0$

$$= \frac{x^2 - 1 + y^2}{(x+1)^2 + y^2} = 0 = x^2 + y^2 = 1 \Rightarrow \sqrt{x^2 + y^2} = 1 \Rightarrow |z| = 1$$

23. α, β are different complex number $|\beta| = 1$ then $\left|\frac{\alpha-\beta}{\alpha+\beta}\right| =$

$|\beta| = 1, Z\bar{z} = |z|^2$

$\beta \cdot \bar{\beta} = |\beta|^2$

$$\beta.\bar{\beta} = 1 = \bar{\beta} = \frac{1}{\beta} \Rightarrow \left|\frac{\alpha-\beta}{\alpha+\beta}\right| = \left|\frac{\alpha-\beta}{1-\frac{\alpha}{\beta}}\right| = \left|\frac{\alpha-\beta}{\frac{\beta-\alpha}{\beta}}\right| = \left|\frac{(\alpha-\beta)\beta}{(\beta-\alpha)}\right| = |\beta| = 1$$

Level – III

24)A,B,C,D 25)B 26)B 27)C 28)a-q,b-r,c-s,d-p 29) B,C

30)B 31)C 32)A 33)a-r,s;b-p,q;c-q,d-s 34)A,B

35)C 36)B 37)C 38)a-s,b-r,c-q,d-p

24. Modules of $(-5,12)$

$|z| = |\bar{z}| = \sqrt{x^2 + y^2}$

$|-5 + 12i| = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$

25. The principal value of argument of $2 + 2\sqrt{3}i$

$z = 2 + 2\sqrt{3}, z \in q_1$

$\text{Arg (z)} = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\frac{2\sqrt{3}}{2} = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$

26. If $z = -5i$ then $\text{Arg}(z) =$

$\text{Arg}(z) = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{(-5)}{0}\right) = -\frac{\pi}{2} \Rightarrow \text{Arg}(z) = -\frac{\pi}{2}$

27. If $z = 10$ then $\text{Arg}(z) = z = 10 + 0i$

$\text{Arg (z)} = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{0}{10}\right) = \tan^{-1}(0) = 0$

28. a. The modules of $(-5,12)$ is $\sqrt{15^2 + 12^2} = \sqrt{169} = 13(2)$

b. Modules of $15 + 8\text{i} = \sqrt{15^2 + 8^2} = \sqrt{225 + 64} = \sqrt{289} = 17(3)$

c. Modules of $(3 - 4i)(4 + 5i)$

$|z_1 \cdot z_2| = |z_1| \cdot |z_2| = |3 - 4i||4 + 5i| = \sqrt{3^2 + (-4)^2} \cdot \sqrt{4^2 + 5^2} = \sqrt{25} * \sqrt{41} = 5\sqrt{41}(4)$

d. Modules of $\frac{5+4i}{1+i} = \left|\frac{z_1}{z_2}\right| = \frac{\sqrt{5^2+4^2}}{\sqrt{1^2+1^2}} = \frac{\sqrt{25+16}}{\sqrt{2}} = \sqrt{\frac{41}{2}}(1)$

29. If amp $|z_1 \cdot z_2| = 0$ and $|z_1| = |z_2| = 1$ then

amp $|z_1|$ + amp $|z_2| = 0$

amp $|z_1| = -amp|z_2| =$ amp $|\bar{z}_2|$

$|z_1| = |z_2| = 1|z_1| = |\bar{z_2}| = z_1 = \bar{z_2}$

$z_1 z_2 = z_2\bar{z_2} = |z_2|^2 = 1$

30. The modules of $3 - 4i$ is $|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5$

31. The modules of $6 + 5i$ is $|6 + 5i| = \sqrt{6^2 + 5^2} = \sqrt{36 + 25} = \sqrt{61}$

32. The value of $\left|\frac{(3-i\sqrt{2})^2}{1+2i}\right|$ is equal to

$$\left|\frac{(3 - i\sqrt{2})^2}{1 + 2i}\right| = \left|\frac{(9 - 2 - 6\sqrt{2}i}{1 + 2i}\right| = \left|\frac{7 - 6\sqrt{2}i}{1 + 2i}\right| = \frac{\sqrt{7^2 + 36^*2}}{\sqrt{1^2 + 2^2}} = \frac{\sqrt{49 + 72}}{\sqrt{5}} = \frac{11}{\sqrt{5}}$$

33. a. if $ab > 0$ then either a, b both positive or both a, b are negative. Hence $z = a + ib$ lies in either first or third quadrant then argument of z is $\tan^{-1}\left|\frac{b}{a}\right|$ or $-\pi + \tan^{-1}\left|\frac{b}{a}\right|$

b. if $ab < 0$, then either a and b have opposite signs, then z is either second or fourth quadrant, then argument of z is $-\tan^{-1}\left|\frac{b}{a}\right|$ or $\pi - \tan^{-1}\left|\frac{b}{a}\right|$

c. if $a < 0, b > 0$ then z lies in second quadrant, then argument of z is $\pi - \tan^{-1}\left|\frac{b}{a}\right|$ d. if a < 0, b < 0 then z lies in third quadrant then argument of z is $-\pi - \tan^{-1}\left|\frac{b}{a}\right|$

36. Complex number $\frac{z_1}{z_2}$ is ⇒ From 2, $\frac{z_1}{z_2}$ is purely imaginary

37. one of the possible argument of complex number $i\frac{z_1}{z_2}$ is possible arguments are o and π

38. a. the value of $i^i \Rightarrow i^i = \left(\text{cis}\ \frac{\pi}{2}\right)^i$

b. the value of $\log^i \Rightarrow \log\ e^i = \log\ e^{cis\frac{\pi}{2}} = \frac{i\pi}{2}$

c. $\log(\log\ i) = \log(\log\ i) = \log\left(\log\ e^{\frac{i\pi}{2}}\right) = \log\ \frac{i\pi}{2} = \log\ i + \log\ \frac{\pi}{2} = \frac{i\pi}{2} + \log\ \frac{\pi}{2}$

d. If $z = i\log(2-\sqrt{3})$ then $\cos\ z = z = i\log(2-\sqrt{3})$

$\cos\ z = \frac{e^{iz} + e^{-iz}}{2} \Rightarrow iz = -\log(2-\sqrt{3}) = \log\left(\frac{1}{2-\sqrt{3}}\right) = \log(2+\sqrt{3})$

$e^{iz} = e^{\log(2+\sqrt{3})} = 2+\sqrt{3}$

$e^{-iz} = \frac{1}{2+\sqrt{3}} * \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{4-2} = \frac{2-\sqrt{3}}{1} = 2-\sqrt{3}$

$\cos\ z = \frac{2+\sqrt{3}+2-\sqrt{3}}{2} = \frac{4}{2} = 2$

Worksheet – 3

Level – I

1.	A	3.	C	5.	C	7.	A	9.	A
2.	C	4.	B	6.	A	8.	B		

1. ω is a cube root of unity then ω^6 is $\omega^6 = (\omega^3)^2 = 1^2 = 1$
2. The value of $\omega + \omega^2 + \omega^3$ is $\omega^3 = 1\ 1 + \omega + \omega^2 = 0$
3. The value of $(1+\omega^2)^2$ is

   $1 + \omega^2 = -\omega$

   $(1+\omega^2)^2 = (-\omega)^2 = \omega^2$
4. $1 + \frac{1}{\omega} + \frac{1}{\omega^2} \Rightarrow \frac{1}{\omega} = \omega^2, \frac{1}{\omega^2} = \omega \Rightarrow 1 + \frac{1}{\omega} + \frac{1}{\omega^2} = 1 + \omega^2 + \omega = 0$
5. $(1+\omega)(1+\omega^2) = 1 + \omega = \omega^2, 1 + \omega^2 = -\omega \Rightarrow (1+\omega)(1+\omega^2) = -\omega^2 \times -\omega = \omega^3 = 1$
6. $(\omega^2 + \omega - 1)^3 - (\omega^2 - \omega + 1)^3 = \omega + \omega^2 = -1, 1 + \omega^2 = -\omega$

   $(\omega^2 + \omega - 1)^3 - (\omega^2 - \omega + 1)^3 = (-1-1)^3 - (-\omega - \omega)^3$

   $= (-2)^3 - (-2\omega)^3 = -8 - (-8\omega^3) = -8 + 8 = 0$
7. The value of $(\cos\ 3\alpha + \ i\sin\ 3\alpha)^5(\cos\ 4\alpha - \ i\sin\ 4\alpha)^6 =$

$(\cos\ 3\alpha + i\sin\ 3\alpha)^5 = \cos\ 15\alpha + i\sin\ 15\alpha$

$(\cos\ 4\alpha - i\sin\ 4\alpha)^6 = \cos\ 24\alpha - i\sin\ 24\alpha = \dfrac{1}{\cos\ 24\alpha + i\sin\ 24\alpha}$

$(\cos\ 3\alpha + i\sin\ 3\alpha)^5(\cos\ 4\alpha - i\sin\ 4\alpha)^6 = \dfrac{\cos\ 15\alpha + i\sin\ 15\alpha}{\cos\ 24\alpha + i\sin\ 24\alpha}$

$= \cos\ (15\ \mathrm{d} - 24\alpha) + i\sin\ (15\ \mathrm{d} - 24\alpha)$

$= \cos\ (-9\alpha) + i\sin\ (-9\alpha) = \cos\ 9\alpha - i\sin\ 9\alpha$

8. $\dfrac{(\cos\ \theta+i\sin\ \theta)^5(\cos\ 3\theta-i\sin\ 3\theta)^6}{(\cos\ 2\theta+i\sin\ 2\theta)^3(\cos\ 4\theta-i\sin\ 4\theta)^5} =$

$= \dfrac{(\cos\ 5\theta + i\sin\ 5\theta)(\cos\ 18\theta - i\sin\ 18\theta)}{(\cos\ 6\theta + i\sin\ 6\theta)(\cos\ 20\theta - i\sin\ 20\theta)} = \dfrac{(\cos\ 5\theta + i\sin\ 5\theta)(\cos\ 20\theta + i\sin\ 20\theta)}{(\cos\ 6\theta + i\sin\ 6\theta)(\cos\ 18\theta + i\sin\ 18\theta)}$

$= \dfrac{\cos\ (5\theta + 20\theta) + i\sin\ (5\theta + 20\theta)}{\cos\ (6\theta + 18\theta) + i\sin\ (6\theta + 18\theta)}$

$= \dfrac{\cos\ 25\theta + i\sin\ 25\theta}{\cos\ 24\theta + i\sin\ 24\theta} = \cos\ (25\theta - 24\theta) + i\sin\ (25\theta - 24\theta) = \cos\ \theta + i\sin\ \theta = cis\theta$

9. $(\cos\ \theta + i\sin\ \theta)^4 + (\cos\ \theta - i\sin\ \theta)^4 \Rightarrow \cos\ 4\theta + i\sin\ 4\theta + \cos\ 4\theta - i\sin\ 4\theta = 2\cos\ 4\theta$

Level – II

1) A 2) C 3) A 4) A 5) B 6) D 7) D 8) A

9) A 10) D 11) C 12) C 13) B 14) B 15) D 16) D

17) B 18) A 19) A 20) A 21) B 22) B

1. The value of $(1 + \sqrt{\omega} + \omega)(1 - \sqrt{\omega} + \omega)\ (1 + \omega + \sqrt{\omega})(1 + \omega - \sqrt{\omega}) = (1 + \omega)^2 - (\sqrt{3})^2 = (-\omega^2)^2 - \omega = \omega^4 - \omega = \omega - \omega = 0$

2. $(1 + \omega^2)^3 - \omega = (-\omega)^3 - \omega = \omega^2$

3. $\left(\cos\ \frac{\pi}{3} + i\sin\ \frac{\pi}{3}\right)^6 = \cos\ \left(6 \times \frac{\pi}{3}\right) + i\sin\ \left(6 \times \frac{\pi}{3}\right) = \mathrm{co}\ 2\pi + i\sin\ 2\pi$

4. $\dfrac{(\cos\ \theta+i\sin\ \theta)^6(\cos\ 3\theta-i\sin\ 3\theta)^4}{(\cos\ 3\theta+i\sin\ 3\theta)^2(\cos\ 4\theta+i\sin\ 4\theta)^{-3}} = \dfrac{(\cos\ 6\theta+i\sin\ 6\theta)(\cos\ 12\theta-i\sin\ 12\theta)}{(\cos\ 6\theta+i\sin\ 6\theta)(\cos\ 12\theta-i\sin\ 12\theta)} = 1$

5. If ω is a complex cube root of unity then the value of $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$ ---2n terms

$(1 + \omega)(1 + \omega^2) = 1 + \omega^2 + \omega + \omega^3 = 1 + \omega + \omega^2 + 1 = 0 + 1 = 1$

$(1 + \omega^4)(1 + \omega^8) = (1 + \omega)(1 + \omega^2) = 1$

$(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$ ….2n terms $= 1$

6. If ω is a cube root of unity then $(1+\omega-\omega^2)(1-\omega+\omega^2)=(1+\omega-\omega^2)(1-\omega+\omega^2)=(1+\omega-\omega^2)(1+\omega^2-\omega)=(-\omega^2-\omega^2)(-\omega-\omega)=(-2\omega^2)(-2\omega)=4\omega^3=4$

7. If α,β are the complex cube root of unity then $\alpha^{100}+\beta^{100}+\frac{1}{\alpha^{100}\times\beta^{100}}=\alpha=\omega,\beta=\omega^2$

$$\omega^{100}+(\omega^2)^{100}+\frac{1}{\omega^{100}\times(\omega^2)^{100}}=\omega^{100}+\omega^{200}+\frac{1}{\omega^{300}}=\omega+\omega^2+1=0$$

8. If ω is a complex cube root of unity then $\sin\left\{(\omega^{10}+\omega^{23})\pi-\frac{\pi}{4}\right\}$

$$\sin\left\{(\omega^{10}+\omega^{23})\pi-\frac{\pi}{4}\right\}=\sin\left\{(\omega+\omega^2)\pi-\frac{\pi}{4}\right\}$$
$$=\sin\left(\pi-\frac{\pi}{4}\right)=-\sin\left(\pi+\frac{\pi}{4}\right)=-\left(\frac{-1}{\sqrt{2}}\right)=\left(\frac{1}{\sqrt{2}}\right)$$

9. If $x^6+1=0$ then $x=$

$x^6+1=0\Rightarrow x^6=-1$

$x=(-1)^{1/6}=\text{cis}\,(2k+1)\frac{\pi}{6}, k=0,1,2,3,4,5$

10. If $x^2+x+1=0$ then the value of $\left(x+\frac{1}{x}\right)^2+\left(x^2+\frac{1}{x^2}\right)^2+\left(x^{27}+\frac{1}{x^{27}}\right)^2$ is $\omega\&\omega^2$ are the roots of cube root of unity

$\left(\omega+\frac{1}{\omega}\right)^2+\left(\omega^2+\frac{1}{\omega^2}\right)^2+\cdots\ldots+\left(\omega^{27}+\frac{1}{\omega^{27}}\right)^2$

$=(\omega+\omega^2)^2+(\omega^2+\omega)+\cdots\ldots+(1+1)^2=(-1)^2+(-1)^2+\cdots..2^2=1+1+\cdots.+4$

$=9(4)+18=36+18=54$

11. The value of $\left(\frac{-1+i\sqrt{3}}{2}\right)^{3n}+\left(\frac{-1-i\sqrt{3}}{2}\right)^{3n}=$

$$\left(\frac{-1+i\sqrt{3}}{2}\right)^{3n}+\left(\frac{-1-i\sqrt{3}}{2}\right)^{3n}=\omega^{3n}+(\omega^2)^{3n}=\omega^{3n}+\omega^{6n}=1+1=2$$

12. If $x=a+b, y=a\omega+b\omega^2$ and $z=a\omega^2+b\omega$ where ω,ω^2 are the imaginary cube roots of 1 , then xyz =

$xyz=(a+b)(a\omega+b\omega^2)(a\omega^2+b\omega)$

$=(a^2\omega+ab\omega^2+ab\omega+b^2\omega^2)(a\omega^2+b\omega)$

$=(a^2\omega-ab+b^2\omega^2)(a\omega^2+b\omega)$

$=a^3\omega^3+a^2b\omega^2-a^2b\omega^2-ab^2\omega+ab^2\omega+b^3\omega^3$

$=a^3+b^3$

13. $\frac{(\cos\alpha+i\sin\alpha)^5}{(\sin\beta+i\cos\beta)^3}$

$$\frac{(\cos\alpha+i\sin\alpha)^5}{(\sin\beta+i\cos\beta)^3}=\frac{\cos 5d+i\sin 5\alpha}{\cos 3\beta-i\sin 3\beta}=(\cos 5\alpha+i\sin 5\alpha)(\cos 3\beta+i\sin 3\beta)$$

$= \cos(5\alpha + 3\beta) + i\sin(5\alpha + 3\beta) = -\sin(5\alpha + 3\beta) + i\cos(5\alpha + 3\beta)$

14. If α, β are the roots of $x^2 + x + 1 = 0$ then $\alpha^{28} + \beta^{28} =$

$\omega^{28} + (\omega^2)^{28} = \omega^{28} + \omega^{56} = \omega + \omega^2 = -1$

15. If $\left[\frac{1+\cos\theta+i\sin\theta}{\sin\theta+i(1+\cos\theta)}\right]^4 = \cos n\theta + i\sin n\theta$ then $n = \left[\frac{1+\cos\theta+i\sin\theta}{\sin\theta+i(1+\cos\theta)}\right]^4 = \cos n\theta + i\sin n\theta$

$\left[\frac{2\cos^2\frac{\theta}{2}+i\times 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}+i\times 2\cos^2\frac{\theta}{2}}\right]^4 = \cos n\theta + i\sin n\theta$

$$\left[\frac{2\cos\frac{\theta}{2}\left[\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right]}{2\cos\frac{\theta}{2}\left[\sin\frac{\theta}{2}+i\cos\frac{\theta}{2}\right]}\right]^4 = \cos n\theta + i\sin n\theta$$

$$\frac{\cos 2\theta + i\sin 2\theta}{\sin 2\theta + i\cos 2\theta} = \cos n\theta + i\sin n\theta$$

$$\frac{\cos 2\theta + i\sin 2\theta}{\cos 2\theta - i\sin 2\theta} = \cos n\theta + i\sin n\theta$$

$(\cos 2\theta + i\sin 2\theta)(\cos 2\theta + i\sin 2\theta) = \cos n\theta + i\sin n\theta$

$\cos 4\theta + i\sin 4\theta = \cos n\theta + i\sin n\theta$

$\therefore n = 4$

16. $(-1 + i\sqrt{3})^{30} + (-1 - i\sqrt{3})^{30} =$

$$2^{30}\left(\frac{-1+i\sqrt{3}}{2}\right)^{30} + 2^{30}\left(\frac{-1-i\sqrt{3}}{2}\right)^{30} = 2^{30}[\omega^{30} + \omega^{60}] = 2^{30}(1+1) = 2\times 2^{30} = 2^{31}$$

17. If n is a positie integer then $(1 + i)^n + (1 - i)^n =$

$$1 + i = \sqrt{2}\left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right) = \sqrt{2}\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right]$$

$$(1+i)^n = (\sqrt{2})^n\left[\cos\frac{n\pi}{4} + i\sin\frac{n\pi}{4}\right], (1-i)^n = (\sqrt{2})^n\left[\cos\frac{n\pi}{4} - i\sin\frac{n\pi}{4}\right]$$

$$(1+i)^n + (1-i)^n = (\sqrt{2})^n\left[\cos\frac{n\pi}{4} + i\sin\frac{n\pi}{4} + \cos\frac{n\pi}{4} - i\sin\frac{n\pi}{4}\right]$$

$$= 2^{\frac{n}{2}} \times 2\cos\frac{n\pi}{4} = 2^{\frac{n}{2}+1}\cos\frac{n\pi}{4} = 2^{\frac{n+2}{2}}\cos\frac{n\pi}{4}$$

18. If $z = \frac{\sqrt{3}+i}{2}$ then $(iz)^{59} = iz = i\left(\frac{\sqrt{3}+i}{2}\right) = \frac{-1+\sqrt{3}i}{2} = \omega$

$$(iz)^{59} = \omega^{59} = \omega^2 = \frac{-1-\sqrt{3}i}{2}$$

19. The value of $\sqrt{-1\sqrt{-1\sqrt{-1} + \cdots .. to\infty}}$

Let $x = \sqrt{-1-x} \Rightarrow x^2 = (\sqrt{-1-x})^2 = 1 - x \Rightarrow x^2 + x + 1 = 0$

$$x = x = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}i}{2} = \omega, \omega^2$$

20. If the cube roots of unity are $1, \omega, \omega^2$ then the roots of an equation $(x-1)^3 + 8 = 0$ are $(x-1)^3 + 8 = 0 \Rightarrow (x-1)^3 = -8$

$(x-1)^3 = (-2)^3$

$x - 1 = (-2)x\frac{1}{1^3} = -2, -2\omega, -2\omega^2$

$x = 1-2, 1-2\omega, 1-2\omega^2 = -1, -1-2\omega, -1-2\omega^2$

21. If α, β are the roots of the equation $x^2 - 2x + 4 = 0$ and then $\alpha^5 + \beta^5 = x^2 - 2x + 4 = 0$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(4)}}{2} = \frac{2 \pm \sqrt{4-16}}{2} = \frac{2 \pm 2\sqrt{3}i}{2} = \frac{2(1 \pm \sqrt{3}i)}{2}$$

$$= -2\left(\frac{-1 \pm \sqrt{3}i}{2}\right) = -2\omega, -2\omega^2$$

$\alpha^5 + \beta^5 = (-2\omega)^5 + (-2\omega^2)^5 = -32\omega^5 + (-32\omega^{10}) = -32\omega - 32\omega^2 = -32(\omega + \omega^2)$

$= -32(-1) = 32$

22. If ω is a complex cube root of unity and a, b, c are real numbers such that $\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = 2\omega^2$ and $\frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = 2\omega$ then $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}$

is $\omega = \frac{1}{\omega^2}, \omega^2 = \frac{1}{\omega} \& \frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = \frac{2}{\omega^2}$

ω and ω^2 are the roots of $x\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = \frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} = \frac{2}{x}$

$$\frac{(b+x)(c+x) + (a+x)(c+x) + (a+x)(b+x)}{(a+x)(b+x)(c+x)} = \frac{2}{x} \Rightarrow \frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = 2$$

Level – III

23) B,C 24) A 25) D 26) D 27) a-q;b-s; c-s; d-p

28) B,C 29) A 30) A 31) A 32) a-r,b-p,c-q,d-s 33) A,D 34) B

35) D 36) A 37) a-r,b-p,c-s,d-q

23. If $1, \omega, \omega^2$ are the cube roots of unity then $(2-\omega)(2-\omega^2)(2-\omega^{13})(2-\omega^{17})$

$(2-\omega)(2-\omega^2) = 4 - 2\omega^2 - 2\omega + \omega^3 = 5 - 2(\omega + \omega^2) = 5 - 2(-1) = 7$

$(2-\omega)(2-\omega^2)(2-\omega^{13})(2-\omega^{17}) = 7 \times (2-\omega)(2-\omega^2) = 7 \times 7 = 7^2 = 49$

24. $\omega^5 + \omega^4 + 1$ is equal to $1 + \omega + \omega^2 = 0$

25. If ω is a complex cube root of unity then $(1-\omega+\omega^2)^6(1-\omega^2+\omega)^6 =$

$(1+\omega^2-\omega)^6 + (1+\omega-\omega^2)^6 = (-\omega-\omega)^6 + (-\omega^2-\omega^2)^6$

$= (-2\omega)^6 + (-2\omega^2)^6 = 64\omega^6 + 64\omega^{12} = 64 + 64 = 128$

26. $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}=\frac{\omega(a+b\omega+c\omega^2)}{\omega(c+a\omega+b\omega^2)}=\frac{a\omega+b\omega^2+c}{\omega(c+a\omega+b\omega^2)}=\frac{1}{\omega}=\omega^2$

27. If $\cos\alpha+\cos\beta+\cos\gamma=0=\sin\alpha+\sin\beta+\sin\gamma$

$\cos 2\alpha+\cos 2\beta+\cos 2\gamma$

$(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta)(\cos\gamma+i\sin\gamma)$

$=(\cos\alpha+\cos\beta+\cos\gamma)+i(\sin\alpha+\sin\beta+\sin\gamma)=0+i.0$

$(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta)(\cos\gamma+i\sin\gamma)=0$

Let $x=\cos\alpha+i\sin\alpha=cisd, y=\cos\beta+i\sin\beta=\text{cis}\ \beta, z=\cos\gamma+i\sin\gamma=cis\gamma$

then $x+y+z=0$

$x^2+y^2+z^2=-2(xy+yz+zx)=-2xyz\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$

$=-2xyz[\cos\alpha-i\sin\alpha+\cos\beta-i\sin\beta+\cos\gamma-i\sin\gamma]$

$==-2xyz[(\cos\alpha+\cos\beta+\cos\gamma)-(\sin\alpha+\sin\beta+\sin\gamma)]=-2xyz(0-i.0)=0$

$x^2+y^2+z^2=0$

$(\cos\alpha+i\sin\alpha)^2+(\cos\beta+i\sin\beta)^2+(\cos\gamma+i\sin\gamma)^2=0$

$\cos 2\alpha+i\sin 2\alpha+\cos 2\beta+i\sin 2\beta+\cos 2\gamma+i\sin 2\gamma=0$

$(\cos 2\alpha+\cos 2\beta+\cos 2\gamma)+i(\sin 2\alpha+\sin 2\beta+\sin 2\gamma)=0$

$\cos 2\alpha+\cos 2\beta+\cos 2\gamma=0$

$\sin 2\alpha+\sin 2\beta+\sin 2\gamma=0$

$2\cos^2\alpha-1+2\cos^2\beta-1+2\cos^2\gamma=0$

$2(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)-3=0$

$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=\frac{3}{2}$

$1-\sin^2\alpha+1-\sin^2\beta+1-\sin^2\gamma=\frac{3}{2}$

$3-(\sin^2\alpha+\sin^2\beta+\sin^2\gamma)=\frac{3}{2}$

$\sin^2\alpha+\sin^2\beta+\sin^2\gamma=\frac{3}{2}$

28. Conjugate of ω is $\frac{1}{\omega}\&\omega^2$

29. $\frac{(\cos\theta+i\sin\theta)^5}{(\cos 2\theta-i\sin 2\theta)^6}=\frac{(\cos\theta+i\sin\theta)^5}{(\cos 2\theta-i\sin 2\theta)^6}=\frac{\cos 5\theta+i\sin 5\theta}{\cos 12\theta-i\sin 12\theta}$

$=(\cos 5\theta+i\sin 5\theta)(\cos 12\theta+i\sin 12\theta)=\cos 17\theta+i\sin 17\theta$

30. If $x=\cos\alpha+i\sin\alpha, y=\cos\beta+i\sin\beta$ then $x^my^n+\frac{1}{x^my^n}$

$x^m=(\cos\alpha+i\sin\alpha)^m=\cos m\alpha+i\sin m\alpha$

$y^n=(\cos\beta+i\sin\beta)^n=\cos n\beta+i\sin n\beta$

$x^my^n=(\cos m\alpha+i\sin m\alpha)(\cos n\beta+i\sin n\beta)$

$=\cos(m\alpha+n\beta)+i(\sin(m\alpha-n\beta)$

$$\frac{1}{x^m y^n} = \frac{1}{\cos(m\alpha + n\beta) + i\sin(m\alpha - n\beta)} = \cos(m\alpha + n\beta) - i\sin(m\alpha - n\beta)$$

$$x^m y^n + \frac{1}{x^m y^n} = 2\cos(m\alpha + n\beta)$$

31. $\left(\frac{\sin\frac{\pi}{8}+i\cos\frac{\pi}{8}}{\sin\frac{\pi}{8}-i\cos\frac{\pi}{8}}\right)^8 \Rightarrow \frac{\sin\pi+i\cos\pi}{\sin\pi-i\cos\pi} = \frac{0+i(-1)}{0-i(-1)} = \frac{-i}{i} = -1$

32. (a) $(a+b)(a\omega + b\omega^2)(a\omega^2 + b\omega)$

$$= (a+b)(a^2 + ab\omega^2 + ab\omega + b^2)$$
$$= (a+b)(a^2(\omega^2+\omega)ab + b^2) = (a+b)(a^2 - ab + b^2)$$
$$= a^3 + b^3$$

(b) $(a+b)^2 + (a\omega + b\omega^2)^2 + (a\omega^2 + b\omega)^2$

$$= (a^2 + 2ab + b^2) + (a^2\omega^2 + b^2\omega + 2ab) + (a^2\omega + b^2\omega^2 + 2ab)$$
$$= a^2(1 + \omega + \omega^2) + b^2(1 + \omega + \omega^2) + 3ab = 6ab$$

(c) $(a+b+1)(a+b\omega+\omega^2)(a+b\omega^2+\omega)$

$= (a+b+1)(a^2 + ab\omega^2 + a\omega + ab\omega + b^2 + b\omega^2 + a\omega^2 + b\omega + 1)$
$= (a+b+1)(a^2 + b^2 - ab - a - b + 1)$
$= a^3 + ab^2 - a^2b - a^2 - ab + a + a^2b + b^3 - ab^2 - ab - b^2 + b + a^2 + b^2 - ab - a - b + 1$
$= a^3 + b^3 + 1 - 3ab$

(d) $(a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$

$$= (a+b+c)(a^2 + ab\omega^2 + ac\omega + ab\omega + b^2 + bc\omega^2 + bc\omega + c^2)$$
$$= (a+b+c)(a^2 + b^2 + c^2 + ab(\omega+\omega^2) + bc(\omega+\omega^2) + ac(\omega+\omega^2)$$
$$= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$
$$= a^3 + b^3 + c^3 - 3abc$$

33. All the values of $(1+i)^{\frac{1}{2}}$ are

$$(1+i)^{\frac{1}{2}} = \left(\sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)\right)^{\frac{1}{2}} = 2^{\frac{1}{4}}\left(\text{cis}\ \frac{\pi}{4}\right)^{\frac{1}{2}} = 2^{\frac{1}{4}}\text{cis}\left(\frac{2k\pi + \frac{\pi}{4}}{2}\right), \text{k} = 0,1$$

$k = 0 \Rightarrow 2^{\frac{1}{4}}\text{cis}\ \frac{\pi}{8}$

$k = 1 \Rightarrow 2^{\frac{1}{4}}\text{cis}\ \frac{9\pi}{8}$

34. The value of $(-8i)^{\frac{1}{3}}$ are

$$(-8i)^{\frac{1}{3}} = z \Rightarrow -8i = z^3$$
$$z^3 + 8i = 0 \Rightarrow z^3 + (2i)^3 = 0$$
$$(z+2i)(z^2 + 2zi + (2i)^2) = 0$$

$(z + 2\text{i})(z^2 + 2zi - 4) = 0$

$z + 2\text{i} = 0$

$z = +2i$

$z^2 + 2i - 4 = 0$

$z = \dfrac{-2i \pm \sqrt{(-2i)^2 - 4(1)(-4)}}{2}$

$z = \dfrac{-2i \pm \sqrt{-4+16}}{2} = \dfrac{-2i \pm \sqrt{12}}{2} = \dfrac{-2i \pm 2\sqrt{3}}{2} = \pm\sqrt{3} - i$

35. All values of $(1 + i\sqrt{3})^{\frac{1}{4}}$ are

$(1 + i\sqrt{3})^{\frac{1}{4}} = \left(2\left(\dfrac{1}{2} + i\dfrac{\sqrt{3}}{2}\right)\right)^{\frac{1}{4}}$

$= 2^{\frac{1}{4}}\left(\text{cis}\ \dfrac{\pi}{3}\right)^{\frac{1}{4}} = 2^{\frac{1}{4}}\left(\text{cis}\ \dfrac{2k\pi + \frac{\pi}{3}}{4}\right), \text{k} = 0,1,2,3 = 2^{\frac{1}{4}}\text{cis}\ \left(\dfrac{6k\pi + \pi}{12}\right), \text{k} = 0,1,2,3$

$= 2^{\frac{1}{4}}\text{cis}\ (6k+1)\dfrac{\pi}{12}, \text{k} = 0,1,2,3$

36. All values of $(1 + i\sqrt{3})^{\frac{1}{3}}$ are

$(1 + i\sqrt{3})^{\frac{1}{3}} = \left(2\left(\dfrac{1}{2} + \dfrac{i\sqrt{3}}{2}\right)\right)^{\frac{1}{3}}$

$= 2^{\frac{1}{3}}\left(\text{cis}\ \dfrac{\pi}{3}\right)^{\frac{1}{3}} = 2^{\frac{1}{3}}\text{cis}\left(\dfrac{2k\pi + \frac{\pi}{3}}{3}\right), \text{k} = 0,1,2 = 2^{\frac{1}{3}}\text{cis}\ (6k+1)\dfrac{\pi}{9}, \text{k} = 0,1,2$

37. If $1, \alpha_1, \alpha_2, \ldots\ldots\alpha_{n-1}$ are n roots of uity then

(a) $(1-\alpha_1)(1-\alpha_2)(1-\alpha_3)\ldots\ldots\ldots(1-\alpha_{n-1}) = n$

(b)If n is odd $(1+\alpha_1)(1+\alpha_2)(1+\alpha_3)\ldots\ldots\ldots(1+\alpha_{n-1}) = 1$

(c) If n is even $(1+\alpha_1)(1+\alpha_2)(1+\alpha_3)\ldots\ldots\ldots(1+\alpha_{n-1}) = 0$

(d) $(3-\alpha_1)(3-\alpha_2)(3-\alpha_3)\ldots\ldots\ldots(3-\alpha_{n-1})$

$1, \alpha_1, \alpha_2, \ldots\ldots\alpha_{n-1}$ are roots of $\text{x}^{\text{n}-1} = 0$

$\text{x}^{\text{n}} - 1 = (\text{x}-1)(x-\alpha_1)(x-\alpha_2)\ldots\ldots(x-\alpha_{n-1})$

$\dfrac{x^n - 1}{x-1} = (x-\alpha_1)(x-\alpha_2)\ldots\ldots(x-\alpha_{n-1})$

$(x-\alpha_1)(x-\alpha_2)\ldots\ldots(x-\alpha_{n-1}) = 1 + x + x^2 + \cdots . + x^{n-1}$

Put x = 3

$(3-\alpha_1)(3-\alpha_2)\ldots\ldots(3-\alpha_{n-1}) = \dfrac{3^n - 1}{3-1} = \dfrac{3^n - 1}{2}$

Worksheet – 4

Level – I

1) A 2) B 3) C 4) A 5) C 6) C

1. $\cos 5\theta = \cos^5\theta - 5c_2\cos^{5-2}\theta\sin^2\theta + 5c_4\cos^{5-4}\theta\sin^4\theta$

$= \cos^5\theta - \frac{5\times 4\times 3!}{3!\times 2\times 1}\cos^3\theta\sin^2\theta + \frac{5\times 4!}{1!\times 4!}\cos\theta\sin^4\theta$

$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$

2. $\sin 4\theta = 4_{c_1}\cos^{4-1}\theta\sin\theta - 4_{c_3}\cos^{4-3}\theta\sin^3\theta$

$= \frac{4\times 3!}{3!\times 2!}\cos^3\theta\sin\theta - \frac{4\times 3!}{3!\times 1!}\cos\theta\sin^3\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$

3. $\sin 5\theta = 5_{c_1}\cos^{5-1}\theta\sin\theta - 5_{c_3}\cos^{5-3}\theta\sin^3\theta + 5_{c_5}\cos^{5-5}\theta\sin^5\theta$

$= 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta$

$= 5(1-\sin^2\theta)^2\sin\theta - 10(1-\sin^2\theta)\sin^3\theta + \sin^5\theta$

$= 5(1+\sin^4\theta - 2\sin^2\theta)\sin\theta - 10(1-\sin^2\theta)\sin^3\theta + \sin^5\theta$

$= \sin\theta[5 + 5\sin^4\theta - 10\sin^2\theta - 10\sin^2\theta + 10\sin^4\theta + \sin^4\theta]$

$= \sin\theta[16\sin^4\theta - 20\sin^2\theta + 5]$

$\frac{\sin 5\theta}{\sin\theta} = 16\sin^4\theta - 20\sin^2\theta + 5$

4. $\cos 4\theta = \cos^4\theta - 4_{c_2}\cos^2\theta\sin^2\theta + 4_{c_4}\sin^4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$

5. $\cot 4\theta = \frac{\cot^4\theta - 4_{c_2}\cot^2\theta + 4_{c_4}\cot^0\theta}{4_{c_1}\cot^3\theta - 4_{c_3}\cot\theta} = \frac{\cot^4\theta - 6\cot^2\theta + 1}{4\cot^3\theta - 4\cot\theta}$

6. $\cos 6\theta = \cos^6\theta - 6_{c_2}\cos^4\theta\sin^2\theta + 6_{c_4}\cos^2\theta\sin^4\theta - 6_{c_6}\sin^6\theta$

$= \cos^6\theta - \frac{6\times 5\times 4!}{2\times 1\times 4!}\cos^4\theta\sin^2\theta + \frac{6\times 5\times 4!}{2\times 1\times 4!}\cos^2\theta\sin^4\theta - \sin^6\theta$

$= \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta$

Level – II

1)A 2)A 3)A 4)B 5)A 6)5 7)B 8)D 9)A 10)D 11)B 12)A

13)A 14)D 15)B 16)C 17)A 18)6 19)B 20)B

21)A

1. $\sin 4\theta = 4_{c_1}\cos^{3\theta}\sin\theta - 4_{c_3}\cos\theta\sin^3\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$

$\frac{\sin 4\theta}{\sin\theta} = \frac{\sin\theta[4\cos^3\theta - 4\cos\theta\sin^2\theta]}{\sin\theta} = 4\cos^3\theta - 4\cos\theta(1-\cos^2\theta)$

$= 4\cos^3\theta - 4\cos\theta + 4\cos^3\theta = 8\cos^3\theta - 4\cos\theta$

2. $\text{Tan } 5\theta =$

$$\text{Tan } 5\theta = \frac{5c_1\text{Tan } \theta - 5c_3\text{Tan}^3 \theta + 5c_5\text{Tan}^5 \theta}{1 - 5c_2\text{Tan}^2 \theta + 5c_4\text{Tan}^4 \theta}$$

$$\text{Tan } 5\theta = \frac{5\text{Tan } \theta - 10\text{Tan}^3 \theta + \text{Tan}^5 \theta}{1 - 10\text{Tan}^2 \theta + 5\text{Tan}^4 \theta}$$

3. $\text{Tan } 4\theta =$

$$\text{Tan } 4\theta = \frac{4c_1\text{Tan } \theta - 4c_3\text{Tan}^3 \theta}{1 - 4c_2\text{Tan}^2 \theta + 4c_4\text{Tan}^4 \theta} \Rightarrow \text{Tan } 4\theta = \frac{4\text{Tan } \theta - 4\text{Tan}^3 \theta}{1 - 6\text{Tan}^2 \theta + \text{Tan}^4 \theta}$$

4. $\cot 5\theta = \cot 5\theta = \frac{\cot^5 \theta - 5c_2\cot^3 \theta + 5c_4\cot \theta}{5c_1\cot^4 \theta - 5c_3\cot^2 \theta + 5c_5\cot^0 \theta} \Rightarrow \cot 5\theta = \frac{\cot^5 \theta - 10\cot^3 \theta + 5\cot \theta}{5\cot^4 \theta - 10\cot^2 \theta + 1}$

5. $\sin 6\theta =$

$\sin 6\theta = 6c_1\cos^5 \theta\sin \theta - 6c_3\cos^3 \theta + 6c_5\cos \theta\sin^5 \theta$

$\sin 6\theta = 6\cos^5 \theta\sin \theta - 20\cos^3 \theta + 6\cos \theta\sin^5 \theta$

6. $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta - n\cos \theta$, then n =

$\cos 5\theta = \cos^5 \theta - 5c_2\cos^3 \theta\sin^2 \theta + 5c_4\cos \theta\sin^4 \theta$

$\cos 5\theta = \cos^5 \theta - 10\cos^3 \theta(1 - \cos^2 \theta) + 5\cos \theta(1 - \cos^2 \theta)^2$

$\cos 5\theta = \cos^5 \theta - 10\cos^3 \theta + 10\cos^5 \theta + 5\cos \theta(1 + \cos^4 \theta - 2\cos^2 \theta)$

$\cos 5\theta = \cos^5 \theta - 10\cos^3 \theta + 10\cos^5 \theta + 5\cos \theta + 5\cos^5 \theta - 10\cos^3 \theta$

$\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$

n = 5

7. $\text{Tan } 6\theta =$

$$\text{Tan } 6\theta = \frac{6c_1\text{Tan } \theta - 6c_3\text{Tan}^3 \theta + 6c_5\text{Tan}^5 \theta}{1 - 6c_2\text{Tan}^2 \theta + 6c_4\text{Tan}^4 \theta - 6c_6\text{Tan}^6 \theta}$$

$$\text{Tan } 6\theta = \frac{6\text{Tan } \theta - \frac{6\times5\times4}{3\times2\times1}\text{Tan}^3 \theta + 6\text{Tan}^5 \theta}{1 - \frac{6\times5}{2\times1}\text{Tan}^2 \theta + \frac{6\times5}{2\times1}\text{Tan}^4 \theta - \text{Tan}^6 \theta}$$

$$\text{Tan } 6\theta = \frac{6\text{Tan } \theta - 20\text{Tan}^3 \theta + 6\text{Tan}^5 \theta}{1 - 15\text{Tan}^2 \theta + 15\text{Tan}^4 \theta - \text{Tan}^6 \theta}$$

8. Last terms in the expansion of $\sin 8\theta, \sin 9\theta$ are

$\sin 8\theta = 8c_1\cos^7 \theta\sin \theta - 8c_3\cos^5 \theta\sin^3 \theta + 8c_5\cos^3 \theta\sin^5 \theta - 8c_7\cos \theta\sin^7 \theta$

$\sin 8\theta = 8\cos^7 \theta\sin \theta - 56\cos^5 \theta\sin^3 \theta + 56\cos^3 \theta\sin^5 \theta - 8\cos \theta\sin^7 \theta$

last term in the expansion of

$\sin 8\theta$ is $-8\cos \theta\sin^7 \theta$

$\sin 9\theta = 9c_1\cos^8 \theta\sin \theta - 9c_3\cos^6 \theta\sin^3 \theta + 9c_5\cos^4 \theta\sin^5 \theta - 9c_7\cos^2 \theta\sin^7 \theta$

$+ 9c_9\sin^9 \theta$

last term in the expansion of

$\sin\ 9\theta$ is $9c_9\sin^9\ \theta = \sin^9\ \theta$

9. $2^4\cos^5\ \theta =$

$2^4\cos^5\ \theta = \cos\ 5\theta + 5c_1\cos\ 3\theta + 5c_2\cos\ \theta$

$2^4\cos^5\ \theta = \cos\ 5\theta + 5\cos\ 3\theta + 10\cos\ \theta$

10. $32\cos^6\ \theta =$

$32\cos^6\ \theta = \cos\ 6\theta + 6c_1\cos\ 4\theta + 6c_2\cos\ 2\theta + 6c_3\cos\ 0$

$32\cos^6\ \theta = \cos\ 6\theta + 6\cos\ 4\theta + \frac{6\times5}{2\times1}\cos\ 2\theta + \frac{6\times5\times4}{3\times2\times1}$

$32\cos^6\ \theta = \cos\ 6\theta + 6\cos\ 4\theta + 15\cos\ 2\theta + 20$

If n is even, last term is $\frac{1}{2}\mathrm{nc}_{\frac{n}{2}} = \frac{1}{2}6\mathrm{c}_3 = \frac{1}{2}\times 20 = 10$

$32\cos^6\ \theta = \cos\ 6\theta + 6\cos\ 4\theta + 15\cos\ 2\theta + 10$

11. Last terms in the expansion of $\cos\ 11\theta, \sin\ 11\theta$ are

$\cos\ 11\theta = \cos^{11}\ \theta - 11c_2\cos^9\ \theta\sin^2\ \theta + 11c_4\cos^7\ \theta\sin^4\ \theta$ … ….

n = 11.so last term of $\cos\ n\theta = \mathrm{n}(-1)^{\frac{n-1}{2}}\sin^{n-1}\ \theta\cos\ \theta$

$\cos\ 11\theta = 11(-1)^5\sin^{10}\ \theta\cos\ \theta = -11\sin^{10}\ \theta\cos\ \theta$

last term of $\sin\ n\theta$ is $(-1)^{\frac{n-1}{2}}\sin^n\ \theta$ if n is odd

last term of $\sin\ 11\theta = (-1)^5\sin^{11}\ \theta$

12. $2^5\sin^6\ \theta =$

n = 6 is even

$2^5i^6\sin^6\ \theta = 6c_0\cos\ 6\theta - 6c_1\cos\ 4\theta + 6c_2\cos\ 2\theta - 6c_3$

$-2^5\sin^6\ \theta = \cos\ 6\theta - 6\cos\ 4\theta + \frac{6\times5}{2\times1}\cos\ 2\theta - 10$

$2^5\sin^6\ \theta = 10 - 15\cos\ 2\theta + 6\cos\ 4\theta - \cos\ 6\theta$

13. $\cos^2\ \theta\sin^5\ \theta =$

let $\mathrm{X} = \cos\ \theta + \mathrm{i}\sin\ \theta, \frac{1}{\mathrm{X}} = \cos\ \theta - \mathrm{i}\sin\ \theta$

$(2\cos\ \theta)^2(2\mathrm{i}\sin\ \theta)^5 = \left(x + \frac{1}{x}\right)^2\left(x - \frac{1}{x}\right)^5$

$= \left(\mathrm{x} - \frac{1}{\mathrm{x}}\right)^3\left(\mathrm{x}^2 - \frac{1}{\mathrm{x}^2}\right)^2$

$= \left(x^3 - \frac{1}{x^3} - 3x^2\frac{1}{x} + 3x\frac{1}{x^2}\right)\left(x^4 + \frac{1}{x^4} - 2\right)$

$= \left(x^3 - \frac{1}{x^3} - 3x + \frac{3}{x}\right)\left(x^4 + \frac{1}{x^4} - 2\right)$

$= x^7 + \frac{1}{x} - 2x^3 - x - \frac{1}{x^7} + \frac{2}{x^3} - 3x^5 - \frac{3}{x^3} + 6x + 3x^3 + \frac{3}{x^5} - \frac{6}{x}$

$= 2i\sin\ 7\theta - 3 \times 2i\sin\ 5\theta + 2i\sin\ 3\theta + 5 \times 2i\sin\ \theta$

$2^7\text{x}(+\text{i})\cos^2\ \theta\sin^5\ \theta = 2\text{i}[\sin\ 7\theta - 3\sin\ 5\theta + \sin\ 3\theta + 5\sin\ \theta]$

$2^6\cos^2\ \theta\sin^5\ \theta = \sin\ 7\theta - 3\sin\ 5\theta + \sin\ 3\theta + 5\sin\ \theta$

$\cos^2\ \theta\sin^5\ \theta = \frac{1}{64}[\sin\ 7\theta - 3\sin\ 5\theta + \sin\ 3\theta + 5\sin\ \theta]$

14. If $16\sin^5\ \theta = a\sin\ 5\theta + b\sin\ 3\theta + \text{csin}\ \theta$ then $a + b + c = \text{n}$ is odd

$2^{\text{n}-1}\text{i}^{\text{n}-1}\sin^{\text{n}}\ \theta = \text{nc}_0\text{sinn}\ \theta - \text{nc}_1\sin\ (\text{n} - 2)\theta + \cdots \ldots + (-1)^{\frac{\text{n}-1}{2}}\text{nc}_{\frac{\text{n}-1}{2}}\sin\ \theta$

$\text{n} = 5$

$2^4i^4\sin^5\ \theta = 5c_0\sin\ 5\theta - 5c_1\sin\ 3\theta + (-1)^2 5c_2\sin\ \theta$

$16\sin^5\ \theta = \sin\ 5\theta - 5\sin\ 3\theta + 10\sin\ \theta$

$\text{a} = 1, \text{b} = -5, \text{c} = 10$

$a + b + c = 1 - 5 + 10 = 11 - 5 = 6$

15. $\cos^4\ \theta\sin^3\ \theta$

let $\text{x} = \cos\ \theta + \text{isin}\ \theta$

$$\frac{1}{x} = \cos\ \theta - i\sin\ \theta$$

$$(2\cos\ \theta)^4(2\text{isin}\ \theta)^3 = \left(x + \frac{1}{x}\right)^4\left(x - \frac{1}{x}\right)^3$$

$$(2\cos\ \theta)^4(2\text{isin}\ \theta)^3 = \left(x + \frac{1}{x}\right)\left(\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)\right)^3$$

$$(2\cos\ \theta)^4(2\text{isin}\ \theta)^3 = \left(x + \frac{1}{x}\right)\left(x^2 - \frac{1}{x^2}\right)^3$$

$$(2\cos\ \theta)^4(2\text{isin}\ \theta)^3 = \left(x + \frac{1}{x}\right)\left(x^6 - \frac{1}{x^6} - 3x^2 + \frac{3}{x^2}\right)$$

$$(2\cos\ \theta)^4(2i\sin\ \theta)^3 = x^7 - \frac{1}{x^5} - 3x^3 + \frac{3}{x} + x^5 - \frac{1}{x^7} - 3x + \frac{3}{x^3}$$

$$(2\cos\ \theta)^4(2\sin\ \theta)^3 = \left(x^7 - \frac{1}{x^7}\right) - 3\left(x^3 - \frac{1}{x^3}\right) + \left(x^5 - \frac{1}{x^5}\right) - 3\left(x - \frac{1}{x}\right)$$

$2^7x\cos^4\ \theta x i^3\sin^3\ \theta = 2i\sin\ 7\theta + 2i\sin\ 5\theta - 6i\sin\ 3\theta - 6i\sin\ \theta$

$2^7x\cos^4\ \theta x(-i)\sin^3\ \theta = -2i[\sin\ 7\theta - \sin\ 5\theta + 3\sin\ 3\theta + 3\sin\ \theta]$

$2^6x\cos^4\ \theta x\sin^3\ \theta = 3\sin\ \theta + 3\sin\ 3\theta - \sin\ 5\theta - \sin\ 7\theta$

$\cos^4\ \theta\sin^3\ \theta = \frac{1}{64}[3\sin\ \theta + 3\sin\ 3\theta - \sin\ 5\theta - \sin\ 7\theta]$

16. $2^6\sin^5\ \theta\cos^2\ \theta =$

let $x = \cos\ \theta + i\sin\ \theta \Rightarrow \frac{1}{x} = \cos\ \theta - i\sin\ \theta$

$(2\cos\ \theta)^2(2i\sin\ \theta)^5 = \left(x+\frac{1}{x}\right)^2\left(x-\frac{1}{x}\right)^5$

$2^7 x\cos^2\ \theta x\sin^5\ \theta = \left(x-\frac{1}{x}\right)^3\left[\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)\right]^2$

$2^7 x\cos^2\ \theta x\sin^5\ \theta = \left(x^3-\frac{1}{x^3}-3x+\frac{3}{x}\right)\left(x^4+\frac{1}{x^4}-2\right)$

$2^7 x\cos^2\ \theta x\sin^5\ \theta = x^7+\frac{1}{x}-2x^3-x-\frac{1}{x^7}+\frac{2}{x^3}-3x^5-\frac{3}{x^3}+6x+3x^3+\frac{3}{x^5}-\frac{6}{x}$

$2^7 x\cos^2\ \theta x\sin^5\ \theta = \left(x^7-\frac{1}{x^7}\right)-3\left(x^5-\frac{1}{x^5}\right)+\left(x^3+\frac{1}{x^3}\right)+5\left(x-\frac{1}{x}\right)$

$2^7 x\cos^2\ \theta x\sin^5\ \theta = 2i\sin\ 7\theta - 3\times 2i\sin\ 5\theta + 2i\sin\ 3\theta + 5\times 2i\sin\ \theta$

$2^7 x\cos^2\ \theta x\sin^5\ \theta = 2\mathrm{i}[\sin\ 7\theta - 3\sin\ 5\theta + \sin\ 3\theta + 5\sin\ \theta]$

$2^6\cos^2\ \theta\sin^5\ \theta = \sin\ 7\theta - 3\sin\ 5\theta + \sin\ 3\theta + 5\sin\ \theta$

17. $8\sin^4\ \theta =$

$\mathrm{n} = 4$ is even

$2^{n-1}\mathrm{i}^n\sin^n\ \theta = \mathrm{nc}_0\cos\ \theta\theta - \mathrm{ncc}_1\cos\ (\mathrm{n}-2)\theta + \cdots \ldots . + \frac{1}{2}(-1)^{\frac{n}{2}}\mathrm{nc}_{\frac{n}{2}}$

$2^3 i^4\sin^4\ \theta = 4c_0\cos\ 4\theta - 4c_1\cos\ 2\theta + \cdots \ldots + \frac{1}{2}(-1)^2 4c_2$

$2^3\sin^4\ \theta = \cos\ 4\theta - 4\cos\ 2\theta + 3$

18. $\sin\ 6\theta = 6\cos^5\ \theta\sin\ \theta - 20\cos^3\ \theta\sin^3\ \theta + \mathrm{k}\cos\ \theta\sin^5\ \theta$ then $\mathrm{k} =$

$\sin\ 6\theta = 6c_1\cos^5\ \theta\sin\ \theta - 6c_3\cos^3\ \theta\sin^3\ \theta + 6c_5\cos\ \theta\sin^5\ \theta$

$\sin\ 6\theta = 6\cos^5\ \theta\sin\ \theta - 20\cos^3\ \theta\sin^3\ \theta + 6\cos\ \theta\sin^5\ \theta$

$\mathrm{k} = 6$

19. $\cos^4\ \theta + \sin^4\ \theta =$

n is even

$2^{n-1}\cos^n\ \theta = \cos\ n\theta + \mathrm{nc}_1\cos\ (\mathrm{n}-2)\theta + \cdots \ldots . + \frac{1}{2}\mathrm{nc}_{\frac{n}{2}}$

$\mathrm{n} = 4$

$2^3\cos^4\ \theta = \cos\ 4\theta + 4c_1\cos\ 2\theta + \frac{1}{2}4c_2$

$2^3\cos^4\ \theta = \cos\ 4\theta + 4\cos\ 2\theta + 3$

$\cos^4\ \theta = \frac{1}{8}[\cos\ 4\theta + 4\cos\ 2\theta + 3]$

$\mathrm{n} = 4$ is even

$2^{n-1}\mathrm{i}^n\sin^n\ \theta = \mathrm{nc}_0\cos\ \theta\theta - \mathrm{nc}_1\cos\ (\mathrm{n}-2)\theta + \cdots \ldots . + \frac{1}{2}(-1)^{\frac{n}{2}}\mathrm{nc}_{\frac{n}{2}}$

$2^3 i^4 \sin^4\ \theta = 4c_0 \cos\ 4\theta - 4c_1 \cos\ 2\theta + \cdots \ldots . + \frac{1}{2}(-1)^2 4c_2$

$8\sin^4\ \theta = \cos\ 4\theta - 4\cos\ 2\theta + 3$

$\sin^4\ \theta = \frac{1}{8}[\cos\ 4\theta - 4\cos\ 2\theta + 3]$

$\cos^4\ \theta + \sin^4\ \theta = \frac{1}{8}[2\cos\ 4\theta + 6] = \frac{1}{4}[\cos\ 4\theta + 3]$

20. $\cot\ 6\theta \cot\ \theta = \cot\ 6\theta \cot\ \theta = \frac{\cot^6\ \theta - 6c_2 \cot^4\ \theta + 6c_4 \cot^2\ \theta - 6c_6}{6c_1 \cot^5\ \theta - 6c_3 \cot^3\ \theta + 6c_5 \cot\ \theta} x \cot\ \theta$

$\cot\ 6\theta \cot\ \theta = \frac{\cot^6\ \theta - 15\cot^4\ \theta + 15\cot^2\ \theta - 1}{6\cot^5\ \theta - 20\cot^3\ \theta + 6\cot\ \theta} \times \cot\ \theta$

$\cot\ 6\theta \cot\ \theta = \frac{\cot^6\ \theta - 15\cot^4\ \theta + 15\cot^2\ \theta - 1}{\cot\ \theta(6\cot^4\ \theta - 20\cot^2\ \theta + 6)} \times \cot\ \theta$

$\cot\ 6\theta \cot\ \theta = \frac{\cot^6\ \theta - 15\cot^4\ \theta + 15\cot^2\ \theta - 1}{6\cot^4\ \theta - 20\cot^2\ \theta + 6}$

21. $\cos^4\ \theta =$

n = 4 is even

$2^{n-1}\cos^n\ \theta = \cos\ n\theta + nc_1 \cos\ (n-2)\theta + \cdots \ldots \ldots + \frac{1}{2} nc_{\frac{n}{2}}$

$2^3 \cos^4\ \theta = \cos\ 4\theta + 4c_1 \cos\ 2\theta + \cdots \ldots \ldots + \frac{1}{2} 4c_2$

$8\cos^4\ \theta = \cos\ 4\theta + 4\cos\ 2\theta + 3$

$\cos^4\ \theta = \frac{1}{8}[\cos\ 4\theta + 4\cos\ 2\theta + 3]$

Level – III

22)B,C 23)A 24)B 25)B 26)a-q, b-r, c-p, d-s

27)A,C 28)A 29)B 30)B 31)a-q, b-s, c-p, d-r 32)B,C

33)C

34)C 35)D

22. $\frac{\sin\ 4\theta}{\sin\ \theta} = x\cos^3\ \theta - y\cos\ \theta$

$\sin\ 4\theta = 4c_1 \cos^3\ \theta \sin\ \theta - 4c_3 \cos\ \theta \sin^3\ \theta \Rightarrow \frac{\sin\ 4\theta}{\sin\ \theta} = \frac{\sin\ \theta[4\cos^3\ \theta - 4\cos\ \theta \sin^2\ \theta]}{\sin\ \theta}$

$\frac{\sin\ 4\theta}{\sin\ \theta} = \frac{\sin\ \theta[4\cos^3\ \theta - 4\cos\ \theta + 4\cos^3\ \theta]}{\sin\ \theta} \Rightarrow \frac{\sin\ 4\theta}{\sin\ \theta} = 8\cos^3\ \theta - 4\cos\ \theta \Rightarrow x = 8, y = 4$

23. $\sin\ 5\theta =$

$\sin 5\theta = 5c_1\cos^4\theta\sin\theta - 5c_3\cos^2\theta\sin^3\theta + 5c_5\sin^5\theta$

$\sin 5\theta = 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta$

24. Last term in the expansion of $\sin 8\theta$

$\sin 8\theta = 8c_1\cos^7\theta\sin\theta - 8c_3\cos^5\theta\sin^3\theta + 8c_5\cos^3\theta\sin^5\theta - 8c_7\cos\theta\sin^7\theta$

Last term is $-8c_7\cos\theta\sin^7\theta = -8\cos\theta\sin^7\theta$

25. $\cos 7\theta =$

$\cos 7\theta = \cos^7\theta - 7c_2\cos^5\theta\sin^2\theta + 7c_4\cos^3\theta - 7c_6\cos\theta\sin^6\theta$

$\cos 7\theta = \cos^7\theta - 21\cos^5\theta\sin^2\theta + 35\cos^3\theta - 7\cos\theta\sin^6\theta$

a. $\frac{\sin 5\theta}{\sin\theta}$

$\sin 5\theta = 5c_1\cos^4\theta\sin\theta - 5c_3\cos^2\theta\sin^3\theta + 5c_5\sin^5\theta$

$\sin 5\theta = 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta$

$\frac{\sin 5\theta}{\sin\theta} = \frac{\sin\theta[5\cos^4\theta - 10\cos^2\theta\sin^2\theta + \sin^4\theta]}{\sin\theta}$

$\frac{\sin 5\theta}{\sin\theta} = 5\cos^4\theta - 10\cos^2\theta + \sin^4\theta \Rightarrow \frac{\sin 5\theta}{\sin\theta} = 5\cos^4\theta - 10\cos^2\theta + (1-\cos^2\theta)^2$

$\frac{\sin 5\theta}{\sin\theta} = 5\cos^4\theta - 10\cos^2\theta + 1 + \cos^4\theta - 2\cos^2\theta \Rightarrow \frac{\sin 5\theta}{\sin\theta} = 6\cos^4\theta - 12\cos^2\theta + 1$

2. $\frac{\sin 4\theta}{\sin\theta} \Rightarrow \sin 4\theta = 4c_1\cos^3\theta\sin\theta - 4c_3\cos\theta\sin^3\theta$

$\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta \Rightarrow \frac{\sin 4\theta}{\sin\theta} = \frac{\sin\theta[4\cos^3\theta - 4\cos\theta\sin^2\theta]}{\sin\theta}$

$\frac{\sin 4\theta}{\sin\theta} = 4\cos^3\theta - 4\cos\theta\sin^2\theta \Rightarrow \frac{\sin 4\theta}{\sin\theta} = 4\cos^3\theta - 4\cos\theta(1-\cos^2\theta)$

$\frac{\sin 4\theta}{\sin\theta} = 4\cos^3\theta - 4\cos\theta + 4\cos^3\theta \Rightarrow \frac{\sin 4\theta}{\sin\theta} = 8\cos^3\theta - 4\cos\theta$

3. $\frac{\sin 6\theta}{\sin\theta} \Rightarrow \sin 6\theta = 6c_1\cos^5\theta\sin\theta - 6c_3\cos^3\theta\sin^3\theta + 6c_5\cos\theta\sin^5\theta$

$\sin 6\theta = 6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta$

$\frac{\sin 6\theta}{\sin\theta} = \frac{\sin\theta[6\cos^5\theta - 20\cos^3\theta\sin^2\theta + 6\cos\theta\sin^4\theta]}{\sin\theta}$

$= 6\cos^5\theta - 20\cos^3\theta\sin^2\theta + 6\cos\theta\sin^4\theta$

$= 6\cos^5\theta - 20\cos^3\theta\sin^2\theta + 6\cos\theta(1-\cos^2\theta)^2$

$= 6\cos^5\theta - 20\cos^3\theta\sin^2\theta + 6\cos\theta(1+\cos^4\theta - 2\cos^2\theta)$

$= 32\cos^5\theta - 32\cos^3\theta + 6\cos\theta$

4. $\frac{\sin 7\theta}{\sin\theta} \Rightarrow \sin 7\theta = 7c_1\cos^6\theta\sin\theta - 7c_3\cos^4\theta\sin^3\theta + 7c_5\cos^2\theta\sin^5\theta - 7c_7\sin^7\theta$

$\sin 7\theta = 7\cos^6\theta\sin\theta - 35\cos^4\theta\sin^3\theta + 21\cos^2\theta\sin^5\theta - \sin^7\theta$

$\frac{\sin 7\theta}{\sin\theta} = 7\cos^6\theta - 35\cos^4\theta\sin^2\theta + 21\cos^2\theta\sin^4\theta - \sin^6\theta$

$$\frac{\sin 7\theta}{\sin \theta} = 7(1-\sin^2\theta)^3 - 35(1-\sin^2\theta)^2\sin^2\theta + 21(1-\sin^2\theta)\sin^4\theta - \sin^6\theta$$

$$\frac{\sin 7\theta}{\sin \theta} = 7(1-\sin^6\theta - 3\sin^2\theta + 3\sin^4\theta) - 35(1+\sin^4\theta - 2\sin^2\theta)\sin^2\theta$$

$$+ 21\sin^4\theta - 21\sin^6\theta - \sin^6\theta$$

$$\frac{\sin 7\theta}{\sin \theta} = 7 - 56\sin^2\theta + 112\sin^4\theta - 64\sin^6\theta$$

27. Last terms in the expansion $\sin 8\theta, \sin 9\theta \Rightarrow n = 8$ is even

last terms of $\sin n\theta = n(-1)^{\frac{n-2}{2}}\cos\theta \cdot \sin^{n-1}\theta$

$\sin 8\theta = 8(-1)^3\cos\theta \cdot \sin^7\theta = -8\cos\theta \cdot \sin^7\theta$

$n = 9$ is odd

Last terms of $\sin n\theta = (-1)^{\frac{n-1}{2}} \cdot \sin^n\theta$

$\sin 9\theta = (-1)^4 \cdot \sin^9\theta = \sin^9\theta$

28. $\cos^6\theta =$

$$2^{n-1} \cdot \cos^n\theta = \cos n\theta + {}^nc_1\cos(n-2)\theta + {}^nc_2\cos(n-4)\theta + \cdots . + \frac{1}{2}\,{}^nc_{\frac{n}{2}}$$

$$2^5 \cdot \cos^6\theta = \cos 6\theta + {}^6c_1\cos 4\theta + {}^6c_2\cos 2\theta + \frac{1}{2}\,{}^6c_3$$

$$32\cos^6\theta = \cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10$$

$$\cos^6\theta = \frac{1}{32}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)$$

29. $\sin^6\theta = n = 6$ is even

$$2^{n-1}i^n\sin^n\theta = {}^nc_0\cos n\theta - {}^nc_1\cos(n-2)\theta + {}^nc_2\cos(n-4)\theta \ldots + \frac{(-1)^{\frac{n}{2}}}{2}\,{}^nc_{\frac{n}{2}}$$

$$2^5i^6\sin^6\theta = {}^6c_0\cos 6\theta - {}^6c_1\cos 4\theta + {}^6c_2\cos 2\theta + \frac{(-1)^3}{2}\,{}^6c_3$$

$$-2^5\sin^6\theta = \cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10$$

$$\sin^6\theta = \frac{1}{32}(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta)$$

30. $\cos^6\theta + \sin^6\theta =$

$$\cos^6\theta + \sin^6\theta = \frac{1}{32}(12\cos 4\theta + 20) = \frac{1}{32} \times 4(3\cos 4\theta + 5) = \frac{1}{8}(3\cos 4\theta + 5)$$

31. a) $\sin 6\theta = 6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 3a$ then $a = \sin 6\theta = {}^6c_1\cos^5\theta\sin\theta - {}^6c_3\cos^3\theta\sin^3\theta + {}^6c_5\cos\theta\sin^5\theta$

$= 6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta$

$3a = 6\cos\theta\sin^5\theta$

$a = 2\cos\theta\sin^5\theta$

b) $\cos\ 5\theta = a\cos\ \theta + b\cos^3\ \theta + 16\cos^5\ \theta$

$\cos\ 5\theta = \cos^5\ \theta - {}^5c_2\cos^3\ \theta\sin^2\ \theta + {}^5c_4\sin^4\ \theta\cos\ \theta$

$= \cos^5\ \theta - 10\cos^3\ \theta\sin^2\ \theta + 5\sin^4\ \theta\cos\ \theta$

$= \cos^5\ \theta - 10\cos^3\ \theta(1 - \cos^2\ \theta) + 5\cos\ \theta(1 - \cos^2\ \theta)^2$

$= \cos^5\ \theta - 10\cos^3\ \theta + 10\cos^5\ \theta + 5\cos\ \theta(1 + \cos^4\ \theta - 2\cos^2\ \theta)$

$= \cos^5\ \theta - 10\cos^3\ \theta + 10\cos^5\ \theta + 5\cos\ \theta + 5\cos^5\ \theta - 10\cos^3\ \theta$

$= 5\cos\ \theta - 20\cos^3\ \theta + 16\cos^5\ \theta$

$\therefore a = 5, b = -20$

c) $2^6\cos^3\ \theta\sin^4\ \theta = \cos\ 7\theta - \cos\ 5\theta + a\cos\ 3\theta + b\cos\ \theta$

Let $x = \cos\ \theta + i\sin\ \theta \Rightarrow \frac{1}{x} = \cos\ \theta - i\sin\ \theta \Rightarrow (2\cos\ \theta)^3(2i\sin\ \theta)^4 = \left(x + \frac{1}{x}\right)^3\left(x - \frac{1}{x}\right)^4$

$2^7x\cos^3\ \theta x i^4\sin^4\ \theta = \left(x - \frac{1}{x}\right)\left(\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)\right)^3$

$= \left(x - \frac{1}{x}\right)\left(x^2 - \frac{1}{x^2}\right)^3 = \left(x - \frac{1}{x}\right)\left(x^6 - \frac{1}{x^6} - 3x^2 + \frac{3}{x^2}\right)$

$= x^7 - \frac{1}{x^5} - 3x^3 + \frac{3}{x} - x^5 + \frac{1}{x^7} + 3x - \frac{3}{x^3}$

$= \left(x^7 + \frac{1}{x^7}\right) - \left(x^5 + \frac{1}{x^5}\right) - 3\left(x^3 + \frac{1}{x^3}\right) + 3\left(x + \frac{1}{x}\right)$

$= 2\cos\ 7\theta - 2\cos\ 5\theta - 3 \times 2\cos\ 3\theta + 3 \times 2\cos\ \theta$

$2^7x\cos^3\ \theta\sin^4\ \theta = 2(\cos\ 7\theta - \cos\ 5\theta - 3\cos\ 3\theta + 3\cos\ \theta)$

$2^6\cos^3\ \theta\sin^4\ \theta = \cos\ 7\theta - \cos\ 5\theta - 3\cos\ 3\theta + 3\cos\ \theta\ \ \therefore a = -3, b = 3$

d) Last term in the expansion of $\sin\ 9\theta = n = 9$ is odd

Last term is $(-1)^{\frac{n-1}{2}}\sin^n\ \theta = (-1)^4\sin^9\ \theta = \sin^9\ \theta$

32. $\cos^3\ \theta\sin^4\ \theta = \frac{1}{k}(\cos\ 7\theta - \cos\ 5\theta - 3\cos\ 3\theta + p\cos\ \theta)$ then k and p are

$2^6\cos^3\ \theta\sin^4\ \theta = \cos\ 7\theta - \cos\ 5\theta - 3\cos\ 3\theta + 3\cos\ \theta$

$\cos^3\ \theta\sin^4\ \theta = \frac{1}{64}(\cos\ 7\theta - \cos\ 5\theta - 3\cos\ 3\theta + 3\cos\ \theta) \Rightarrow k = 64, p = 3$

33. Expansion of $\cos\ 5\theta$ in terms of $\cos\ \theta$

$\cos\ 5\theta = \cos^5\ \theta - {}^5c_2\cos^3\ \theta\sin^2\ \theta + {}^5c_4\cos\ \theta\sin^4\ \theta$

$= \cos^5\ \theta - 10\cos^3\ \theta\sin^2\ \theta + 5\cos\ \theta\sin^4\ \theta$

$= \cos^5\ \theta - 10\cos^3\ \theta(1 - \cos^2\ \theta) + 5\cos\ \theta(1 - \cos^2\ \theta)^2$

$= \cos^5\ \theta - 10\cos^3\ \theta + 10\cos^5\ \theta + 5\cos\ \theta(1 + \cos^4\ \theta - 2\cos^2\ \theta)$

$= \cos^5\ \theta - 10\cos^3\ \theta + 10\cos^5\ \theta + 5\cos\ \theta + 5\cos^5\ \theta - 10\cos^3\ \theta$

$= 16\cos^5\ \theta - 20\cos^3\ \theta + 5\cos\ \theta$

34. Last term in the expansion of $\cos 9\theta$ is $n = 9$ is odd

Last term is $(-1)^{\frac{n-1}{2}} {}^{n}c_{n-1} \cos\theta \sin^{n-1}\theta = (-1)^{4}\, {}^{9}c_{8} \cos\theta \sin^{8}\theta = 9\cos\theta \sin^{8}\theta$

35. $64\cos^{7}\theta = 2^{n-1}\cos^{n}\theta = \cos n\theta + {}^{n}c_{1}\cos(n-2)\theta + {}^{n}c_{2}\cos(n-4)\theta + \cdots$

$n = 7$

$2^{6}\cos^{7}\theta = \cos 7\theta + {}^{7}c_{1}\cos 5\theta + {}^{7}c_{2}\cos 3\theta + {}^{7}c_{3}\cos\theta$

$64\cos^{7}\theta = \cos 7\theta + 7\cos 5\theta + 21\cos 3\theta + 35\cos\theta$

7. THEORY OF EQUATIONS

worksheet – 1

Level – I

1.	D	4.	D	7.	B	10.	A		
2.	C	5.	A	8.	D				
3.	D	6.	B	9.	A				

1. Obtained by synthetic division

4	1	-6	3	26	-24
	0	4	-8	-20	24
	1	-2	-5	6	0

Quotient is $x^3 - 2x^2 - 5x + 6$; Re mainder is 0

2. Obtained by synthetic division

2	-3	5	-3	7	-9	
1	0	2	-1	10	4	0
3	0	0	6	-3	30	12
	2	-1	10	4	41	3

Quotient is $2x^3 - x^2 + 10x + 4$

Remainder is $41x + 3$

3. Required equation is $(x-1)(x+1)(x-3) = 0 \Rightarrow x^3 - 3x^2 - x + 3 = 0$

4. Required equation is $(x-1)(x^2 - 4x + 13) = 0 \Rightarrow x^3 - 5x^2 + 17x - 13 = 0$

5. Required equation is $\left(x^2 - 2\sqrt{3}x + 5\right)\left(x^2 + 2\sqrt{3}x + 5\right) = 0$

$$\Rightarrow (x^2+5)^2 - (2\sqrt{3}x)^2 = 0x^4 + 10x^2 + 25 - 12x^2 = 0 \Rightarrow x^4 - 2x^2 + 25 = 0$$

6. $\alpha^2 + \beta^2 + \gamma^2 = (\alpha\beta\gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = (-1)^2 - 2(1) = 1 - 2 = -1$

7. $x^2 - 3x^2 + 2x = 0 \Rightarrow x(x^2 - 3x + 2) = 0 \Rightarrow x(x-1)(x-2) = 0 \Rightarrow x = 0,1,2$

8.

	2	1	3	-4
	−	2	10	12
1	5	6	K + 12	

$x^2 + 5x + 6 = 0 \Rightarrow (x+2)(x+3) = 0 \Rightarrow x = -2, -3$

9. Required Equation $x^3 - x^2(-2+1+3) + x(-2+3-6) - 6 = 0$

$x^3 - 2x^2 - 5x - 6 = 0$

10. By synthetic division

x = −2	3	-1	2	-2	-4
	0	-6	14	-32	68
	3	-7	16	-34	64

Quotient: $3x^3 - 7x^2 + 16x - 34$

Remainder: 64

Level – II

1) 2	2) 3	3) 1	4) 2	5) 3	6) 3	7) 1	8) 1
9) 1	10) 2	11) 3	12) 1	13) 3	14) 1	15) 4	16) 2
17) 2	18) 4						

1. The equation having the roots 1,2,3 is $(x-1)(x-2)(x-3) = 0$

$x^3 - 6x^2 + 11x - 6 = 0$

2. The equation whose roots are $\alpha = -1, \beta = 2+\sqrt{3}, \gamma = 2-\sqrt{3}$

$$(x+1)\left(x-(2+\sqrt{3})\right)\left(x-(2-\sqrt{3})\right) = 0 \Rightarrow x^3 - 3x^2 - 3x + 1 = 0$$

3. $x = -32\ 7\ 2\ -3\ 2x - 1 = 0 \Rightarrow x = \frac{1}{2}$

$x = -1$	2	1	-1
		0	-2
		2	-1

$$0\ -6\ -3\ 3\ x = -3 - 1, \frac{1}{2}$$

4. $\sum \alpha_1\alpha_2\alpha_3 = S_3 = \frac{-d}{a} = \frac{4}{7}$

5. $S_1 = \frac{5}{2}\ S_2 = -\frac{3}{2}\ S_3 = -\frac{1}{2}$

$$\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}=\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma}=\frac{S_1}{S_3}=\frac{\frac{5}{2}}{\frac{-1}{2}}=-5$$

6. $(2) = 2(2)^5 - 3(2)^4 + 5(2)^3 - 7(2)^2 + 3(2) - 4$
$= 64 - 48 + 40 - 28 + 6 - 4 = 110 - 80 = 30$

7. $(-1)^3 - 3(-1) - 2 = 0 \Rightarrow -1$ is a root of $x^3 - 3x - 2 = 0$

$x^3 - 3x - 2 = 0 \Rightarrow (x+1)(x^2 - x - 2) = 0$
$\Rightarrow (x+1)(x+1)(x-2) = 0 \Rightarrow x = -1, -1, 2$

8.

-3	1	-13	15	189	
		-	-3	48	-189
	7	1	-16	63	0
		-	7	-63	
	9	1	-9	0	
		-	9		
		1	0		

9. $f(x) = x^4 - 4x^2 - x + 2 \Rightarrow f'(x) = 4x^3 - 8x - 1$

$2 - x - 4x^2 + x^3 + x^4\ \ -1 - 8x + 0x^2 + 4x^3\ \ -\frac{1}{2} - \frac{17}{4}x$

$-1 + \frac{1}{2} + 2x^2 + 0x^3 - \frac{1}{2}x^4$

$-\frac{17}{2}x - 2x^2 + 4x^3 + \frac{1}{2}x^4$

$-\frac{17}{2}x + \frac{17}{4}x^2 + 17x^3 + 0x^4 - \frac{17}{4}x^5$

$\sum \frac{1}{\alpha^2} = S_2$, coefficient of x in $\frac{f'(x)}{f(x)} = \left(\frac{17}{4}\right) = \frac{17}{4}$

10. $S_1 = 8,\ S_\lambda - 8\ S_1 + 23(2) = 0 \Rightarrow S_2 - 64 + 46 = 0 \Rightarrow S_2 = 18$
$S_3 - 8\ S_2 + 23\ S_1 - 28(3) = 0$

$S_3 = 44.$

11. $-1 + 2 + \alpha = \frac{-1}{2} \Rightarrow \alpha = -\frac{3}{2}$

12.

	1	-3	-1	3
-1	-	-1	4	-3
	1	-4	3	0

Quotient = $x^2 - 4x + 3$

13. The no. of changes of sign 1 ⇒ Number of positive roots is at most one.

14. $f(x) = x^9 + 5x^8 - x^3 + 7x + 2 \Rightarrow f(-x) = -x^9 + 5x^8 + x^3 - 7x + 2$

Number of changes of signs in $f(-x)$ is $3 \Rightarrow f(x) = 0$ has atmost 3 negative roots

15. $f(x) = x^7 + 5x^6 - x^3 + 7x + 2 \Rightarrow f(-x) = -x^7 + 5x^6 + x^3 - 7x + 2$

Number of changes of signs in f(x) is three ⇒ f(x) = 0 has at most three negative roots.

∴ f(x) = 0 has at most five real roots ⇒ f(x) = 0 has at least two imaginary roots

16 α, β, γ are the roots of $x^3 + 2x^2 - 3x - 1 = 0 \Rightarrow \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ are the roots of

$$x^3 + 3x^2 - 2x - 1 = 0, \alpha^{-2} + \beta^{-2} + \gamma^{-2} = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)^2$$

$$-2\left(\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}\right) = (-3)^2 - 2(-2) = 13$$

17. $f(x) = x^4 - x^3 + 2x^2 + kx + 17 = 0 \alpha + \beta = \gamma + \delta = \frac{1}{2}$

Product of two roots $(\alpha + \beta)(\gamma + \delta) + \alpha\beta + \gamma\delta = 2$

$$\frac{1}{4} + \alpha\beta + \gamma\delta = 2 - \frac{1}{4} \Rightarrow \alpha\beta + \gamma\delta = \frac{7}{4}$$

$$\alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) = -k \Rightarrow \frac{1}{2}\left(\frac{7}{4}\right) = -k;\ k = \frac{-7}{8}$$

18. $f(x) = ax^2 + bx^2 + cx + d = 0$ one factor is $x^2 - x + 1$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(1)}}{2 \times 1} \quad s_3 = \frac{-d}{a}$$

$$= \frac{1 \pm \sqrt{-3}}{2}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$= \frac{1 + \sqrt{3}i}{2}$$

$$(-w)(-w^2)r = \frac{-d}{a}$$

$$= \frac{-(-1 \pm \sqrt{3}i)}{2}$$

$$w^3 r = \frac{-d}{a}$$

$$x = -w, -w^2$$

$$r = \frac{-d}{a}$$

Two roots are $\alpha = -w, \beta = -w^2$, let 3rd root $\gamma = \frac{-d}{a}$.

Level – III

19) 1,2,3,4 20) 1 21) 2 22) 3 23) 4

24) a-r; b-q; c-s; d-p 25) 4 26) 1,2,3 27) 1 28) 1 29) 1 30) 2
31) a-s; b-r; c-q; d-p 32)

19. Let the roots on $a-d, a, a+d$

$s_1 = 3a = 3 \Rightarrow a = 1 \Rightarrow f(a) = 0 \Rightarrow f(1) = 0 \Rightarrow 1 - 3 + k + 3 = 0 \Rightarrow k = -1$

20. $s_3 = \frac{-d}{a} = \frac{-1}{3}$

21. $x^3 - 12x^2 + 39x + k = 0, a = 1, b = -12, c = 39, d = k$

Condition for roots to be in AP is

$2b^3 + 27a^2 d = 9abc \Rightarrow 2(-12)^3 + 27(1)(k) = 9(-12)$
$-3456 + 27k = -756 \Rightarrow k = -28.$

22. $x^3 - 7x^2 + 14x + k = 0$ condition $\Rightarrow ac^3 = b^3 d$

$1(14)^3 = (-7)^3 k \Rightarrow 14 \times 14 \times 14 = -7 \times (-7)(-7)k \Rightarrow k = -8$

23. $x^3 + 3ax^2 + 3bx + c = 0 \Rightarrow$ condition of roots to be in H.P $\Rightarrow 2c^3 + 27ad^2 = 9bcd$ $2(3b)^3 + 27(1)c^2 = 9(3a)(3b)(c) \Rightarrow 27(2b^3 + c^2) = 81abc$ $2b^3 + c^2 = 3abc \Rightarrow 2b^3 = 3abc - c^2 \Rightarrow 2b^3 = c[3ab - c]$

24. a) $(x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6 = 0$

b) $(x-5)\left(x+\frac{3}{2}\right)\left(x-\frac{4}{3}\right) = 6x^3 - 29x^2 - 17x + 60 = 0$

c) Required equation $(x-1)(x^2 - 4x + 13) = 0 \Rightarrow x^3 - 5x^2 + 17x - 13 = 0$

d) Required equation $x(x-1)(2x+3)(2x+5) = (x^2 - x)(4x^2 + 16x + 15) = 0$

$$4x^4 + 12x^3 - x^2 - 15x = 0$$

25. $\alpha + \beta + \gamma = 6 \Rightarrow 1 + 1 + \alpha = 6 \Rightarrow d = 4.$

26. $x^4 - 5x^2 + 6 = 0 \Rightarrow (x^2 - 2)(x^3 - 3) = 0 \Rightarrow x = \pm\sqrt{2}, \pm\sqrt{3}.$

27. $s_1^2 - 2\, s_2 = (-1)^2 - 2(1) = 1 - 2 = -1.$

28. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{s_2}{s_3} = \frac{q}{r}$ 29. $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} = \frac{5/2}{1/2} = 5$

29. $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = 10^2 - 2(6) = 100 - 12 = 88.$

31 .

a) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{q}{r}$

b) $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} = \frac{s_1}{s_3} = \frac{p}{r}$

c) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{(\alpha\beta+\beta\gamma+\gamma\alpha)^2}{(\alpha\beta\gamma)^2} - 2\alpha\beta\gamma(\alpha + \beta + \gamma) = \frac{q^2 - 2pr}{r^2}$

d) $\frac{1}{\alpha^2\beta^2}+\frac{1}{\beta^2\gamma^2}+\frac{1}{\gamma^2\alpha^2}=\frac{\alpha^2+\beta^2+\gamma^2}{(\alpha\beta\gamma)^2}=\frac{(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)}{(\alpha\beta\gamma)^2}$

$$=\frac{(-p)^2-2q}{(-r)^2}=\frac{p^2-2q}{r^2}$$

32. $(x-2)(x+3)(x-5)=0 \Rightarrow x^3-4x^2-11x+30=0.$

8. Sets

Worksheet – 1

Level – I

1) A	2)D	3)B	4)C	5) B	6)D
7) C	8) C	9)D	10)C	11)A	12) A
13)C	14)B	15)B			

1. $1^2 = 1, 2^2 = 4, 3^2 = 9, 4^2 = 16, 5^2 = 25, 6^2 = 36$, all are < 40.
2. The collection of ten most talented in your class.
3. (i) The collection of some stars in the sky is not a set.
   (ii) The collection of 1 vowels in the English alphabets is a set of because one can determine early the given letters is vowel.
   (iii) "Some prime numbers" is not a set, but all prime numbers is a set.
   (iv) "The number of brilliant students in your class is not a set because one cannot determine whether a student is brilliant or not.
4. {x/x is odd natural numbers &x < 15}

5. ϕ 7. 8 9. {1,2,3,4,5,6}

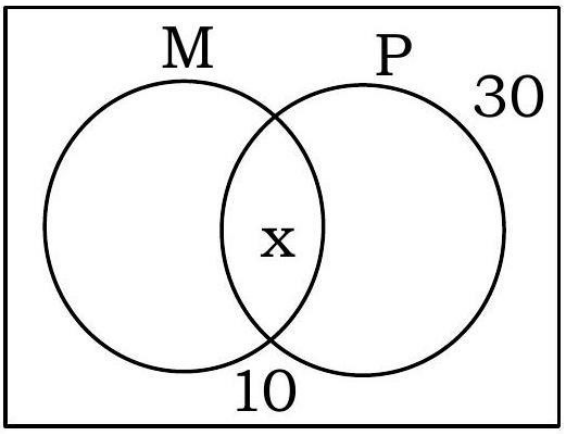
11. {1,2,3}
12. A
13. 5
14. $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 16 + 12 - 8 = 28 - 8 = 20$
15. $n(A) + n(B) = n(A \cup B)$

Level – II

1) D	2)C	3) B	4)B	5) B	6) C	7)D
8)D	9)B	10)D	11)B	12) B	13)B	
14)C	15)A	16)A	17)C	18)B	19)A	
20)C						

1. The collection of all natural numbers less than 100, all the months starting with ' j ' is also a set.
2. $A = (2,3,4,5,6)\ B = \{2,3\}\ n(A) > n(B)$
3. $A = \{1,3,7,15,31\} B = \{2,4,8,16\}$
   $A \Delta B = (A \cup B) - (A \cap B) = \{1,2,3,4,7,8,15,16,31\} - \{\} \Rightarrow = \{1,2,3,4,7,8,15,16,31\}$
4. $A = \{3,9,27,81,243,729\} \Rightarrow B = \{9,81,729,6561\} \Rightarrow A \cap B = \{9,81,729\}$
5. 2^n
6. The set of all integers except Zero
7. $A - (B \cap C) = \{2,3,4,5,6\} - \{4,5\} = \{2,3,6\}$
8. $A \Delta B = (A \cup B) - (A \cap B) = \{1,2,3,4,5\} - \{4,5\} \Rightarrow A \Delta B = \{1,2,3\}$
9. $\{3,4,5,6, \ldots \ldots 100\} \Rightarrow A = \{y/y = x + 2, x \in N\}$

10. Given $A = \{a, b\} \Rightarrow$ power set of A is $p(A) = \{\{a\}, \{b\}, \{a, b\}\phi\}$
11. Here N is natural number $\Rightarrow$ Given set $\{x; x \in N; x^2 = 1\}$
    put $x = 1 \Rightarrow 1^2 = 1$, So, $1 \in$ Given set if we put $x = 2 \Rightarrow 2^2 \neq 1$, So, $2 \notin$ Given set so, we observe that the above condition $x^2 = 1$, is satisfied for $x = 1$ so, the above set contains only one element.
12. $x^2 + 1 = 0 x = \sqrt{-1} \Rightarrow \therefore$ x is imaginary



Total No. of students $n(\mu) = 30\ n(M) = 10\ n(P) = 15$
$n(M \cup P) = 30 - 10 = 20 \Rightarrow$ Let $n(M \cap P) = X$
$n(M \cup P) = n(M) + n(P) - n(M \cap P) \Rightarrow 20 = 10 + 15 - X \Rightarrow X = 5$

14. $n(\mu) = 200, n(F) = 120 n(C) = 70 \Rightarrow n(F \cap C) = n(F) + n(C) - n(F \cap C)$
    $= 120 + 70 - 60 = 130$
    $\therefore$ No. of students neither football nor cricket
    $= n(\mu) - n(F \cup C) = 200 - 130 = 70$
15. $n(T \cup C) = 30 n(C) = 15 n(T) = ?$
    No. of people those who do not take teat $n(C - T) = 12$
    $\therefore n(T \cup C) = n(T) + n(C - T) \Rightarrow 30 = n(T) + 12 \Rightarrow n(T) = 18$
16. A = { multiplies of 3}, B = { multiply of 4} $\Rightarrow$ L.C.M of 3 and 4 is 12, $A \cap B = 12n$
17. $n(A \cup B) = n(A) + n(B) \Rightarrow = 3 + 6 = 9$
18. $X = \{2,10\}, Y = \{-7,2\} \Rightarrow X - Y = \{10\}$
19. $n(\mu) = 60, n(A) = 21, n(B) = 43$ Greatest value of $n(A \cup B) = n(\mu) = 60$,
    Least value of $n(A \cup B) = n(B) = 43$
20. $n(A \cup B)^c = n(\mu) - n(A \cup B) \Rightarrow n(A \cup B) = n(A) + n(B) - n(A \cap B) = 50 + 20 - 10 = 60$
    $n(A \cup B)^c = n(\mu) - n(A \cup B) = 100 - 60 = 40$

Level – III

21)A,D	22)A,B,C	23)A	24)C	25)A
26)D	27)C	28)D	29)C	30) a-q, b-p, c-s,d-r
31) A,B,C	32)A	33) A,D	34) B	35) A 36) A

21. $A^1 = \{x \in R: x \leq 0 \text{ or } x \geq 2\} \Rightarrow B^1 = \{x \in R: x \leq 1 \text{ or } x > 3\} \Rightarrow A \cap B = \{x \in R: 1 < x < 2\}$
22. Conceptual
23. Both statements I and II are true people who are living in A.P are also Indians.
24. $A = \{3,5,7,9,11\}\ B = \{1,4,7,10\}$
    $A \cup B = \{1,3,4,5,7,9,10,11\}$
25. $Y = \{3,5,9\}$
26. $A = \{-2,-1,0,1,2\}\ B = \{0,1,2,3\}\ C = \{1,2\}$
27. $n(A \cup B) = n(A) + n(B) = 10 + 12 = 22$
28. $n(A \cup B) = n(A) + n(B) = 20 + 30 = 50$
29. $n(A \Delta B) = n(A \cup B) - n(A \cap B)$
    $(\because n(A \cup B) = 100 + 70 - 20 = 150)$
    $= 150 - 20 = 130$
30. $1.n(A - B) = m - p$
2. $n(A \cup B) = m + n - p$
3. $n(A \Delta B) = m + n - 2p$
4. $n(B - (B \cap A)) = n - p$
31. A-B, $A \cap B^1$, $A - (A \cap B)$
32. Both statements I and II are true.
33. Let A = set of person, who drink tea
    B = set of persons who drink coffee
    $A \cap B$ = set of persons who drink tea and coffee both
    $\therefore n(A \cap B) = n(A) - n(A - B) = 33 - 16 = 17$
    No. of persons who drink coffee but not tea = $n(B - A)$
    $n(B - A) = n(B) - n(A \cap B) = 33 - 17 = 19$
    $\therefore$ 17 drink tea and coffee both
    No. of persons who drink coffee but not tea = 19
34. $A = \{2,3,5,7,9,11\}\ B = \{2,3,5,$
    $A \cap B = \{2,3,5\}$
35. $A - B = \{7,9,11\}$
    $(A \cap B)^1 = \{4,6,8,10,7,9,11\}$

9. RELATIONS

Worksheet – 1

Level – I

1) D	2) C	3) B	4) D	5) C	6) B
7) C	8) A	9) B	10) B		

1. $n(A \times B) = 6$
2. $\{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)\}$
3. $2^{2m} = 2^6 \Rightarrow 2m = 6 \Rightarrow m = 3$
4. $2n^2$
5. $\therefore (x, y) = (2,3): x + 1 = 3, \Rightarrow x = 2 \Rightarrow y - 2 = 1, y = 3$
6. $\{1,2,3,4\}$ 7. $\{(1,1)(2,4)(3,9)(4,16)\}$
7. 64
8. $n(A) = 2, n(B) = 2$: No of relations $= 2^{n(A) \times n(B)} = 2^{2\times 2} = 2^4$
9. $R = \{(1,2), (2,3), (3,4), (4,5), (5,6)\} = \{2,3,4,5,6\}$

Level – II

1) A	2) D	3) D	4) B	5) A	6)B	7)D
8)C	9)C	10)B	11)B	12)D	13)B	
14)C	15)A	16)C	17)B	18)A	19)D	
20) A,B,C,D	21)A,B,C	22)A	23)A	24)B	25)C	
26)a-r,b-t,c-s,d-p		27) A	28)A	29)C	30)A	31)A,B,C
32)A	33)D	34)A	35)6			

36)a-r,b-s,c-q,d-p

1. We have $A = \{2,3\}\ B = \{2,4\}\ C = \{4,5\}$
   $\therefore B \cap C = \{4\} \Rightarrow Ax(B \cap C) = \{(2,4), (3,4)\}$
2. $(Y \times A) n (Y \times B) = Y \times (A \cap B) = Y \times \phi = \phi$
3. $A = \{3,2\}, B = \{2\}, C = \{4,5\}, (A - B) \times (B - C) = 3 \times 2 = \{(3,2)\}$
4. Domain $\in$ A and range $\in$ B
5. Given $A = \{1,2,3\}\ B = \{2,4\}$
   R is relation from A to B which satisfies $x = y \Rightarrow R = \{2,2\}$
6. $(-1,0) \in A \times A$ and $(0,1) \in A \times A \Rightarrow (-1,0) \in A \times A \Rightarrow -1,0 \in A\ -1,0,1 \in A$
   Hence $A = \{-1,0,1\}$
7. $\therefore 3x - y = 0 \Rightarrow y = 3x \Rightarrow \therefore R = \{(1,3), (2,6), (3,9), (4,12)\}$
8. Given ' x ' greater than ' y ' and R is a relation from A to B
   $R = \{(2,1), (3,1)\}$ Range of $R = \{1\}$
9. $Z = \text{Re}(z)$ only if $Z \in R$
10. $(2,2) \notin f$
    f is not reflexive

if $(a,b) \in f$, then $(b,a) \in f$ $\therefore$ f is symmetric
$(1,2) \in f, (2,1) \in f$ but 2 is not equal to 1
$\therefore f$ is not antisymmetric also f is not equivalence

11. $R = \left\{(1,1), \left(2,\frac{1}{2}\right), \left(3,\frac{1}{3}\right), \left(4,\frac{1}{4}\right), \left(5,\frac{1}{5}\right)\right\}$ domain of $R = \{1,2,3,4,5\}$
12. We have $(a, b) \in R \Leftrightarrow |a^2 - b^2| < 16$
$\therefore a = 1 \Rightarrow |1 - b^2| < 16 \Rightarrow |b^2 - 1| < 16$
$\Rightarrow -15 < b^2 < 17 \Rightarrow 0 < b^2 < 17$
$\Rightarrow b = 1,2,3,4.\ a = 2$
$\Rightarrow |4 - b^2| < 16 \Rightarrow |b^2 - 4| < 16 \Rightarrow -12 < b^2 < 20$
$\Rightarrow 0 < b^2 < 20$
$\Rightarrow b = 1,2,3,4. a = 3 \Rightarrow |9 - b^2| < 16 \Rightarrow |b^2 - 9| < 16$
$\Rightarrow -7 < b^2 < 25 \Rightarrow 0 < b^2 < 25$
$\Rightarrow b = 1,2,3,4.\ a = 4 \Rightarrow |16 - b^2| < 16$
$\Rightarrow |b^2 - 16| < 16 \Rightarrow 0 < b^2 < 32 \Rightarrow b = 1,2,3,4,5$
$a = 5$
$\Rightarrow |b^2 - 25| < 16 \Rightarrow 9 < b^2 < 41 \Rightarrow b = 4,5.$
Then R = {(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4)(3,1), (3,2)(3,3), (3,4), (4,1), (4,2), (4,3), (4,4), (4,5), (5,5)}
13. No element in A is greater than any element of B $\Rightarrow \phi$
14. A = {1,2,3, … … 10}
$\Rightarrow R = \{(1,2), (2,4), (3,6), (4,8), (5,10)\}$
R = {(x, y)/(x, y) ∈ AxA: y = 2x}
15. (1,2), (1,3), (1,5), (2,3), (2,5), (3,5), ∈ R
16. $\{0, \pm 3, \pm 4, \pm 5\}$
17. R = {(1,7), (2,5), (3,5)} Domain {1,2,3} Range {7,5}
18. For any element x; presesnt in x, y cases arises while making subsests y and z.
Case1: x; ∈ y, x: ∈ z ⇒ y ∩ z ≠ ϕ ⇒ Case2: x; ∈ y, x: ∈ Z ⇒ y ∩ z ≠ ϕ
Case3: x; ∈ y, x: ∈ z ⇒ y ∩ z ≠ ϕ ⇒ Case4: x; ∈ y, x: ∈ z ⇒ y ∩ z ≠ ϕ
For any element, numbers of ways = 3 for which y ∩ $z \neq \phi$
Total number of ways = 3 × 3 × 3 × 3 × 3
[∵ No. of elements in set x = 5] = $(3)^5$
19. R = {(1,1), (1,2), (2,1), (2,2), (2,3), (3,2), (3,3)}
20. Since $(a,2), (b,3), (c,2), (d,3), (e,2)$ are elements of A X B $a, b, c, d, e \in A$ and 2,3 $\in B$
Its gives that n(A) = 5 and n(B) = 2 ⇒ a, b, c, d, e ∈ A and n(A) = 5 ⇒ A = {a, b, c, d, e} 2,3 ∈ B and n(B) = 2 ⇒ B = {2,3}
21. $a + 3b = 12 \Rightarrow a = 12 - 3b$
putting b = 1,2,3 we get a = 9,6,3 for b = 4,
we get a = 0 ∈ N for b > 4, a ∈ N $\therefore R = \{(9,1)(6,2)(3,3)\}$
22. n(A × B) = n(A) × n(B) = 3 × 5 = 15
23. $A = \emptyset, B = \{1,2,3,4,5\} A \times B = \emptyset n(A \times B) = 0$

24. $\mathrm{n(A \times B) = n(A) \times n(B) = 3 \times 3 = 9}$
25. $A - B = \{1,3,5\}, \mathrm{B - A} = \{2,4,6\}$
    $n((A - B) \times (B - A)) = 3 \times 3 = 9$
26. a) $\mathrm{A \times (B \cup C) = (A \times B) \cup (A \times C)}$
    b) $A \times (B \cap C) = (A \times B) \cap (A \times C)$
    c) If $A \subseteq B$ and $C \subseteq D = (A \times C) \subseteq (B \times D)$
    d) $A \times (B - C) = (A \times B) - (A \times C)$
27. $\mathrm{b} - \frac{2}{3} = \frac{1}{3}; \mathrm{b} = \frac{1}{3} + \frac{2}{3} = \frac{3}{3} = 1$
28. $\mathrm{R} = \{(1,3), (1,5), (2,3), (2,5), (3,5), (4,5)\}$
29. $R^{-1} = \{(3,1), (5,1), (3,2), (5,2), (5,3), (5,4)\}$
30. $\mathrm{R} = \{1,2,3,4\}$
31. We have aRb if $a, b \in N$ and $a + 2b = 11$
    The smallest natural number is ' 1 '
    $b = 1 \Rightarrow a + 2(1) = 11 \Rightarrow a = 9$
    $\mathrm{b} = 2 \Rightarrow \mathrm{a} + 2(2) = 11 \Rightarrow \mathrm{a} = 7$
    $b = 3 \Rightarrow a + 2(3) = 11 \Rightarrow a = 5$
    $b = 4 \Rightarrow a + 2(4) = 11 \Rightarrow a = 3$
    $b = 5 \Rightarrow a + 2(5) = 11 \Rightarrow a = 1$
    $b = 6 \Rightarrow a + 2(6) = 11 \Rightarrow a = -1 \in N$
    $\therefore \mathrm{R} = \{(9,1), (7,2), (5,3), (3,4), (1,5)\}$
    $\Rightarrow \therefore$ Domain of $\mathrm{R} = \{9,7,5,3,1\}$
    Range of $R = \{1,2,3,4,5\}$
    $R^{-1} = \{(1,9), (2,7), (3,5), (4,3), (5,1)\}$
32. $\{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (1,3), (1,4), (1,5), (2,4)\}$
33. $\{(1,1), (2,2), (3,3), (4,1), (5,5), (2,1), (3,1), (4,1), (5,1), (4,2)\}$
34. $\{(3,5), (5,3), (4,4)\}$
35. Anti symmetric relations $= 2^n . 3^{\frac{n(n-1)}{2}}$
    $n = 3$
    $m = \dfrac{n(n-1)}{2} = \dfrac{3(2)}{2} = 3, \Rightarrow m + n = 3 + 3 = 6$
36. a) Domain = {7,8,9 ....
    b) Range = {1,2,3 .....
    c) Domain = {1,2,3,4,6,12}
    d) Range = {1,2,3,4,5}